

AMP 201: LIVESTOCK AND POULTRY PRODUCTION MANAGEMENT (2+1)

LECTURE NOTES



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UNIT I: INTRODUCTION TO LIVESTOCK MANAGEMENT

LIVESTOCK IN INDIA

Livestock farming is an integral part of crop farming and contributes substantially to household nutritional security and poverty alleviation through increased household income. Indian agriculture is an economic symbiosis of crop and livestock production with cattle as the foundation. Dairy animals produce milk by converting the crop residues and by products from crops which otherwise would be wasted. Dairy sector contributes by way of cash income, draught power and manure. Livestock provides for human needs by way of 1.Food, 2.Fiber, 3.Fuel, 4.Fertilizer, 5.Skin and 6.Traction. Indian agriculture marches on the back of the bullock. 70 per cent of the livestock are owned by small and marginal farmers. 76 per cent of the milk is produced by weaker sections of society. India has vast resource of livestock and poultry, which play a vital role in improving the socio-economic conditions of the rural masses. There is no other country in the world having varied domestic animal biodiversity like India. Livestock Census has been conducted in the country periodically since **1919-20**. So far 19 such censuses have been conducted by the **Ministry of Fisheries, Animal Husbandry and Dairying**, in participation with State Governments and UT Administrations. The household-level data through online transmission from the field has been used for the first time in 20th Livestock Census.

As per the 20th Livestock census (2019), India's livestock sector is one of the largest in the world, with a share of 11.6% of world livestock population. It has 57 per cent of the world's buffalo population and 16 per cent of the cattle population in the world. It ranks first in respect of cattle and buffalo population, third in sheep and second in goat population in the world. The livestock population shows high degree of diversity in its composition. Out of total livestock in the country, 35.9 per cent cattle, 20.4 per cent buffaloes, 13.8 per cent sheep, 27.8 per cent goats and only 1.6 per cent were pigs. All other animals are less than 0.23 per cent of the total livestock. However, only Camel and Donkey population is showing declining trends over last four Livestock Censuses i.e., 17th (2003), 18th (2007), 19th (2012) and 20th (2019) Livestock Census.

The species-wise breakup of livestock population in India

S.No.	Species	Livestock population (in millions)		
		2012	2019	
		Number	Number	Growth %
1	Cattle	190.90	192.49	0.83

2	Buffalo	108.70	109.85	1.06
3	Yaks	0.08	0.06	-25.00
4	Mithuns	0.30	0.38	26.66
4	Sheep	65.07	74.26	14.13
5	Goat	135.17	148.88	10.14
6	Pig	10.29	9.06	-12.03
7	Horses & ponies	0.63	0.34	-45.58
8	Mules	0.20	0.08	-57.09
9	Donkeys	0.32	0.12	-61.23
13	Camels	0.40	0.25	-37.05
Total livestock		512.06	535.78	4.63

Poultry	2012 (in millions)	2019 (in millions)	% growth
Backyard poultry	217.49	317.07	45.78
Commercial Poultry	511.72	534.74	4.50
Total Poultry	729.21	851.81	16.81

According to the National Council of Applied Economic Research (NCAER) cattle, dung in India has a fuel value equivalent to 35 million tonnes of coal or 68 million tonnes of wood. An estimated one-third of the dung, amounting to some 300 million tonnes, was used as fuel in rural houses. Another 340 million tonnes go back to the soil as organic fertilizer. The available energy from animal power is estimated at around 60,000 million kilo watt hours, valued at between Rs. 60,000 and 100,000 million from 70 million bullocks, 8 million buffaloes and one million horses and another million camels. To generate this amount of energy by modern industrial process would cost three times as much. Thus in modern - era livestock not only produces the protein rich food for human beings (milk, meat and egg) but also provides sustainable income to the rural poor through draught power, fuel and provides the only natural organic manure the dung for sustainable agriculture.

Livestock Population (2012 & 2019) of major states

S.No	States	2012	2019	% of change
1	Uttar Pradesh	68.7	67.8	-1.35
2	Rajasthan	57.7	56.8	-1.66
3	Madhya Pradesh	36.3	40.6	11.81
4	West Bengal	30.3	37.4	23.32

5	Bihar	32.9	36.5	10.67
6	Andhra Pradesh	29.4	34.0	15.79
7	Maharashtra	32.5	33.0	1.61
8	Telangana	26.7	32.6	22.21
9	Karnataka	27.7	29.0	4.70
10	Gujarat	27.1	26.9	-0.95

Livestock population in Tamil Nadu

Tamil Nadu ranks 1st in poultry, 4th in sheep, 7th in goat, 13th in cattle and 14th in buffalo population in the country. Tamil Nadu ranks second in egg production, with a share of 17.5% of the total eggs produced in the country. During 2019-20, Tamil Nadu produced 20.02 billion eggs with a per capita availability of 265 eggs per annum. The total meat production in India in 2019-20 was 8.6 million tonnes with contribution of poultry meat by 50.5%. Tamil Nadu produced 0.66 million tonnes of meat in 2019-20 with the share of 7.72% in total meat production and ranks sixth in the nation.

Species	2012 (in millions)	2019 (in millions)
Cattle	8.81	9.51
Buffaloe	0.78	0.51
Sheep	4.78	4.50
Goat	8.14	9.89
Pigs	0.18	0.07
Livestock	22.72	24.49
Poultry	117.3	120.8

LIVESTOCK AND ROLE IN ECONOMY

Livestock population dynamics:

India has huge livestock population with 199 million cattle, 105 million buffaloes, 140 million goats, 71 million sheep, 11 million pigs and 648 million poultry birds. The composition of livestock population in broad groups like bovine (cattle and buffaloes), ovine (sheep and goats), pigs and poultry, however, has changed over the last two decades. Cattle population that had been increasing until 1992 has started declining between 1997 to 2003 and again increasing between 2003 and 2007.

Contribution of livestock to Indian economy

Livestock plays a major role in our economy and contributed about 4.11 per cent of GDP, 25 per cent to the Agricultural Gross Domestic Product. It provided employment to 8.8 per cent of population in India. Livestock contributed 16% to the income of small farm

households as against an average of 14% for all rural households. Over the last two decades, livestock sector has grown at an annual rate of 5.6 percent, which is higher than the growth of agricultural sector (3.3 per cent). This suggests that livestock is likely to emerge as an engine of agricultural growth in the coming decades. It is also considered as one of the potential sector for export earnings. The importance of livestock goes beyond its food production function. Livestock sector also makes significant contributions towards conservation of environment. Livestock sector supplements income from crop production and other sources and absorbs income shocks due to crop failure. It generates a continuous stream of income and employment and reduces seasonality in livelihood patterns particularly of the rural poor. In value terms, milk continues to dominate livestock production structure and its share has increased between 1980 and 1999 from 43% to 57%. In 1997- 1999 milk has become the number one farm commodity exceeding the value of paddy. In contrast the share of draught power declined from 33% to 10%. Meat and meat products share in 1997-98 exceed the share of draft power. Milk production reached 100.9 million tonnes at the end of 10th plan (2006-2007) as compared to 53.9 million tonnes in 1990-91. Species wise cattle outputs (milk, meat and draught power etc.) dominated the shares in early eighties but by 1997 the share of buffalo exceeded share of cattle and now has the highest share (42%) in total livestock value from all species.

Livestock development for rural development

The role of livestock in the lives of small farmers and village poor is many folds as shown below,

1. It buffers the risks due to crop failure under employment.
2. Enhances family nutrition status, allows some domestic consumption of essential nutrients.
3. Provides regular cash income.
4. Render services-draft, manure and fuel.
5. Helps enhance socio-economic status.

Role of Animal Husbandry / Livestock:

Animal husbandry plays an important role in augmenting rural income, health and employment as follows:

1. Provides food like milk, meat and other value added products, thus preventing poverty and helping to have self-sufficiency in food security.
2. Provides valuable organic manure for agriculture, thus improving soil fertility and productivity.
3. Provides draught animal power for different agricultural operations and also for transportation, thus helping to conserve fossil fuel.
4. Provides a platform for ensuring sustainability (biological and economic), thus bringing about a change in the social status (society).

5. Provides skin and hides for leather products, thus helping to make additional income and earn foreign exchange.
6. Provides other by-products like hoof and horns, thus promoting rural handicraft industries.
7. Provides wool and other natural fibres of importance, thus helping to improve income and promote industrial growth.
8. Provides blood as a by-product, thus helping to produce haematinics and other pharmacological products.
9. Provides gainful rural employment for many people, thus helping to have full-time self-employment.
10. Provides a means of storing wealth and also a way to meet social obligations, thus helping to have security and overcome difficult times.
11. Provides a means of controlling weeds by grazing, thus helping to keep the premises clean
12. Overall, Animal husbandry can be considered as a major instrument for effecting a desirable change in the society.

MILK PRODUCTION

World level contribution:

India is the world's largest milk producer, with 23 percent of global production. India ranks first in milk production (146.31 million tonnes) followed by USA (93.5 million tonnes), China (45 million tonnes), Pakistan (42 million tonnes) and Brazil (35 million tonnes) during the year 2010. The total milk production worldwide is 720.98 million metric tonnes. Every year, since 2001, **June 1** is observed as **World Milk Day** by the Food and Agriculture Organization (FAO) of the United Nations to acknowledge the importance of milk as a global food and to celebrate the dairy sector.

National level contribution:

India's milk production in 2019-20 was 198.44 million tonnes with per capita availability of 406 grams per day. Tamil Nadu ranked 11th in 2019-20 with a share 8.76 million tonnes contributed about 4.41% in the country. The per capita availability was 316 grams per day. Milk production in Tamil Nadu has increased by 4.75% over the previous year. **Dr. Varghese Kurien**, widely renowned as the "**Father of White Revolution**" in India, was the first chairman of NDDB. In India, the birthday of **Dr. Varghese Kurien**, on **November 26**, is observed as **National Milk Day**. In three decades (the 1980s, 1990s and 2000s), the daily milk consumption in the country rose from a low of 107 grams per person in 1970 to 427 grams per person in 2020-21 as against the world average of 322 grams per day during 2021.

Role of dairying in India:

1. More than 70 per cent of population depends on income from agriculture and animal husbandry is an adjunct to crop agriculture and cattle and buffaloes are kept for milk

production, motive power of various farm operations. The animals are generally maintained on agricultural by-products.

2. Cattle and buffalo convert larger quantity of agricultural wastes and by-products into milk and provide gainful employment to large agricultural labour force.

3. The cattle rearing are mainly in the hands of small and marginal farmers whose land holding is meager, being $\frac{1}{2}$ to 2 acres. Dairy farming provides sustainability to the landless labourers, small and marginal farmers.

4. Dairying contributes about 65 per cent to the total income of landless labourers and 35 per cent to that of marginal and small farmers.

5. Dairy is an important sub-sector of the Indian agriculture accounting for nearly 17 per cent of the value of output from agriculture and allied activities. India is the largest producer of milk in the world with estimated production of about 146.31 million tonnes followed by the USA, although in terms of milk yield, the performance of Indian dairy sector is dismal.

7. Milk production over the last decades had increased by about 4 to 5 per cent per annum. The per capita availability of milk has improved from 128 g/day in 1980 to 406 g/day in 2019-20.

8. The milk production was almost doubled from 1990-91 (55.7 million tonnes) to 2006- 2007 (100.9 million tonnes) which showed the improvement in milk production.

9. The milk is the single largest item which is estimated to fetch Rs.450 billion ways ahead of rice and wheat. The estimated value of milch animals alone is around Rs.35 billion. Dairy animals also contribute to hides/skin and dung valued at Rs.60 billion.

11. Draught animal power (DAP) investment in animal energy sector is more than Rs.35 billion. Currently DAP contributes 57% of total farm energy requirement through about 72 million bullocks.

12. Estimated value of a good part of dung is used as FYM. In terms of nutrient nitrogen when its value is estimated in terms of fertilizer the value comes to about Rs.3, 300 crores, (World Bank report on energy sector in India).

13. Apart from manurial value biogas can be produced from livestock dung and poultry droppings. 32 kg of cow dung/20 kgs of pig feces/12 kgs of poultry droppings can produce 1 m³ to 34 m³ of bio-gas. The calorific value of bio-gas is 500 to 700 BTU per cft in comparison to natural gas, which is about 850 BTU/cft.

14. 1 m³ of slurry fed to biogas plant produces on an average 0.15 to 0.20 m³ of biogas daily. Based on equivalent effective heat produced 2 m³ biogas plant replaces in a month fuel equivalent of 26 kgs of LPG contained in standard gas cylinder or 37 litres of kerosene or 88 kgs of charcoal or 210 kgs of fuel wood or 740 kgs of animal dung.

In conclusion it can be summarized that the dairy farming once a subsidiary occupation has now become central to lives of millions of families. It is symbiotically associated to crop

farming sustained largely on crop residues and by-products. In low rainfall areas like arid and semi-arid & hill areas, dairying serves as food security and draught preventing. Dairying in India is thus an environmental friendly activity. It maintains the quality of environment through milk, bio-fertilizer, farm power and fuel energy.

DIFFERENT LIVESTOCK DEVELOPMENT PROGRAMMES OF GOVERNMENT OF INDIA

Centrally-Sponsored Dairy Development Schemes

Dairy entrepreneurship development scheme Intensive
Dairy Development Programme

Centrally-Sponsored Animal Husbandry Schemes

- ☐ Fodder and Feed Development Scheme
- ☐ Pig Development Scheme
- ☐ Poultry Venture Capital Fund
- ☐ Establishment and Modernization of Rural Slaughter Houses
- ☐ Salvaging and Rearing of Male Buffalo Calves
- ☐ Utilization of Fallen Animals
- ☐ Integrated Development of Small Ruminants (Sheep, Goat) and Rabbits
- ☐ Conservation of Threatened Breeds of Small Ruminants, Rabbits, Pigs, Pack Animals and Equines
- ☐ Livestock Insurance Scheme

Other schemes

- ☐ NABARD Venture Capital Fund Scheme for Dairy/Poultry
- ☐ Govt of India Scheme - Dairy Entrepreneurship Development Scheme
- ☐ Govt of India Scheme - Small Ruminants and Rabbits
- ☐ Uttarakhand Regional Economic Development Programme
- ☐ SBI to offer concessional rates to dairy, poultry farmers till 31 Mar 2011

Scheme activities of Tamil Nadu

State Sponsored Schemes

- ☐ Kalnadai Padhukappu Thittam
- ☐ Capacity Building for Poverty Reduction Scheme
- ☐ Scheme for free distribution of milch cows
- ☐ Scheme for free distribution of goats / sheep
- ☐ State fodder development scheme
- ☐ Centrally sponsored fodder development schemes

World Bank Assisted Projects

- ☐ Tamil Nadu Irrigated Agricultural Modernization and Water Bodies Restoration and Management (TNIAMWARM project)

Centrally Sponsored Schemes

- ☐ Rural Backyard Poultry Development
- ☐ Hill Area Development programme (HADP)
- ☐ National Agricultural Development Programme (NADP)
- ☐ National Mission for Protein Supplementation (NMPS).
- ☐ Accelerated Fodder Development Programme
- ☐ National Project on Rinderpest Eradication (NPRE)
- ☐ Foot and Mouth Disease Control Programme
- ☐ National Animal Disease Reporting System (NADRS)
- ☐ National Control Programme of Peste Des Petits Ruminants (NCP PPR)
- ☐ National Control Programme for Brucellosis (NCPB)
- ☐ Assistance to States for Control of Animal Diseases (ASCAD)
- ☐ Poultry Development
Assistance to State Poultry/Duck Farms,
Rural Backyard Poultry Development,
Establishment of Poultry Estates
- ☐ Livestock Insurance
- ☐ Conservation of Threatened Livestock Breeds

SYSTEMS OF LIVESTOCK PRODUCTION

Livestock production systems:

In general, husbandry systems are usually classified as **intensive, semi-intensive and extensive**.

Meat and dual production systems which is common in the tropical countries can be classified as,

1. Extensive (migratory, free range, pasture or range grazing).
2. Semi-intensive (pasture or range grazing, use of supplementary feeding mainly on crop residues and conserved roughage)
3. Intensive (grazing on improved pastures, zero grazing, conserved forage, crop residues and increased use of concentrates).
4. Tethering (small size flocks of 2-10 animals). This is a subsistence family system and the animals live on kitchen remnants crop residues, grazing near inhabited areas and other supplementary feed.

Extensive system:

In this system, the livestock are kept on grassland, all the operations are in open place. Small shelters are made for young animals during extreme weather conditions.

1) Nomadic and transhumant system are the most difficult to improve, because they involve continuous movement, not only of the whole flock, but also of all its owners. There is, therefore, no possibility of dividing the flock by age, sex or stage of reproductive cycle with other goats. It is essential to change the place of tethering every day for obtaining fresh herbage and a variety of different feed plants by the animal.

I. Intensive System:

In intensive system all the operations are confined in one place and animal movement is restricted. Poultry, pig, rabbit are more suitable for this system. In developed countries dairy also maintained by intensive system. Intensive production systems involve either grazing on crops or cultivated pastures at a very high stocking density or zero-grazing. The control of goats by fencing is both difficult and expensive. Fences need to be at least 1.5m high and made of strong wire netting, closely placed taut wires or wooden rails.

II. Mixed Farming:

Along with crop husbandry one or more component of livestock or poultry maintained. Mixed farming is the economical rearing of different types of livestock & poultry in the farm along with

- Better utilization of farm by products.
- Utilization of unconventional feed and fodder
- Recycling : Farm Yard Manure – Dung – Gas – Slurry – Soil fertility
- Bring constant income to the family throughout the year
- Indirectly enhances standard of living.

Integrated farming system – (IFS)

In the integrated farming system the defects of mixed farming is overcome by proper planning, monitoring and execution of work according to size of the farm, farm resources, Agro climatic etc. In this type, the type of livestock species or poultry enterprises are selected based on the availability of feed, fodder, water resources of the farm.

III. Specialized farm

1. Sole income is derived from one species – Cattle, Buffalo, goat, pig or poultry

- White cattle
- Black cattle
- Sheep
- Goat
- Poultry

2. Pure Breed

- Breeding policy
- Income from sale of breeding bulls (eg.) Work Bullock (Kangayam)

COMMON TERMS USED IN ANIMAL HUSBANDRY

Abomasum: Abomasum is the fourth compartment, or true stomach, of the ruminant animal

Artificial insemination : Artificial insemination is the placing of sperm in the female reproductive tract by other than natural means

Bitch : An adult female dog

Boar : An adult uncastrated male pig used for breeding

Bobby calf : A male calf about 1 week old

Bos indicus : Bos indicus is humped cattle found in tropical countries

Bos Taurus : Bos taurus is the domestic cattle originating from either the Aurochs or the Celtic shorthorn

Bovine : Bovine is a generic name for cattle

Breed : Breed is a group of animals with distinct shapes and colors which produce offspring with similar shapes and colors

Breeding Bull or Stud Bull : An adult male used for breeding

Broken-mouth: Broken-mouth is a mouth having teeth missing. Usually applied to sheep or goats and occurs with old age

Browse: Browse is the shoots, twigs, and leaves of brush plants found growing on rangeland. Browse is a fodder obtained from eating leaves and twigs of bushes

Buck : Buck is a male of such species as goats, deer, rabbits, etc, Buck is an intact mature, male goat

Buckling : Buckling is an intact, immature male goat

Buffalo bull calf : A male young one

Buffalo calf : A young one of either sex

Buffalo heifer calf : A female young one

Bull : Bull is a male of bovine of any age that has not been castrated

Bull calf : A male young one under 1 year of age

Buller: A female which is always in heat or estrus. It is also known as a nymphomaniac animal.

Bullock or Steer : An adult castrated male used for carting and tillage

Burdizzo : Burdizzo is an instrument used for bloodless castration which clamps off the tissue connecting the testis to the rest of the body

Calf : Calf is a beef animal under one year of age, Calf is a young bovine of either sex

Calving: The act of giving birth to a young one

Calving interval: Calving interval is the length of time from one calving to the next

Cannibalism: Cannibalism is the act of eating another member of the same species.

Seen in poultry and swine

Capon: Capon is a male chicken whose reproductive organs have been removed or rendered inactive while the individual is still young

Caponization: A process in which the testes are removed from the adult male. This is done to make a good table bird (for eating purpose).

Carcass : Carcass is the dressed body of an animal or deal animal

Carpet : Carpet wool is a coarse wool used for making carets

Carrying capacity : Carrying capacity is the number of animals that can be grazed on a pasture during the grazing season

Cast :Cast is to make the animal fall on the ground

Castrate :Castrate is to remove the testes of the male or to render them inactive; alternatively, an animal whose testes have been removed or rendered inactive

Cattle :Cattle is the animals of the family bovine, genus bos

Chevon: Chevon is the meat from goats

Colostrums :Colostrums is the milk produced the first few days after parturition

Concentrate :Concentrate is the feed containing less than 18 percent crude fiber when dry; grains and protein supplements are concentrates

Conception rate :Conception rate is the percentage of a group o animals that become pregnant when bred

Cow :Cow is a female bovine that has had one or more calves; or an older female that has not had a calf but has matured

Creep :

Creep is an enclosure to which only the young of the species have access so they maybe fed separately from the adult stock

Creep feed :Creep feed is used to provide special feed for the young; also, the feed provided for the young within a special enclosure

Crisscrossing :Crisscrossing is the mating crossbred females to a sire belonging to one of the parent breeds of the female; also called backcrossing

Crone :An old broken mouthed ewe which has been retained in a breeding flock because of her excellent breeding performance.

Cud :Cud in ruminants, a ball-like mess of feed that is brought up from the stomach to be rechewed

Cull :Cull is used to dispose of the poorer animals in a herd or flock

Dam :Dam is the mother of an animal

Deadweight :Deadweight is the weight of an animal after it has been slaughtered and the offal, head and hide removed

Dehorn :Dehorn is used to remove chemically or mechanically the horns of livestock

Dewlap :Dewlap is a hanging fold of skin under the neck of animals, especially some breeds of cattle and goats

Disbud :Disbud is used to remove or prevent growth of the horn buds in young livestock

Dock :Dock is used to remove all or part of the tail

Doe: Doe is a female of those species in which the male is called a buck; for example; goats, deer, rabbits, etc . Doe is a female goat, rabbit or antelope

Double rig: Both the testicles are retained in the abdomen. It is also known as **double ridgling** or “Cryptorchid”

Draft animal :Draft animal is an animal used for pulling loads

Drench :Drench is the medicine in a liquid form administered to the animal through the mouth

Elastrator :Elastrator is an instrument used to place strong rubber bands over, the scrotum or tail for castration or docking, respectively

Estrus :Estrus is the time during which the female will accept the male for copulation; also referred to as being “in heat”

Ewe :Ewe is a female sheep or lamb ;Ewe is a mature female sheep

Ewe lamb :A female young one

Feed efficiency :Feed efficiency is the ratio of units of feed needed per one unit of production

Feeder calf :Feeder calf is a weaned calf that is under one year of age and is sold to be fed for more growth

Fertilization :Fertilization is the union of a sperm cell with an egg cell

Fleece :Fleece is the total wool coat of a sheep

Flock: Flock is a group of animals (sheep, goats, birds).

Flushing :Flushing increasing the amount of feed to an animal for a short period of time, usually just prior to breeding

Free martin: When twin calves of different sexes are born the bull calf is usually sexually normal but the heifer calf is always sterile, and the external genitalia are abnormal in structure.

Gimmer :A female sheep which is between 1 and 2 shearing

Goatling :A female goat over 1 year but not exceeding 2 years of age

Haylage :Haylage is the low-moisture grass silage

He-buffalo :An adult male buffalo

Heifer :Heifer is a female bovine that has not had a calf or has not matured as a cow

Heterosis :Heterosis improvement in the offspring resulting from favorable combinations of gene pairs; sometimes called hybrid vigor

Inbreeding :Inbreeding is the mating of related animals

Kid :Kid is a goat under one year of age

Kidding :The act of giving birth to a young one

Lamb :Lamb is a young sheep

Lambing :The act of giving birth to a young one

Livestock: **Livestock** means stocks that are live and are used for production of items for commercial and for domestic consumption. The term livestock includes all animals, birds and all living creatures used for producing items for the use of man. The term livestock production or animal production is used to indicate farm animal production.

Mohair :Mohair is the fleece of the Angora goat

Ovulation :Ovulation is the release of the egg cell from the ovary

Ovum :Ovum see egg

Parturition :Parturition is the act of giving birth

Pedigree :Pedigree the record of the ancestors of an animal

Pelleting :Pelleting is the grinding a feed into small particles and then forming it into a small, hard form called a pellet

Phenotype :Phenotype is the physical appearance of an animal

Placenta :Placenta in mammals, the structure by which the fetus is nourished in the uterus

Polled :Polled is not having horns

Puberty :Puberty is the age at which sexual maturity is reached

Ram lamb :A male young one

Ram or Tup :An adult male sheep used for breeding

Rectum :Rectum is the last part of the large intestine

Rumen :Rumen is the first and largest compartment of the ruminant stomach

Repeatability :Repeatability in dairy cattle, a measure of the confidence that can be placed on the predicted difference being a true measure of a bull's ability to transmit genetic characteristics

Reproduction :Reproduction is the production of offspring

Roughage :Roughage is a feed containing more than 18 percent crude fiber when dry; examples, hay, silage, and pasture

Ruminant :Ruminant is an animal that has a stomach divided into several compartments; for example, cattle, sheep, goats

Rumination :Rumination in ruminants, the process of chewing the cud

Scrotum :Scrotum is the saclike part of the male reproductive system outside the body cavity that contains the testicles and the epididymis

Semen :Semen is the mixture of the seminal and prostate fluid and the sperm

Shearing :Removal of wool

She-buffalo :An adult female buffalo

Slink calf :An aborted calf

Steer :Steer is a male bovine animal that was castrated before reaching sexual maturity

Teat :Teat is the outlet for milk produced in the udder

Testicles :Testicles is the male organs that produce the sperm cells

Testosterone :Testosterone is the male hormone that controls the traits of the male animal

Udder :Udder is the milk producing gland of mammals such as cows

Urine :Urine is the liquid waste collected in the bladder

Uterus :Uterus is the part of the female reproductive system where the fetus grows;

also called the womb

Veal :Veal is the calves younger than three months of age sold for slaughter

Wattle :Wattle is a projection of skin hanging from the chin or throat, especially in poultry and some breeds of goats

Wether or Wether: An adult castrated male sheep.

Yearling :A horse over 1 year and under 2 years of age

Yearling bull :An uncastrated male between one and two years

Yeld or Eild : A barren or non-lactating animal

ZOOLOGICAL CLASSIFICATION OF LIVESTOCK

CLASSIFICATION	ZEBU CATTLE	EXOTIC CATTLE	BUFFALO	GOAT	SHEEP	PIG
Kingdom	Animalia	Animalia	Animalia	Animalia	Animalia	Animalia
Phylum	Chordata	Chordata	Chordata	Chordata	Chordata	Chordata
Class	Mammalia	Mammalia	Mammalia	Mammalia	Mammalia	Mammalia
Sub-Class	Eutheria	Eutheria	Eutheria	Eutheria	Eutheria	Eutheria
Order	Ungulata	Ungulata	Ungulata	Ungulata	Ungulata	Ungulata
Sub - Order	Artiodactyla	Artiodactyla	Artiodactyla	Artiodactyla	Artiodactyla	Artiodactyla
Family	Bovidae	Bovidae	Bovidae	Capridae	Ovidae	Suidae
Genus	Bos	Bos	Bubalus	Capra	Ovis	Sus
Species	indicus	Taurus	bubalis	Hircus	aries	domesticus

Common Animal Husbandry Terms

Details	Cattle	Buffalo	Sheep	Goat	Pig
Species	Bovine	Bovine or Bubaline	Ovine	Caprine	Swine
Groups of animals	Herd	Herd	Flock	Flock or band	Drove or herd or stock
Adult male	Bull	Buffalo bull	Ram or tup	Buck	Boar
Adult female	Cow	She buffalo or buffalo cow	Ewe	Doe	Sow
Young male	Bull calf	Buffalo bull calf	Ram lamb or Tup lamb	Buckling or male kid	Boarling
Young female	Heifer calf	Buffalo heifer calf	Ewe lamb or gimmer lamb	Goatling	Gilt
New born	Calf	Buffalo calf	Lamb	Kid	Piglet or pigling
Castrated male	Bullock or steer	Buffalo bullock	Wether or wedder	Castrated	Hog or stag or barrow
Sterilized female	Spayed	Spayed	Spayed	Spayed	Spayed
Female with its offspring	Calf at foot	Calf at foot	Suckling	Suckling	Suckling
Act of parturition	Calving	Calving	Lambing	Kidding	Farrowing
Act of mating	Serving	Serving	Tupping	Serving	Coupling
Cry	Bellowing	Bellowing	Bleating	Bleating	Grunting
Chromosome number	60	50	58	60	38

Unit II: Dairy Cattle Management

CATTLE AND BUFFALO BREEDS

The Indian breeds of cattle are classified as milch breeds, draught breeds and general utility breeds. They are as follows:

S.No	Type of Breed	Breeds	Salient characters
I	Milch breeds	1.Gir	Ponderous in build, pendulous dewlap and sheath have lateral or curved horns, predominantly red with varying colour pattern
		2.Red Sindhi	
		3.Sahiwal	
		4.Deoni	
II	Draught breeds	5.Nagori	Short horned white or light grey colour, with long coffin shaped skull and slightly convex profile
		6.Bachur	
		7.Kenketha	Lyre horned grey with wide forehead, prominent orbital arches, flat dished profile
		8.Malvi	
		9.Kerigarh	
		10.Hallikar	Mysore type – prominent fore head with long pointed horn raises closer at poll, coffin shaped skull. Grey colour
		11.Amirthamahar	
		12.Khillari	
		13.Kangeyam	
		14.Bargur	
		15.Ponwar	Short horned or slightly lyre horned small black, red, dun colour
		16.Siri	
III	General utility breeds	17.Gaolao	Short horned white or light grey colour, with long coffin shaped skull, convex profile
		18.Krishna valley	
		19.Tharparkar	Lyre horned grey with wide forehead prominent orbital arches, flat or dished profile
		20.Kankrej	

IV	Exotic breeds	21.Shorthorn 22.Ayrshires 23.Jersey 24.Brown Swiss 25.Red Dane 26.Guernsey 27.Holstein– Friesian	
V	Buffalo Breeds	28.Murrah 29.Bhadawari 30.Jaffarabadi 31. Surti 32.Mehsana 33. Nagpuri or Ellichpuri 34. Nili – Ravi 35. Toda	

CHARACTERISTIC FEATURES OF DIFFERENT BREEDS MILCH BREEDS

GIR



Origin: Gir Forests of south Kathiawar in Gujarat

Basic colours of skin are white with dark red or chocolate-brown patches or sometimes black. Most of the Gir animals seen today are purely red.

The typical characteristics of Gir breed are:

- i. A broad convex forehead like a bony shield.
- ii. Long and pendulous ears are folded like a leaf with a notch at the tip. Their inside faces forward and always remains hanging from the base.
- iii. Horns are peculiarly curved, it take a downward and backward curve and again incline a little upward and forward taking a spiral inward sweep giving a '*half moon*' appearance.
- iv. Tail is long and whip like.

Performance parameters

- i. Milk yield ranges from 1200-1800 kg
- ii. Age at first calving 45-54 months and inter calving period from 515 to 600 days

RED SINDHI:



Origin : Mostly found in Karachi and Hyderabad district of Pakistan.

Colour is red with shades varying from dark red to light, strips of white colour also sometime seen on dewlap and on forehead.

The typical characteristics of the breed is

- i. Head is well proportioned with a occasional bulge on the forehead
- ii. Horns are thick at the base and emerge laterally and curve upward.
- iii. Hump is well developed in males.
- iv. Dewlap and sheath are pendulous
- v. Udder is capacious and pendulous.
- vi. Compact frame and dropping quarters.

Performance parameters

- i. Milk yield ranges from 1100-2600 kg
- ii. Age at first calving 39-50 months and inter calving period from 425-540 days

SAHIWAL



Origin: Montgomery district in present Pakistan.

The colour is reddish dun or pale red, sometimes flashed with white patches

The typical characteristics of the breed is

- i. The Sahiwal is a heavy bred with symmetrical body and loose skin. Animals are long, deep, fleshy and comparatively lethargic.
- ii. Horns are short and stumpy.
- iii. Dewlap is large and heavy.
- iv. Hump in males is massive and frequently falls on one side.
- v. Naval flap is loose and hanging. Sheath in males is also pendulous.
- vi. Tail is long and fine with a black switch reaching almost to the ground.

Performance parameters

The average milk yield - 2,725 -3,175 kg.

DEONI



Origin: Marathwada region of Maharashtra state and adjoining part of Karnataka and western Andhra Pradesh states.

Body colour is usually spotted black and white. Ears are grey-white or complete white with black pinna

The typical characteristics of the breed is

- i. Ears are drooping like in Gir and forehead is prominent and slightly bulged similar to that of Gir.
- ii. Horns emerge from the sides of the poll behind and above the eyes.
- iii. Hump is massive and well developed
- iv. Skin is thick, loosely attached with the body.
- v. Udder is moderately developed. Teats are black, cylindrical with rounded tips and are squarely placed.

Other details:

Milk yield 636 -1230 kg

DUAL PURPOSE BREEDS

THARPARKAR



Origin : Tharparkar district of southeast Sind in Pakistan.

The typical characteristics of the breed is

- i. Body colour is white or light grey
- ii. Horns are set well apart curving gradually upward and outward.
- iii. Tharparkar has proved to be one of the best dual-purpose breeds of India. The bullocks are quite suitable for ploughing and casting and the cows are good milch animals (1,800 – 2600 kg).
- iv. Age at first calving ranges from 38-42 months and inter calving period from 430 to 460 days.

HARIANA



Origin: Rohtak, Hisar, Jind and Gurgaon districts of Haryana

Animals were white or light grey colour. In bulls, colour in between fore and hind quarters is relatively dark or dark grey.

The typical characteristics of the breed is

- i. Horns are small
- ii. Long and narrow face, flat forehead and well-marked bony prominence at the centre of the poll.
- iii. Udder is capacious and extends well forward with a well-developed milk vein.

Average milk yield 600 to 800 kg per lactation. The age at first calving is 40-60 months.

KANKREJ



Origin : Southeast Rann of Kutch of Gujarat and adjoining Rajasthan (Barmer and Jodhpur district)

Colour of the animal varies from silver-grey to iron-grey or steel black. In males forequarters, hindquarters and hump are slightly darker than the rest of the body.

The typical characteristics of the breed is

- i. Fore head is broad and slightly dished in the centre.
- ii. Face is short and nose is slightly upturned.
- iii. Ears are large and pendulous and open.
- iv. Horns are strong, and curved outward and upward in a lyre shaped fashion.

They are covered with skin to a longer distance as compared to other breeds.

DRAUGHT BREEDS:

More than 80% of the cultivator in India depends upon bullocks for motive power in the farms about 42% of the cattle on the farms are draught animals. The idle time of the bullocks on the farms is economically used in farm transport and village industry. Draught breeds are of four types

HALLIKAR



Origin: From the former princely state of Vijayanagarm, presently part of Karnataka

The typical characteristics of the breed is

- i. The colour is grey
- ii. They are compact, muscular and medium size animal.
- iii. Their horns emerge in proximity to each other near the top of the poll and are carried backward, straight for nearly half their length and then with a forward bend.
- iv. The forequarters are well developed and legs are strong.
- v. The breed is best known for its draught capacity and especially for its trotting ability.
- vi. These animals were used in the war by *Tipu Sultan* for carrying cannons

AMIRITMAHAL



Origin: Hassan, Chikmagalur and Chitradurga district of Karnataka

The typical characteristics of the breed is

- i. Amritmahals are grey cattle but their shade varies from almost white to near black.
- ii. Head is long and tapering towards muzzle. Forehead is narrow bulging out with a furrow in the middle.
- iii. Horns are long and emerge from the top of the pole fairly close together in backward and upward direction, turn in and end in sharp black points.

KANGAYAM



Origin: Kangayam, Dharapuram, Perundurai, Erode, Bhavani and part of Gobichettipalayam taluk of Erode and Coimbatore district. The kangayam breed was developed by the efforts of the late Pattagar of Palayakottai, Sri N. Nallathambi Sarkarai Manradiar.

The typical characteristics of the breed is

- i. Colour : Coat is red at birth, but changes to grey at about 6 months of age. Bulls are grey with dark colour in hump, fore and hind quarters. Bullocks are grey. Cows are grey or white or grey. However, animals with red, black, fawn and broken colours are also observed. Horns, muzzle, eyelids, tail switch and hooves are black.
- ii. They are short with stout legs and strong hooves.
- iii. The horns are spread apart, nearly straight with a slight curve backward.
- iv. The eyes are dark and prominent with black rings around them.
- v. The dewlap is thin. The sheath is well tucked up to the body.
- vi. The average milk yield 600 to 700 kg in a lactation.

BARGUR



The typical characteristics of the breed is

- i. Found around Bargur hills in Bhavani taluk of Erode district.
- ii. Bargur cattle are of brown colour with white markings. Some white or dark brown animal are also seen.
- iii. Animals are well built, compact and medium in size.

- iv. Muzzle is moderate and black in colour.
- v. Horns are of light brown colour, moderate length, closer at the roots inclining backward, outward and upward with a forward curve and sharp at the tip.

UMLACHERRY



Origin : Thanjavur, Thiruvarur and Nagappattinam districts of Tamil Nadu.

Synonyms: Jathi madu, Mottai madu, Molai madu, therkathi madu.

The typical characteristics of the breed is

- i. Umblacherry calves are generally red or brown at birth with all the characteristic white marking on the face, on limbs and tail.
- ii. The colour changes to grey at about 6 months of age. In adult females, the predominant coat colour is grey with white marking on the face and legs.
- iii. All the legs below hocks have white marks either socks or stockings.
- iv. Horns are very small, curving outward and inward and sometimes spreading laterally.
- v. The practice of dehorning bullocks is peculiar in Umblacherry cattle. Horn buds are removed at 6 months of age by singeing with red hot iron.
- vi. Ears are pruned and hot iron branding is done.

PULIKULAM / JELICUT BREED



Pulikulam is a distinct cattle breed and the present population is about 45000. This breed is seen in Madurai and Theni districts. They are also raised in Cumbum valley and the Periyar river, where there are grazing grounds of vast extent. These animals depend extensively on forest grazing. This is a quick trotting (5-6 miles per hour) breed. They are comparatively small, but active and capable of much endurance. Selected bulls are utilized

as Jallikattu or bull baiting purposes. Farmers are benefited through penning, which provides rich organic manure for agricultural crops. This breed can only be conserved in the agro-eco system where forest grazing is available.

ALAMBADI

Alambadi bulls are dark grey, almost black and cows grey or white. They have the typical backward curving horns of Mysore type cattle. They are active, useful draught animals but not fast trotter. This breed is restricted to Salem and Coimbatore district of Tamil Nadu and part of Bangalore district in Karnataka and closely resembles Hallikar breed.



D. Exotic breeds of Cattle:

In order to improve the milk yielding capacity of the indigenous cows, exotic cattlebreeds have been extensively used in India.

Important foreign breeds are described below:

Jersey

- i. It is the smallest of the dairy types of cattle developed on island of Jersey, U.K.
- ii. In India this breed has acclimatized well and is widely used in cross breeding with indigenous cows.
- iii. Because of their small and compact size they are more suitable for cross-breeding with zebu cattle.
- iv. The typical colour of Jersey cattle is reddish fawn.
- v. Dished forehead and compact and angular body.
- vi. These are economical producers of milk with 5.3% fat and 7 % SNF.
- vii. The highest record of milk yield was 11381 kg and fat 544 kg in a lactation period of 365 days.
- viii. In India this breed has acclimatized well and has brought about increased production of milk (2.5 times or 50 in the first generation progeny).
- ix. The age of maturity and inter calving period in the progeny have been considerably reduced.



Holstein Friesian:

- i. This breed was developed in the northern parts of Netherlands, especially in the province of Friesland.
- ii. They are ruggedly built and they possess large udder.



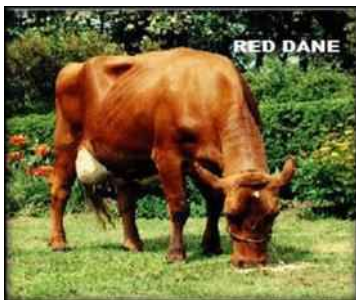
- iii. They are the largest dairy breed and mature cows weigh as much as 700kg.
- iv. They have typical marking of black and white that make them easily distinguishable.
- v. This is by far the best dairy breed.
- vi. The average production of cow is 6000 to 7000 kg per lactation. However, the fat content in their milk is rather low (3.45 per cent).

Brown Swiss

- i. The mountainous region of Switzerland is the place of origin of Brown Swiss breed. It is rather less refined than other dairy breeds, but is famous in its home tract for its rugged nature and good milk production.
- ii. In India excellent crossbred cattle have been obtained by crossing this breed with recognized Indian breeds of cattle; the *Karanswiss* cattle of NDRI, Karnal.



Red Dane



and above.

- i. The typical body colour of this Danish breed is red, reddish brown or even dark brown.
- ii. It is also a heavy breed; mature males weighing up to 950 kg and mature female weigh 600 kg.
- iii. The lactation yield of Red Dane cattle varies from 3000 to 4000 kg with a fat content of 4 per cent

E. Breeds of Buffaloes

The buffalo species originated in India. The buffaloes are normally classified into river and swamp types though both are called *Bubalus bubalis*. Most of the animals in India river type though swamp type are also found in eastern parts of India.

India is considered as the home tract of some of the best buffalo breeds. Because of preference of buffaloes for milk. Indian buffaloes are an important source of milk supply today and yield nearly three times as much milk as cows. More than half of the total milk produced (55%) in the country was contributed by the 47.22 million milch buffaloes, whereas the 57.0 million cows contribute only 45% of the total milk yield. There are about 10 indigenous standard breeds of buffaloes.

MURRAH



- i. It is the most important breed of buffaloes whose home is Rohtak, Hisar and Jind of Haryana and Nabha and Patiala districts of Punjab.
- ii. **Synonyms** : Delhi,
- iii. The colour is usually jet black with white markings on tail and face and extremities sometimes found.
- iv. The tightly curved horn is an important character of this breed.
- v. The body size is massive, neck and head are comparatively long. Head of females is short, fine and clear cut.
- vi. Hips are broad and fore and hind quarters are drooping.
- vii. The buffalo cows of this breed are one of the most efficient milk and butter fat producers in India. Butter fat content is 7% Average lactation yield is vary from 1500-2500 kg.
- viii. Age at first calving is 45-50 months and inter calving period is 450-500 days.

NILI RAVI



- i. This breed is found in Sutlej valley in Ferozpur district of Punjab and in the Sahiwal district of Pakistan. (bred around Ravi river)
- ii. Usually the colour is black with white marking on forehead, face, muzzle, legs and tail. The most desired character of the female is the possession of white markings.
- iii. The head is elongate, bulging at top and depressed between eyes. The muzzle is fine. The frame is medium sized.
- iv. The horns are small and coiled tightly. The neck is long, thin and fine.

- v. The milk yield is 1500-1850 kg per lactation and the inter calving period is 500-550 days. Age at first calving is 45-50 months.

JAFFARABADI



- i. These are massive animals found in their pure form in Gir forests. The breeding tract of this breed is Kutch, and Jamnagar districts of Gujarat.
- ii. The head and neck are massive. The forehead is very prominent, wide with a slight depression in the middle.
- iii. The horns are heavy, inclined to droop at each side of the neck and then turning up at point, but less tightly curved than in Murrah (drooping horns).
- iv. The colour is usually black.
- v. The average milk yield is 100 to 1200 kg

SURTI



- i. The breeding tract of this breed is Kaira and Baroda district of Gujarat.
- ii. Coat colour varies from rusty brown to silver-grey. Skin is black or brown.
- iii. The body is well shaped and medium sized; the barrel is wedge shaped.
- iv. The head is long with prominent eyes.
- v. The horns are sickle shaped, moderately long and flat.
- vi. The colour is black or brown
- vii. The peculiarity of the breed is two white collars, one round the jaw and the other at the brisket.
- viii. The milk yield ranges from 900 to 1300 kg.

MEHSANA



- i. Mehsana is a dairy breed of buffalo found in Mehsana town in Gujarat and adjoining Maharashtra state.
- ii. Body is mostly black; a few animals are black-brown in colour.
- iii. The breed is supposed to have been evolved out of crossbreeding between the Surti and the Murrah.
- iv. The body is longer than in Murrah and the limbs lighter.
- v. The head is longer and heavier.
- vi. The horns usually are less curved at the end compared to Murrah breed but are longer and could be of irregular shape.
- vii. The milk yield is 1200-1500 kg. The breed is supposed to have good persistence.
- viii. The inter-calving period ranges between 450-550 days.

TODA



- i. Toda breed of buffaloes is named after an ancient tribe, Toda of Nilgiris of south India.
- ii. Coat colour of the calf is generally fawn at birth. In adult the predominate coat colours are fawn and ash-grey.
- iii. These buffaloes are quite distinct from other breeds and are indigenous to Nilgiri hills.
- iv. The animals have long body, deep and broad chest, and short and strong legs.
- v. The head is heavy with horns set well apart, curving inward outward and forward.
- vi. Thick hair coat is found all over the body. They are gregarious in nature.

NEW CROSSBRED CATTLE STRAINS

Sunandini: Under Indo-Swiss project in Kerala, local non-descript cows were crossed with Brown Swiss bulls. The crosses with 62.5% brown-Swiss inheritance were mated intense followed by selection to synthesize a new breed named Sunandini. Average lactation yield 4351 kg in 305 days.



Frieswal: Friesian x Sahiwal crossbreds with Friesian inheritance between $\frac{3}{8}$ and $\frac{5}{8}$ at military farms are being interbred with semen of $\frac{5}{8}$ Friesian crossbred bulls into a breed formation programme. Average yield 2729 kg



BREEDING - CROSS BREEDING- UPGRADING

Selection and system of breeding constitute the only tools available to the breeder for improvement of animals since new cannot be created though they can recombine into more desirable groupings.

Systems of breeding have been broadly divided as

1. Inbreeding – Breeding of related animals
2. Out breeding – Breeding of unrelated animals

Inbreeding:

a. Close breeding:

This means the mating of full sister to full brother or sire to his daughter or dam to her son. These type of breeding is followed only when the both parents are outstanding. This leads to more homozygosity.

b. Line breeding:

This means the mating of animals of wider degrees of relationship than those selected for close breeding. It promotes uniformity in the character.

Out breeding:

It consists of outbreeding, cross breeding and upgrading.

a. Out Crossing:

Mating of unrelated pure bred animals within the same breed.

b. Cross breeding:

It is the mating of animals of different breeds. It is generally used where the crossed progeny is directly marketed and are needed for breeding and further multiplications. It has become quite common in pigs and in the production of hybrid chickens.

Methods of cross breeding:

1. ***Criss-crossing:*** When the two breeds are crossed alternatively, the method is known as criss-crossing. This method is proposed for utilizing heterosis in both dams and progeny.
2. ***Triple crossing:*** In this system three breeds are crossed in a rotational manner. It is also known as rotational crossing.
3. ***Back crossing:*** Back crossing is mating of a crossbred animal back to one of the pure parent races which were used to produce it. It is commonly used in genetic studies, but not widely used breeders.

Advantages of cross breeding:

1. It is valuable as a mean of introducing desirable characters into a breed in which they have not existed formerly.
-

2. It serves a good purpose in evolving a new breed owing to the fact that it disturbs the balance and brings about recombination in the germ to cause variations.
3. It is an extremely handy tool to study the behavior of characteristics in hereditary transmission.
4. The cross bred animals usually exhibit an accelerated growth and vigour or heterosis.

Disadvantages:

1. It has tendency to break up the established characters and destroy combinations of that characters.
2. Cross breeding requires maintenance of two or more pure breeds in order to produce the cross breeds.

C. Grading: Grading is a form of out crossing, where in bulls of a distinct breed are bred on non-descript cows from generation to generation, so that in course of time a populations essentially resembling the breed from which the Bulls are used.

Non-descript cow x Jersey Bull

F1 50% ND + 50% Jersey x Jersey Bull

F2 25% ND + 750% Jersey x Jersey

Bull

F3 12.5% ND + 87.5% Jersey x Jersey Bull

After 5-6 generations the off springs will have 96.9 & 98.3% of the hereditary characters of 'Pure Breed'. So grading is a process by which a few 'Pure Breed' sires can rather quickly transform local variety of animals into a 'Group' resembling the pure breed.

ECONOMIC TRAITS OF CATTLE AND BUFFALOES

Criteria	Economic traits
1. Age at 1 st calving	: Age in days of the cow or buffalo on the date of 1 st calving.
2. Lactation Length	: Days in milk from the date of calving to the final drying off or cessation of milk (305 days)
3. Lactation Yield	: Milk yield in Kgs from the date of calving to the date of drying (corrected to 305 days)
4. Dry period	: Days from the date of drying to the date of calving
5. Inter calving period	: Days from the date of one calving to the date of next calving (1 st , 2 nd)
6. Peak yield	: The highest daily yield in Kgs during lactation period
7. Average Fat%	: Average Fat %
8. Service period	: The interval between calving and subsequent service resulting in conception
9. Breeding efficiency:	Measured as the No. Services/Conception

Comparison of Economic Traits between Local, Exotic and Crossbred cattle

Particulars	Local	Exotic	Cross breed
Birth weight	20Kg	Jersey – 25-30kg. Friesian – 30-35kg.	25-30 kg
Age at maturity	33 Months	15 months	18-24 months
Age at 1 st calving	42 months	24 months	30 months
Lactation yield	12000 Kg.	3000-6000Kg.	2100-2400
Lactation period	180-210 Days	305 days	240 – 270 days
Dry period	90-120 days	60 days	75 days
Inter calving period	18 months	12-13 months	13-14 months

ESTRUS CYCLE

Generally a heifer attains **puberty** (this is the stage at which animal becomes sexually mature and the secondary sexual characteristics become conspicuous) by 8-12 months period. Smaller cows attain puberty at an early date than larger one. The **estrous cycle** starts from puberty, it is regular cycle occurs once in 21 ± 3 days throughout the reproductive phase. During the cycle a mature ovum is liberated from the ovary, the cervix become receptive to the spermatozoa, the female exhibits behavioural adjustment and attraction to receive the male.

Oestrus cycle is divided into

Proestrus (2 days): Period of building up growth of Graffian Follicle (GF) which helps for the nourishment of ovum fluid contains hormone called ‘oestrogen’. It causes changes in uterus, blood supply. This marks the animal is coming in heat. Graffian follicle in the ovary is growing resulting in increased secretion of follicular fluid which have estradiol. This increase number of cilia and increased vascularity of uterus resulting in increase in thickness of epithelial wall of vagina. The vaginal wall adjustment is well filled to prevent possible damage to the wall when coitus occurs.

Oestrus (1day): This is the period of desire. During which the female is ready to receive male. ‘Graffian Follicle’ will ripe or very turgid. This period is brought to an end by the rupture of the follicle (or) ovulation. Vulva becomes swollen. Vulva and Vagina are congested

Metooestrus (4 days): Implantation of the embryo takes place and Corpus Luteum (CL) will be formed and prevents the growth of graffian follicle thereby arrests oestrus cycle. This is the period when the organ returns to normal non congested condition. During this period the cavity of the GF from which ovum has been expelled becomes recognized and forms a new structure known as C.L.

Diestrum (14 days): Further development of uterus takes place. If the animal has not conceived involution of uterus take place. Longest part of estrus cycle. Implantation of embryo and secretion of uterine milk takes place, which is used for the nourishment of the

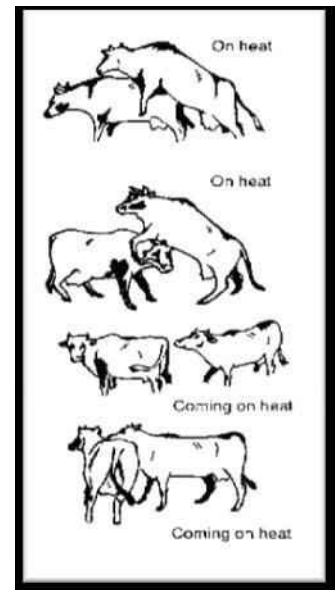
embryo prior to implantation. In case of absence of pregnancy, the animal returns to normal and thus the cycle continues.

Estrus is the period in which the animal exhibits sex desire. This period lasts for 12 to 18 hours. The estrus or heat is followed by ovulation which normally occurs 12-16 hours after the end of estrus.

Signs of estrus / heat in cattle

1. Off feed
2. Sudden drop in milk production
3. The cow may search for male, make deep bellow
4. Restless and excited
5. Bellowing
6. Frequent urination
7. Homosexuality character like mounting over other cow and accepting other cow to mount over
8. During estrus the cow will accept the mounting by a bull or other cow. The behavior of standing quietly while being mounted by other cow/bull is called standing heat and this is the surest sign of heat.
9. Oedema / swelling of genitalia. Mucous discharge from the vulva and it may stick on the tail head.
10. Hair on the tail head may be pushed forward
11. There may be dirt on the side of the abdomen due to mounting by other animal

Signs of heat



The signs of heat that may be shown will be described for each stage.

a. Early heat period

A sexually mature, non-pregnant cow comes in heat every 18 to 24 days. It starts with the development of an ovum in the ovary. At this stage the cow shows early signs of heat. The length of this period varies from 6 to 24 hours.

b. Standing heat period

Early heat changes into standing heat. The length of this phase of heat ranges from 6 to 18 hours. It is shorter under tropical conditions than under more temperate conditions.

c. Late heat period

After the period of standing heat some cows continue to show behavioral signs of heat. This period is called the late heat period and can last for 12 to 24 hours.

About two days after the end of heat, cows may show a bloody mucous discharge from the vulva. This is called metoestral bleeding.

Right time for breeding a cow

Egg/ova from ovary is released about 12-18 hours after the onset of symptoms of heat. Ovum will survive upto 16 hours after the release. Sperm live for 12-14 hours. As a routine practice, if a cow is seen showing signs of early heat in the morning, it may be inseminated in the evening. If such signs are seen in the evening, the cow should be bred next day morning.

In buffaloes the expression of estrus sign is very poor. Particularly during summer period the buffalo show only **silent heat**. Silent heat means the animal could not show any visible sign of estrus. It occurs more frequently in first calver than in older cows, in early lactation than in later lactation. More pronounced in buffaloes during summer.

ARTIFICIAL INSEMINATION

Artificial insemination is the deposition of male reproductive cells (sperm) in the female tract by mechanical means rather than "Natural Service"

ADVANTAGES:

1. Increases usefulness of superior sires to extra ordinary degree.
2. Services of Superior Sires are greatly extended.
3. No need to maintain Breeding Bull. The frozen semen can be stored in the Liquid Nitrogen-196°C.
4. Semen can be quickly and easily transported by air to different continents.
5. Spreading of diseases is absolutely- NIL.
6. Overcomes the difficulty of size and weight between Dam and Sire.
7. Increase the rate of Conception because in the artificial insemination the semen is being deposited in the mid cervix.
8. Outstanding animals located apart can be mated.
9. Helps in better record keeping.
10. Old and heavy sizes bulls, injured / disabled sires can be used.
11. If the sires are used for Natural Service the animal can serve only 50-60 animals/year but under Artificial Insemination the amount of semen secreted by the animal can be used to satisfy the requirements of 1000 animals per year
12. Semen can be stored in the frozen state , so progeny can be obtained even after the transfer, WHY even after death of bull-15-20 years.(atomic, radioactive, X- ray unit)
13. Semen is expanded and no. of animal can be crossed.
14. Frozen semen can be transported to destination once in a month from the semen bank.

DISADVANTAGES.

1. Some bull's semen may not freeze well.
2. If inferior bull semen is frozen and used –Extensive damage is caused.
3. Maintenance of frozen semen bank is not economical for a small area of operation.
4. Requires well trained technical personnel's and special equipments and hygienic measures are to adapted in preparation.
5. Improper cleaning of instruments and unsanitary condition may lead to lower fertility and may be nucleus for the spreading of diseases.

DILUTION OF SEMEN

Average of Sperm/mL	:	1000 million
Total volume of semen/2 ejaculate	:	6 mL (i.e.) 6000 million sperms.
No. of motile Sperms	:	90 %
Total number of motile sperms	:	5400 million
Expected wastage during processing (i.e.) filling and sealing	:	10 %
Net no.of sperms available	:	5400-540= 4860
Minimum No. of sperms required / dose @ this rate no. of doses that could be prepared So, total no. of doses that can be prepared / week	:	30 Million 4860 /30 = 160 doses. 160 x 52 weeks = 8320 doses
Frozen semen required / dose	:	1 mL.

EMBRYO TRANSFER

Most of the applicable embryo transfer technology was developed in the 1970s and 1980s. Embryo transfer (ET) was first performed and recorded by Walter Heape in 1890. He transferred two Angora rabbit embryos into a gestating Belgian doe but it was not until the 1950s that successful embryo transfers were reported in cattle and pigs by Jim Rowson at Cambridge, England. The first commercial embryo transfer is in the early 1970s. Initially, embryos were recovered from valuable donors and transferred to recipient animals using surgical procedures. It was not until non-surgical methods were developed in the late 1970s, that embryo transfer grew in popularity.

Advantages

1. Increase the number of offspring sired from superior females.
2. Results in faster genetic progress.
3. Increase the frequency of desired matings, capitalizing on excellence of a mating.
4. Obtain offspring from old or injured animals incapable of breeding or calving naturally.

5. Increased farm income through embryo sales.
6. Exportation and/or importation of embryos is easier than with live animals.

Disadvantages

1. Can be cost prohibitive and success rates are less than AI.
2. Cost and maintenance of recipient females.
3. Requires a technician with the skills to flush embryos from the reproductive tract.
4. Possible spread of disease through recipients.

ANIMAL COMPONENTS OF ET PROGRAM

Donor: animal that donates embryos

Donor must be good reproductive performance and progeny performance. Animals must be exhibiting regular estrous cycles. At least 60 to 90 days post calving. Eliminate cows with a history of reproductive problems.

Recipient: animal that receives embryo from donor

Recipients are the greatest single cost of an ET program because they need to be at the same stage of the estrous cycle as the donor when she donated the embryos. Females should be healthy, good body condition, and vaccinated for all the common reproductive diseases. Females should have been through at least two normal estrous cycles before use. Pregnancy rates are greatest when day of the estrous cycles of the donor and recipient are within 24 hours of each other. The recipient should exhibit estrus from 24 hours before to 12 hours after the donor was in estrus. Embryos are typically transferred on day 7 of the estrous cycle.

MANAGEMENT COMPONENTS OF ET PROGRAM

Superovulation

This is the process of super stimulating the ovaries with FSH to produce multiple oocytes. There is tremendous variation in the number of embryos recovered after superovulation, which is due to variables like animal age, breed, lactation status, nutritional status, season, and stage of the estrous cycle when FSH treatment is initiated. Expected response is 5 to 12 embryos that are transferable.

Flushing

Most embryos are collected by a non-surgical process between day 6 - 8 of the estrous cycle with day 7 being the most common. A specially designed instrument called a Foley catheter is used for the flushing procedure. The Foley is a 2-way catheter that has one channel for inflation of a balloon at the end of the catheter plus an additional channel for the inflow and outflow of flushing medium.

Evaluation

As the individual embryos are located using the microscope, they are evaluated for their quality and classified numerically as to the potential likelihood of success if transferred to a recipient female. The major Criteria for evaluation include:

- Regularity of shape of the embryo
- Compactness of the blastomeres (the dividing cells within the boundaries of the embryo)
- Color and texture of the cytoplasm (the fluid within the cell wall)
- Overall diameter of the embryo
- Regularity of the zona pellucida (the protective layer of protein and polysaccharides around the single celled embryo)
- Presence of vesicles (small bubble-like structures in the cytoplasm)

Embryos can be maintained at room temperature for 12 - 24 hours at a pH of 7.1 to 7.5 when holding media is changed every few hours. Excessive temperature harms embryos (39°C). It is recommended that embryos be frozen within two hours after flushing.

Embryos are classified by stage of development and graded based on gross morphological appearance. It takes a great deal of experience to do this well.

Stage of Development Classification

Quality Grades or Scores

Excellent (1) - Embryos with few or no recognizable imperfections.

Fair (2) - Embryos that show disarrangement. Usually 1 to 2 days retarded development.

Poor (3) -Embryos with signs of cellular degeneration. Development has been retarded by more than 2 days.

Dead or Degenerating (4) - Embryos contain mostly dead cells and very few live. Tiny cell mass that is disorganized in appearance. Not worth transferring.

After classification, embryos are prepared for the next processing procedure, which may include sexing of embryonic cells and (or) splitting the embryo in half to make twins. Embryos are either transferred immediately into a recipient or frozen for storage.

Freezing

Cryoprotectants like glycerol and ethylene glycol are used to dehydrate cells and protect the cells of the embryo during the freezing and thawing process. Ethylene glycol is the cryoprotectant of choice in cattle since embryos can be frozen in a single step and eventually thawed in a single step and immediately transferred into a recipient straws are plunged into liquid nitrogen (-196°C).

Transferring

Embryos can be transferred in two different manners

1) **Same day transfer** Embryos are transferred into recipients on the same day they are collected from the donor. Transferring fresh embryos results in greater pregnancy rates, but requires a great deal of coordination to get recipients and donors on the same day of the estrous cycle when flushing occurs.

2) **Transfer of Frozen Embryos** Embryos are frozen the day they are flushed and then transferred at a later date. Although pregnancy rates for frozen embryos are slightly less than fresh, the process has numerous advantages including:

- a) Frozen embryos offer important logistical and economic advantages. More cows can be flushed in a day, which facilitates freezing of embryos.
- b) Frozen embryos can be marketed and imported and/or exported more easily.

Pregnancy Rates Resulting from Embryo Transfer

When done by experienced technician, the transfer of fresh embryos yields pregnancy rates of 70 to 80%, while transferring frozen embryos yields pregnancy rates of 50 to 60%.

HOUSING OF DAIRY CATTLE AND

BUFFALOES

SELECTION OF SITE FOR FARM BUILDINGS

Proper housing which is conducive to good health, comfort and protection from inclement weather and which would enable the animals to utilize their genetic ability and feed for optimal production. For construction of farm buildings selection of site is most important.

Before selecting a site the following points are to be considered,

1. Type of soil
2. Availability of land
3. Electricity
4. Availability of water
5. Protection from wind and solar radiation
6. Away from noise and other pollution sources
7. Availability of market facility
8. Transport facilities
9. Drainage system
10. Miscellaneous like, telephone, school, post office and shopping centre etc.,

HOUSE LAYOUT

1. Orientation

In general animal sheds are located with long axis east to west the paddock side facing the north to get direct sunlight during winter and to prevent entry of direct sunlight into the shed during other seasons

2. Length of building

The standard length of building may be of any. It may vary depending upon the number of animals housed.

3. Height of the building

The standard height of the building may differ according to the roofing material and agro-climatic condition.

4. Width of buildings

- | | | |
|------------------------|---|-----------------|
| 1. Single row cow shed | - | 3. 80 to 4.25 m |
| 2. Double row cow shed | - | 7. 90 to 8.70 m |

5. Roof

It is designed to suit the local climatic conditions. Gable with roof ventilators are necessary for hot condition. Monitor roof is suitable for building with smaller width. Locally available materials like thatches, tiles, wood, agricultural byproducts are commonly used as roof materials. Improved materials like asbestos sheet or concrete are also used.

6. Floor

For any type of animal house flooring, the primary thing required is provision of good 'bottom'. It is as important as that of foundation wall. The bottom is called some times are hard core for the floor. Different materials are used for animal house flooring. The choice depends on availability, cost and other quality required for the animal houses. The floor may be cement concrete floor, vitrified paving bricks, stones, wood, building bricks, gravel and synthetic flooring like composition brick, rubber floor.

HOUSING

In India, a great diversity exists in the design of dairy animal shelters. Efficiently designed sheds can help lessen the thermal stress thereby increasing feed intake, milk production and reproductive efficiency. Under varied climatic, geographical and economic conditions prevailing in India, designing an ideal set of building for dairy animals throughout the country is impossible. Hence, practically there are two systems of housing for dairy animals viz.,

- i) Loose housing and
- ii) Conventional barns

The former being widely followed in our country.

i. LOOSE HOUSING

It is a system of housing in which animals are kept loose in an open paddock throughout the day and night except at the time of milking and treatment. In this system, shelter is provided along one side of open paddock under which animals can rest when it is very hot or cold or during rains. Common feed manger and water tank is provided and concentrates are fed at the milking time which is done in a separate milking barn or parlour in which cows are secured at milking time and are milked. The open paddock is enclosed by means of half walls or plain wire fences of convenient height.

Advantages

1. Cost of construction is cheaper.
2. Future expansion is possible.
3. The animals will move freely so that it will get sufficient exercise.
4. The animal can be kept clean.
5. Common feeding and watering arrangement is possible.
6. Clean milk production is possible because the animals are milked in a separate milking barn.
7. Oestrus detection is easy.
8. At least 10-15 per cent more stock than standard can be accommodated for shorter period.

Disadvantages

1. It is not suitable for temperate Himalayan region and heavy rainfall areas.
2. It requires more floor space.
3. There is competition for feed.
4. Attention of individual animal is not possible.
5. A separate milking barn is needed for milking of animals.

ii. CONVENTIONAL BARNs OR STANCHION BARNs

In this system of housing, the animals are confined together on a platform and secured at neck by stanchions or neck chain. The animals are fed as well as milked in the same barn. These barns are completely covered with roofs and the sidewalls are closed with windows or ventilator located at suitable places to get more ventilation and lighting. It is applicable for temperate and heavy rainfall region. The same type of housing can be utilized for tropical region with slight modification.

Advantages

1. The animals and men caring for animals are less exposed to harsh environment.
2. The animals can be kept clean.
3. Diseases are better controlled.
4. Individual care can be given.

5. Separate milking barn is not required.

Disadvantages

1. Cost of construction is more.
2. Future expansion is difficult.
3. Not suitable for hot and humid climatic conditions.

Milking barn

This is a barn where milch animals are milked and is fully covered. It should be located at the centre of the farm with all other farm buildings arranged around it. Depending upon the number of milch animals, there are two types of milking barns, viz.,

- i) Single row system.
- ii) Double row system: head to head or face-in; tail to tail or face-out.

As per ICAR norms, in single row system, 12-16 numbers of animals can be kept. If it is greater than 16, then double row system is preferable. In double row system upto 50 animals can be maintained in a single shed. The distance between two sheds should be greater than 30 feet or it should be twice the height of the building.

There shall be an individual standing in the milking barns and the number of standings required should be 25% of total number of milch animals in the herd and the milking operation should be carried out in batches.

Advantages

Tail to tail system

1. Cleaning and milking of animals are easy.
2. Supervision of milking is also easy.
3. Less chance for transmission of diseases from animal to animal.
4. Animals can get more fresh air from outside.

Head to head system

1. Getting animals into the shed is easy.
2. Feeding of animals is also easy.
3. Disinfection of gutter will be more due to the direct fall of sunrays over the gutter.
4. Animals are better exhibited to visitors.

Disadvantages

1. Milking supervision is difficult.
2. Possibilities of transmission of disease are more.

Floor space requirements

Type of animal	Floor space requirement (m ²)		Maximum number of animals/pen	Height of the shed (cm)
	Covered area	Open area		

Bulls	12.0	24.0	1	175 cm. in medium and heavy rain fall and 220 cm. in dry areas.
Cows	3.5	7.0	50	
Buffaloes	4.0	8.0	50	
Down – calver	12.0	12.0	1	
Young – calves	1.0	2.0	30	
Old – calves	2.0	4.0	30	

Feeding and watering space requirements

Type of animal	Space per animal (cm)	Total manger length in a pen for 100 animals(cm)	Total water tank length in a pen for 100 animals (cm)
Adult cattle & buffaloes	60 – 75	6000 – 7500	600 – 750
Calves	40 – 50	4000 – 5000	400 – 500

FEEDING MANAGEMENT OF CALVES, HEIFER, PREGNANT, LACTATING

AND DRYANIMALS, AND BULLS AND WORK ANIMALS

Care and management of new born calf

Remove the mucus from the nose and mouth and clean it. If the calf does not start breathing, artificial respiration should be used by pressing and relaxing alternatively, the chest walls with hands. Another method is to hold the calf by the rear legs and lift from the floor with the head down. This may be repeated several times and helps in restoring respiration.

As soon as the calf starts breathing, observe as to whether the navel cord is still attached. The navel cord should be disinfected. The navel cord of the calf is tied about 2.5 cm away from the body and cut about one centimetre below the ligature. Apply tincture of iodine to the cut end and repeat it 2-3 days. This will prevent infection. Then, if the cow does not lick the calf dry, or if the weather is cold, the herdsman should wipe the calf to clean and dry.

The next important step to follow is to feed the Colostrum **within 15 minutes** of calving, the calf should be fed with colostrum at the rate- $1/10^{\text{th}}$ of body weight and buffalo calves at the rate $-1/15^{\text{th}}$ of body weight. Colostrum containing low fat, high protein, vitamins and minerals forms a balanced feed for new-born calves. This helps to protect the calf

against various diseases as it contains antibodies. Colostrum also helps to eliminate the material accumulated in the digestive tract before it was born.

If muconium (first faecal matter) is not voided out, mild enema by dissolving soap in a liter of warm water should be given.

Weaning: If weaning at birth is followed care should be taken to see that adequate colostrum is fed for the first 3-4 days. If weaning is practised 4 days after calving, then further ration has to be fed as per the schedule described.

Calf rearing system varies with the facilities available to farmers. They may be reared indoors or outdoors or partly indoors and partly outdoors. The important factors to be considered are:

1. Availability of quality fodder.
2. The humid tropical environment is ideal for the proliferation of internal parasites and it is very difficult to keep the calves free from massive infection if they are grazing.
3. Outside grazing may not provide sufficient nutrients for optimum growth.
4. Climatic stress affects growth and utilization of feed nutrients.

In humid tropics, it may be desirable to keep the calves indoor in day time and outdoor at night. This will reduce parasitic infection also. Thus, it is advantageous to keep new born calf in individual pen for the first 3-4 weeks of age. Calves that are running in batches often suckle or lick each other after feeding and it is a good practice to keep them in their ties for some time after milk feeding. Hair swallowed by the calves after suckling each other often form a hard ball in the abomasum and this is a constant cause of digestive disturbances. Cleaning the mouth of the calves after each milk feeding is a sanitary practice. The calf pens should provide comfort and easy cleaning.

Identification: This is essential for good management, especially in breeding farms. The best method of permanent identification is by tattooing the inside of the ear with indelible ink. Metal ear tags or button with letters and numbers may be inserted in the ear as a means of identification. Neck strap or neck-chain with a number plate attached, make an easy method of identification.

Body weight: of the calf is recorded on a balance along with length, breadth and height for the computation of milk allowance. Well-fed cross bred calves on an average should gain 400 grams a day or 2.5 to 3 kilograms per week.

Removal of supernumerary teats is also important and this has to be carried out before development begins. This is usually done in the first month of age with the help of a short pair of sterile scissors. If the extra teat is at the base of the normal teat, veterinary help may be resorted to remove it.

Dehorning or disbudding: Disbudding is carried out either by the use of hot iron, caustic sticks and electrical dehorning cone. Both the buds are destroyed at the early age (within 3 to 10 days).

Feeding Management: Utensils in which whole milk or milk replacer is fed to calves, must be clean and should be cleaned after each feeding. Severe digestive upsets can result from such contamination of the feeding parts. Either the nipple pail or the open type bucket are satisfactory for feeding milk or milk replacer. It may take less effort to teach a calf to nurse from a nipple pail than to drink from an open pail. Also, a rapid consumption of milk from an open pail may at times cause digestive upsets.

To teach a calf to drink from an open pail, place your fingers in its mouth and after it starts to nurse lower its head into a pail of warm milk or milk replacer. It may be necessary to repeat the process several times. A stubborn calf may need to be backed into a corner and restricted by standing aside its neck. Maintaining the temperature of the milk as removed from the cow is not necessary. However, it should be aimed to feed the milk at this temperature itself. However, cold milk at 35° to 40° F may cause calves to shiver and chill. At any rate, calves should not be overfed.

CARE AND MANAGEMENT OF HEIFERS.

1. Heifers are either reared indoors or outdoors for 9-12 months.
2. In case of outdoors rearing protection must be given from the adverse climatic condition, rain, hot sun, snow, heavy winds biting flies, parasitic infestation etc.
3. Heifers of exotic breeds performance is slow in tropical areas in the outdoors.
4. Small breeds attain age at first breeding in 15 months and large breeds in 18 months.
5. Adequate live weight would be 200-225 kg for smaller breeds and 275 kgs for the larger breeds for breeding.
6. Cross bred heifers show signs of heat as early as 10 months of age but none of them are mated until attain the body weight of 225/275 body weight or a minimum of 14 months age.
7. Age at first calving 25-28 months.
8. Feeding of Heifers.

A. Concentrate feed:

- | | |
|---------------------|-------------|
| a. 3 months- 1 year | : 1 kg. |
| b. Above one year | : 2 kg. |
| c. Pregnant Heifers | : 3-3.5 kg. |

B. Green Fodder

- | | |
|--------------------------|----------|
| a. Leguminous fodder | : 10 kg. |
| b. Non Leguminous fodder | : 25 kg. |

C. Dry fodder : 3 kg.

9. Vaccination Schedule.

Brucell Strain 19 –to prevent abortion due to Brucellosis-Vaccinated at the age of 4-6 months of Calf.

Other Contagious Disease Vaccination are done accordingly prior to the prevalence of disease and prior to rainy season

Foot and Mouth disease : Once in 4 months.

Rinderpest : 1-3 years.

Haemorrhagic Septicaemia : 1 year.

Anthrax : 1 year.

Black Quarters : 1 year.

10. Housing :

a. Outdoor system / Grazing method

- Reared chiefly of grazing
- Care to be taken not to overstock on limited grazing land.
- Rotational grazing.
- Arrangement of shade and drinking water – pasture land
- Concentrate feed is to be provided – Centrally located feed trough.
- Protect from rain.

b. Indoors

- Management in covered area.
- Sufficient concentrate feed and fodder provided.
- Steaming up of heifers.

Feeding grains to pregnant heifers prior calving at 1.5 Kg. per day. It helps in their growth, bear the stress of foetus. It produces more milk after calving and increases lactation length.

Training of heifers

Heifers in early stage should be lead with halter to make them docile. Pregnant Heifers are to be housed along with milking cows at least a month prior to calving. The udder should be washed warm water and mopped with cloth to accustom her to feel the hands in this place. Just few days prior to calving pulling teats slightly may be practiced so that heifer would not excited.

Control of Parasites – Dewormed periodically at 3 –4 months intervals

Grooming is to be practiced to avoid ecto- parasites.

CARE, MANAGEMENT OF PREGNANT ANIMALS

1. Identify pregnant animal after 90 days of artificial insemination if animal does comes to heat again.
2. Provide gentle treatment

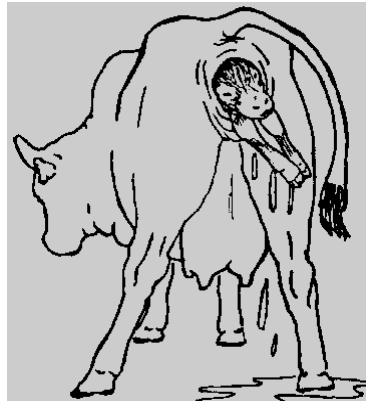
3. First quarter of gestation period are critical
4. In early stages of pregnancy disturbances can cause abortion.
5. Provide concentrate feed 3.5 kg per day.
6. Provide 25 – 35 Kg. Greed fodder per day and 5 Kg. Paddy straw.
7. Minimum 45 – 60 days of dry period is essential.
8. Avoid long distance travel.
9. Avoid slippery condition in the shed.
10. Avoid chasing by dogs, bulls or children.
11. Avoid infighting between pregnant animals.
12. Separate pregnant animals from recently aborted animals or carriers of diseases like Brucellosis.
13. Provide adequate clean drinking water
14. Protect against extremes of climate.
15. A few days before the probable date of calving, cow should be transferred to individual calving pens. Pregnant cow may be transferred to calving pen 1 to 2 weeks before the expected calving date. The number of calving pens required on a farm depends on the number of breedable cows and heifer, generally 5 per cent of this number.
16. Ample amount of drinking water, laxative feed and generous supply of bedding may be provided.
17. The calving pen should be scrupulously cleaned and sterilized before bringing in the cow

Signs of approaching parturition

- i. Cow will leave the herd and seek isolation
- ii. Loss of appetite and distress
- iii. Distention of teat and udder, considerable milk appears in the udder and there may be dripping of milk
- iv. Relaxation of pelvic ligament one day before calving, the ligament on the sides of the tail head is loosened so that hollows appear on either side of the backbone and the tail head is raised and the quarters are dropped.
- v. The vulva become enlarged and flabby
- vi. Animal will be restless and will pace about often trying to kick or scratch the flank region.
- vii. The parturition process has three stages,
 - a. preparatory stage (uterine contraction and dilatation of cervix)
 - b. active expulsive stage

- c. expulsion of foetal membrane.
- viii. Cow will deliver the calf within 12 hours after commencement of first stage.
- ix. Care must be taken to observe expulsion of placenta (after birth). It should be removed immediately so as to avoid cow eating it.

Normal calving



Management immediately after calving

1. Udder and hind quarter should be washed with lukewarm water containing an antiseptic solution of potassium permanganate lotion and dried with clean cloth.
2. Cows may be milked to relieve the pressure of the udder. If day old weaning is not practiced the calf can be allowed to remain with the mother in the calving pen for 7 to 10 days. Otherwise the calves can be removed immediately to calf pen. If the maternal instinct is more, cow's eyes can be blindfolded before the calf is removed.
3. The placenta should be expelled within 12 hours after parturition, if not it should be removed manually. Before manual removal the body temperature should be noted. In case of pyrexia, attempt should be made systemically to reduce fever. Otherwise systemic infection may establish.
4. The cow should be monitored carefully for signs of any metabolic disorders like milk fever, grass tetany, ketosis, and acidosis and should be treated immediately.

CARE AND MANAGEMENT OF LACTATING ANIMALS.

The animals must be protected against inclement weather. Hygiene and sanitation of cattle shed and animals grooming, washing, disinfection etc must be carefully followed. Feeding must have: a) concentrate b) Roughages – Green fodder which include leguminous (1/3) and non-leguminous fodder (2/3) and dry fodder. Thumb rule for concentrate feeding is 450 –500 g Concentrate / Kg. milk production. The concentrate must contain DCP 15%: TDN 75%. The model feed formula may contain maize 33%, groundnut oil cake - 25%, Wheat Bran - 40%, Mineral Mixture- 1% and Salt 1%.

Feed alone constitute 60 per cent of the production cost of milk. Hence, feeding management plays a vital role in farm economy. The nutrient requirement should be determined for maintenance as well as for milk production and to meet the fat percentage in milk and gestation. In general the dry matter from roughage should not exceed 2 per cent of cow's live weight nor should it be less than 1 per cent.

Feeding of dairy cow at different stages of lactation

Quantity of green grass to be give (kg) for animal weighing			Concentrate (kg)
250 kg	300 kg	350 kg	
25	30	35	1.0 kg for every 2.5 kg of milk of average 4% fat percentage, in case of buffalo 1.0 kg for every 2.0 kg of milk produced.

Early lactation

The recently calved high producing cow is unable to eat enough feed to support her milk production. This means that the cow should have enough reserve to store nutrient to be drawn to tide over the period of heavy demand in early lactation, during which period the cow loses weight. This is the period of peak yield. Sometime during this period (usually 6 week), calcium deficiency may occur and in this case the animal must be supplemented with calcium in feed or injected through intravenous route.

Feeding during mid and late lactation

The cow can consume enough feed to meet the various demands for nutrients and the body weight of the cow remains more or less stable. During this period the cow may be fed a well-balanced ration of good quality fodder and concentrate according to the milk yield and fat percentage of milk.

During the late lactation, intake ability of the cow exceeds nutrient needs. This is the time when the cow starts needing extra allowance for the growing foetus. This is also the period when the cow can readily replenish the already depleted body reserve and gain weight very fast. From 7 ½ month to 10 months of lactation, cow may be fed 1-2 kg concentrate feed in addition to their nutrient requirement for maintenance and milk production to replenish the condition lost in early lactation.

Frequency of feeding: Dividing the daily ration into 3 or 4 parts and feeding them in so many installments has been found to be useful in overcoming this problem. This also results in greater digestibility and better utilization of protein apart from preventing explosive release of acids.

Animals must be vaccinated periodically. For Rinderpest, Black quarter, Hemorrhagic septicemia, Anthrax-once in a year and for foot and mouth disease once in four months.

CARE AND MANAGEMENT OF DRY COWS

The day from the cow stops giving milk to the day it calves next, can be considered as dry period. Mostly dry cow does not give milk but it is pregnant. In some cases like infertility, the dry cow may be non-pregnant.

Purpose of dry period

- The mammary gland functions incessantly during the entire lactation period, which results in considerable wear and tear in the secretory tissues of the udder. Dry period give rest to the cow's udder and helps the repair and regeneration of the secretory cells.
- During lactation the high yielders lose condition (body weight), to maintain high milk flow especially greater quantity of nutrient, calcium and phosphorous during lactation. Dry period gives an opportunity for the cow itself to recuperate condition lost during the lactation.
- On the other hand, uninterrupted lactation between calving is known to impair succeeding lactation production. Thus dairy cows need to be dried off before the ensuring calving and should be "steamed up" for the next lactation.

Ideal dry period

The optimum length seems to be approximately six weeks for second lactation and older cows, and 60 days for first lactation cows.

Dry period and udder health

The dry period is an extremely important time for udder health for three reasons.

- First, the dry period is a good time for the long-term treatment of cows with a high cellcount.
- Second, many of the cases of mastitis in the first month after calving are the result of infections during the last weeks of the dry period.
- Third, the first few weeks after calving are a time of greatly reduced disease resistance for the cow: during this period most clinical diseases (including clinical mastitis) occur. The degree to which her resistance is lowered depends heavily on the cow's nutrition, hygiene and care during the dry period, transition period and fresh cow period.

Drying off

Principally, there are three methods of drying cows, viz., abrupt cessation of milking, intermittent milking and incomplete milking.

Abrupt/sudden cessation of milking: Sudden cessation of milking cause builds up of pressure in the udder which has no immediate permanent effect on the rate of secretion and so on drying off for a few days. After 4 or 5 days, involution or regression of the secretory

cells commences suddenly which is completed in another few days. The fluid trapped within the udder is ultimately resorbed. This method is successful for the low milk yielding cows (less than 9 liters).

Intermittent milking: Milking once every second or third day or even less frequently, is not so effective a method and may even result in mastitis. But the method is preferred for cows already having mastitis; the udder is emptied and infused with antibiotic cream at each milking.

Incomplete milking: Emptying the udder gradually smaller and smaller quantities of milk from the udder spread over a week or so, can at best be preferred for very high yielding cow, in which sudden cessation of milk may cause swelling and pain in the udder.

More rational approach to drying high yielding cow is to reduce ration, especially protein, a couple of weeks prior to the proposed date of drying. In all cases, it is a beneficial precautionary step to wipe the teats with alcohol immediately after the last milking and then apply an antiseptic colloidal seal. This will go a long way in preventing mastitis, especially in cows with weak sphincters. This method is known as “*dry cow therapy*”

Feeding of dry cow

More than half of the calf's foetal growth takes place during the last quarter of pregnancy-the dry period. Feeding of cow during this period is often neglected because the cow is not giving any income. Due to this reason, good fodder and concentrates are generally withheld from the dry cow. This will have adverse consequences on her future production.

A dry cow should be fed adequate quantity of well-balanced ration for

- i. Maintenance of the cow
- ii. Forming sufficient reserve for ensuing lactation
- iii. Growth of foetus
- iv. Production of colostrum with high nutritive value.

	Quantity of green grass to be give (kg) for animal weighing			Concentrate (kg)
	250 kg	300 kg	350 kg	
Dry cow	25	30	35	• For non-pregnant cows no concentrate is required. • Pregnant cows should be fed additional quantity of 1.5 kg of concentrate from 7th month of gestation • In case of dry cow, allowance up to 1 kg concentrate can be given if the condition of cow is poor or the fodder quality is inferior.

CARE AND MANAGEMENT OF WORK BULLOCKS

- 60% - 70% of time – allotted to care and management of limbs and neck.
- Avoid over working the bullocks. The work should be evenly distributed.
- Protect the bullocks from rain and inclement weather exposure
- Lean type roof on the side of farmers house.
- Shoe the bullock properly before using them for work on hard ground.
- The hoof should be prepared first and shoe should made to fit the natural shape of hoof.
- Shoeing – road work – once in a month, field work – once in two months.
- Hooves should be hard, black and waxy the two halves should be even. The cleft of hoof should be narrow.
- Grooming is essential as it increases cutaneous respiration, spreads subcutaneous uniformly and parasitic infection is avoided.
- Feeding depends of type work
 1. Normal – 2-4 hours
 2. Heavy - 8 hours (Ploughing, pulling loaded cart

Maintenance – 1.5Kg. Concentrate

Body weight	Normal work	Heavy work
300 Kg.	2 Kg.	2.5Kg.
400 Kg.	2.5 Kg.	3 Kg.
500	3	3.5

In addition 25 Kg of green and 3 to 5 Kg of dry fodder should be given.

NUTRITION

Nutrition involves various chemical reaction and physiological process which transforms food into body tissue (milk, meat, egg, wool) and activities (Work power). Nutrition involves ingestion, digestion, and absorption of the various nutrients and their transport to all the body cell and the removal of unusable elements and waste products of metabolism.

Nutrients are defined as the substances which can sustain or aids in the support of the life. **Lavoiser** - French, a scientist is referred as father of Nutrition. There are two aspects in Animal Nutrition

1. Science of Nutrition – It is the work of Animal Nutritionist
2. Art of feeding of animals.- Good stockman ship.

Feed or feed ingredients that supplies higher-level of major nutrients but contains less than 18 % crude fibre are called as **concentrates** and those having more than 18% crude fibre are classified as **roughage**.

Concentrates containing higher than 18% protein are called **protein rich concentrates** while those containing less than 18% protein are called “**Energy rich concentrates**”.

RATION:

It is the feed allowed for a given animal during a day of 24 Hours.

Balanced Ration.

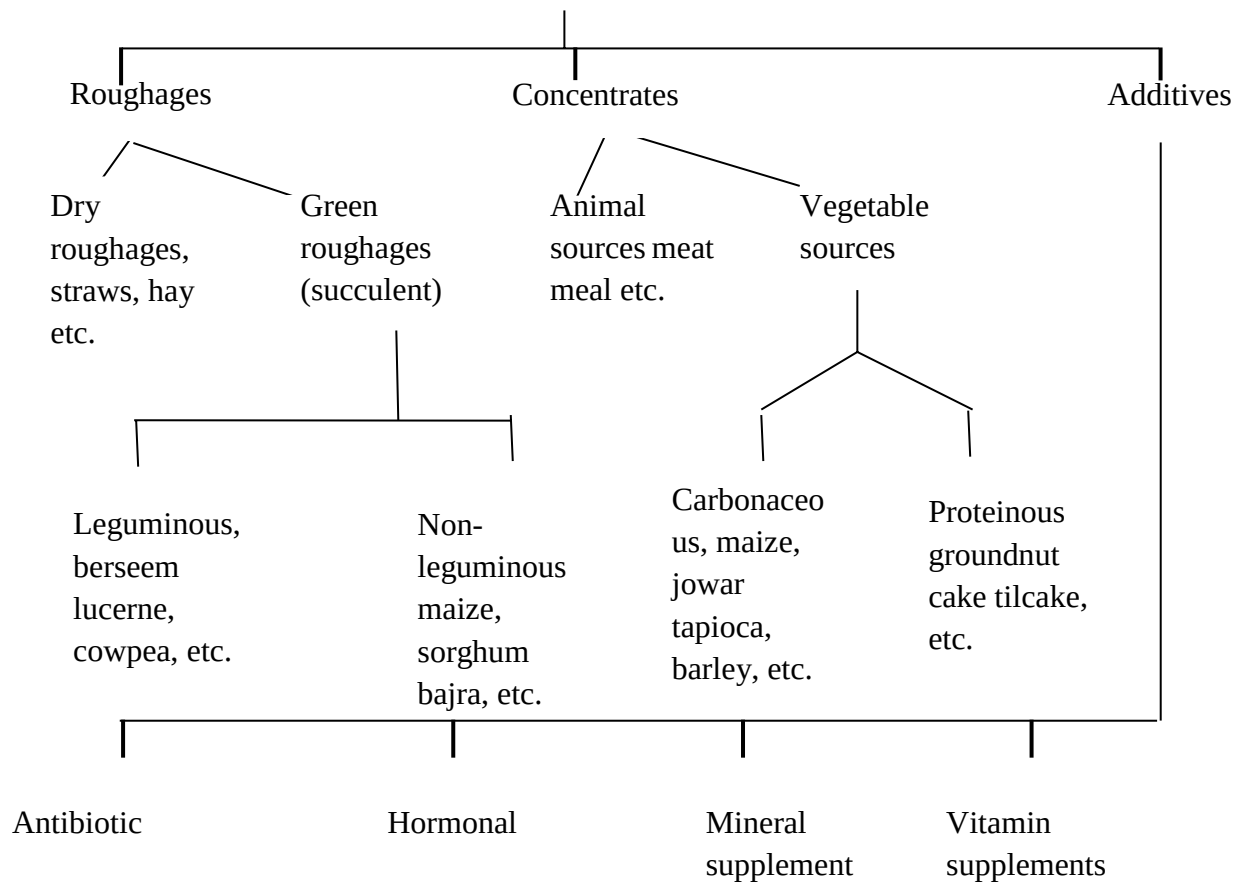
Balanced ration which provides essential nutrients to the animals in such proportion and amount that are required for the proper nourishment of the particular animal.

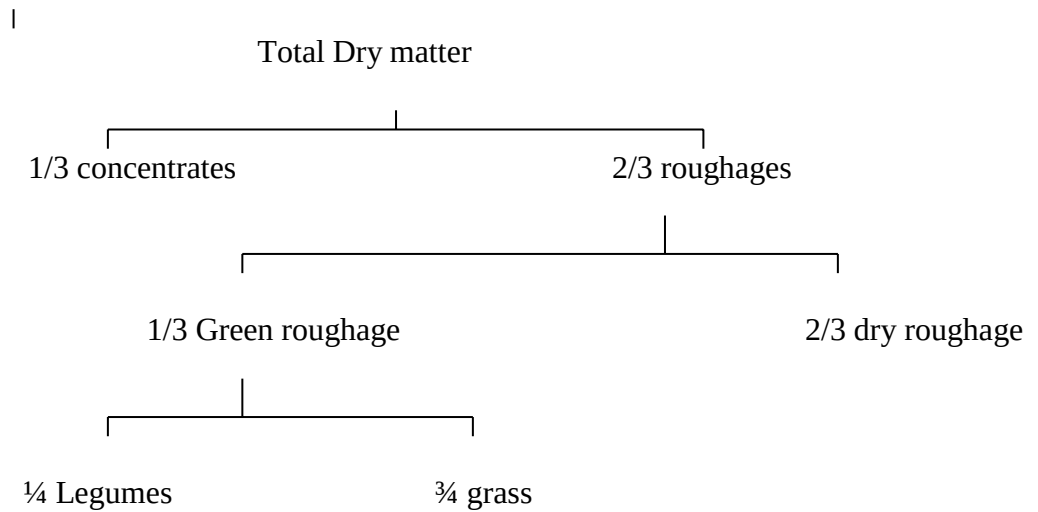
Desirable Characters of a ration.

1. Liberal feeding; Satisfy all the physiological status +waste in preparation + Feeding.
NOT over feeding-Doubly Wasteful.
2. Individual Feeding: Avoid Competition; adequate –individual feeding is always better.
3. Properly Balance: Concentrate; Roughage. a. green fodder i. Leguminous and Non Leguminous fodder. b. Dry fodder.
4. Palatable & Variety: Better and balanced mixture of protein, vitamins and other nutrients.
5. Good and Sound: Low quality-unwholesome ingredients, may contain toxic components-poor quality –reduce feed value.
6. Mineral Mixture: Every Kg milk- 0.7%.-Deficit –depletion cause metabolic disease. Milk contains eg. Ca 3 g and 2.7 g in milk and 3 grams in egg shell.
7. Laxative: otherwise food will be incompletely digested constipation-digestive disorder-utilization –nutrients affected-reduction in production.
8. Bulky: Capacious and satiety.
9. Green Fodder: Source of vitamin-‘A’-reproduction-Bulky- laxative-cost wise cheap-unidentified factors-easily digestible.
10. Avoid change in the diet: Bacterial digestion-Prevalence of specified species-sudden change – digestive disorder.
11. Maintain regularity: Glandular Secretion-essential for digestion.
12. Properly Prepared: Hard grain-Coarsely ground-Cottonseed soaked-coarse fodder – chaffed- sprinkled salt-molasses-increases consumption.
13. Labour and cost; ultimate –aim-profit; 70 % cost of production is attributed to feeding of animals.

Classification feed and fodder

General classification of feeding stuff





Sl.No	Concentrate	Roughage
1.	10 % Moisture and 90% Drymatter	Dry fodder--10 % Moisture and 90% Drymatter Green Fodder---80-90 % Moisture and 10% Dry Matter
2.	Highly Digestible	Comparatively less digestible
3.	Crude fibre less than 18%	More than 18%
4.	Nutritive Value/unit mass is high	Low
5.	Compact in Nature	Bulky
6.	Keeping quality -High	Variable : Dry Fodder – High, Greenfodder- Less/low

Common feed ingredients

Carbohydrate rich concentrates	Protein rich concentrates	Fibre rich concentrates
Maize or Corn	Groundnut cake	Wheat bran
Barley	Soyabean meal	Raw rice bran
Oats	Linseed meal	De-oiled rice bran
Sorghum	Mustard cake	
Bajra	Coconut cake	
Mill by-products like flour, polishings	Cottonseed cake	
Molasses	Sesame meal	
Roots and tubers	Pulse protein	
	Sunflower meal	
	Unconventional plant proteins	

	Meat meal	
	Meat and Bone meal	
	Fish meal	
	Blood meal	
	Feather meal	
	Hatchery by-product meal	

MODEL COMPOSITION OF CONCENTRATE MIXTURE OF YOUNG AND ADULT STOCK

IMPORTANCE OF GREEN FODDER PRODUCTION

India has about 15% of world livestock population with only 2% of world's geographical area. The current feed and fodder resources in India can meet only less than 50% of the requirement of its livestock population. The grazing intensity is very high viz., 2.6 cattle unit per ha as against 0.8 cattle unit per ha in developed countries. We are highly deficient in various livestock products, though we have about one-fourth of the total cattle population of the world. The analysis of this situation reveals that one of the main reasons for the low productivity of our livestock is malnutrition, under-nutrition or both, besides the low genetic potential of the animals.

IMPORTANCE OF GREEN FODDER

- Green fodder is the primary only source of vitamin A for lactation and vitamin A is present in the form of precursor.
- Maintenance & function of the mucous membrane
- is directly related to vision.
- is essential reproduction a. conception, b. early embryonic mortality, c. maintenance of pregnancy, d. shedding of placenta.
- is essential for the respiratory tract
- is essential in the Gastro intestinal tract/ digestive tract-deficiency causes diarrhoea, mal absorption of nutrients etc.,
- is essential for the urinary tract –deficiency causes stones in the kidney , ureter , bladder.
- During lactation 2000 I.U. of Vitamin 'A' is eliminated in every litre of milk-It is to be replenished
- laxative in action
- cheap source of Vitamin 'A'
- source of minerals ,Crude protein, Total digestible nutrients and dry matter
- Unidentified factors.
- Carotene Content of some fodder

- a. Agathi 18.3 mg / 100 dry matter
 - b. Lucerne 15.6 mg / 100 dry matter
 - c. Guinea grass-14.2 mg / 100 dry matter
 - d. Desmodium 7.09 mg / 100 dry matter
- Feed should be available to cows at least 20 hours / day.
 - Feed at least 60 % of ration during night in the hot weather (Summer)
 - Cows will reduce feed intake by about 3.3% for every 2.2 rise in temperature over 24° C
 - High producing cows will eat up to 12 meals / day each averaging 23 minutes.
 - Water should be available Ad-libitum.

NUTRITIVE VALUE OF FODDER CROPS

These are highly digestible (55 – 65%) mostly when harvested at a proper time. The crude protein may range from as little as 3% in very mature forages to over 30% in young heavily fertilized grass (on DM basis). The soluble carbohydrate of grasses ranges in the dry matter from 4-30%. The cellulose and hemicellulose are generally within the range of 20-30% and 10-30% of the dry matter respectively. Grass proteins are particularly rich in arginine, glutamic acid and lysine. Green forages are excellent source of carotene 250mg/kg), the precursor of vitamin A.

Generally leguminous fodder contain 8-12% DCP and 45-60% TDN. The phosphorus content of leguminous fodder are poor. It is advisable to supplement a ration containing a large amount of leguminous fodder with a limited quantity of wheat or rice bran, which is rich in phosphorus. The non-leguminous fodder are having 2.5% DCP and 45-60% TDN on dry matter basis. Green fodder is the primary source of vitamin A. Vit.A is present in the form of precursor. Green fodder contains 100 mg carotenes /Kg when compared with about 20 mg /Kg in silage. Carotene requirement of milch animals is 60 mg for production, 30mg for pregnancy, for growth requirement is 11 mg carotene per 100 Kg live weight.

Vit A is directly related to vision, maintenance and function of mucous membrane, essential for reproduction (for conception, maintenance of pregnancy, shedding of placenta), deficiency leads to diarrhoea, malabsorption of nutrients, incidence of stone in the kidney, ureter & bladder. During lactation 2000 I.U. of Vit.A is eliminated in milk.

VALUE OF TREE FODDER

Trees, which can be grown either in combination with agricultural crops or on separate land usually not fit for agriculture, offer opportunity of producing green nutritious fodder for the livestock. It is seldom realised that in some parts of our country, probably more animals feed on shrubs and trees than on grass or grass legume pasture.

- Trees can produce as much, if not more, green fodder per unit area as agricultural

fodder crops. The more important desirable agronomic features of a tree species are

- Be reasonably easily and reliably established
- Exhibit a good competitive ability against weeds
- Remain regally productive under repeated ability or grazing and browsing.
- Be well adopted to the particular climatic and edaphic features of the environment
- Require, no or little fertilizer
- Be resistant to local pests and diseases
- Have adequate forage production or be reliably vegetatively propagated and
- Have good nutritive value and reasonable palatability and acceptability to animals.

MILKING METHODS

Definition : Milk is defined as whole, fresh, clean lacteal secretion – complete milking of healthy milch animals excluding that obtained 15 days before or 5 days after calving and containing prescribed % of fat and SNF

PRINCIPLES OF REMOVING MILK:

3 PRINCIPLES

1. Natural Technique (calf suckling)
2. Manual Technique (hand milking)
3. Mechanical Technique (machine milking)

Natural Technique:

This method calf is able to draw the milk from the udder. To extract the milk the calf presses the teat with the tongue and pallet on the other side. The tongue encircles the teat and vacuum is created in the mouth by separating the jaws and retracting the tongue nearly 100-200 alternating cycles may be observed per minute. A calf's suckling is the best method of evacuating the milk with least damage to the delicate tissue of mammary gland .The art of milking is a cycle.

1. Active Phase

2. Restive Phase

ACTIVE

PHASE:

- a) Creation of vacuum in the teat canal
- b) Pressure is applied over the teat canal
- c) The base of the teat is apparently occluded with the help of the tip of the tongue with the idea to prevent the back flow of the milk into the gland cistern when the pressure is applied which is followed by restive phase

RESTIVE PHASE:

At this stage 20mm Hg pressure is created at the teat end .in the phase both active

and restive phase are alternated and it has been scientifically proved that the amount of pressure applied over the teat canal by calf is 535mm Hg pressure whereas in the case of hand milking the pressure is 310mm of Hg.

In the mechanical milking pressure on the teat is with the range of 350 mm-400mm Hg. In the case of buffalo's 400mm of Hg of pressure is applied but in the case of cattle it can be restricted to 360-380mm of Hg. It has been proved that cycling rate during nursing is twice as fast as hand or machine milking. Thus the difference along with increased cycling rate facilitates and explains the removal of milk from the udder at a faster rate by a calf when compared to hand or machine milking.

Hand milking: It is commonly practiced in the harvesting of milk. In order of milking of various teats also differ.

1. Teats crosswise left four and right hind or right four and left hind.
2. Fore quarters teat together
3. Hind quarters teat together
4. Teats appearing more distended should be milked first. The milk should only be squeezed and not drawn

STRIP CUP:

It is a device with four circular plates for each quarter which has the quantity of milk normally first few strip of milk are drawn in the respective circles to assess the physiological status of the udder. If there is any change in color, consistency appearance, etc., the milk should be drawn at the end so as to prevent spreading the disease from one quarter to other.

Prevention of Kicking of the cow:

1. Application of milk man's rope.
2. Anti-cow kicker.

Methods of milking

1. Hand milking
2. Machine milking

1. Hand milking

Hand milking is the most common practice in India. Cows are milked from left side. Stripping and Full hand milking are two commonly used methods of milking.

a. Wet hand milking

It is done by lubricating the milker's hand and teat either with water or oil. These make the teats dry and chaffed. Crack and sores may appear which will cause pain to the animal. Instead, apply some antiseptic cream after the end of milking in such animal.

b. Dry hand milking

In this method, the milking operation is practised without lubrication of the milker's hand and teats. It is considered to be the best method as it doesn't cause any chaffing /sore on

the teats.

Different Methods of Hand Milking:

i. Full hand milking

It comprises holding the whole teat in the ring formed by the palm and the forefingers. Simultaneously, the teat is squeezed using middle, ring and the little fingers and hollow of the palm thus forcing the milk out. This process should be repeated in quick succession. Full hand milking removes the milk quicker than stripping. Cows with large teats and buffaloes are milked by full hand method. Full hand method is superior to stripping.



ii. Stripping

It consists of firmly holding the teat at its base between the thumb and the forefingers and drawing down the entire length of the teat pressing it simultaneously to force the milk to flow down in a string. The process is repeated in quick succession. Both the hands may be used, each holding a different teat stripping alternatively. Stripping is practised in cows with very small teats. It causes more irritation and teat injuries due to repeated sliding of the fingers. In spite of this few



strippings are done to milk the cow completely especially at the end of the milking process. The last drawn milk is called stripping which is richer in fat.

iii. Knuckling: This method is painful to the animal and hence not recommended

iv. Fisting: This method is not normally practiced.

v. Pinching: This method is painful to the animal and hence not recommended

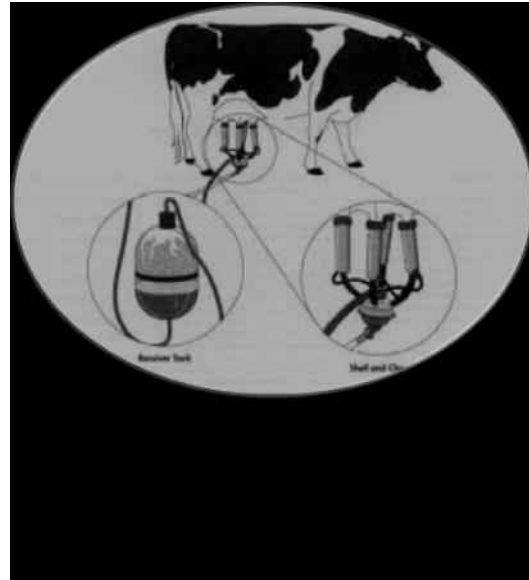
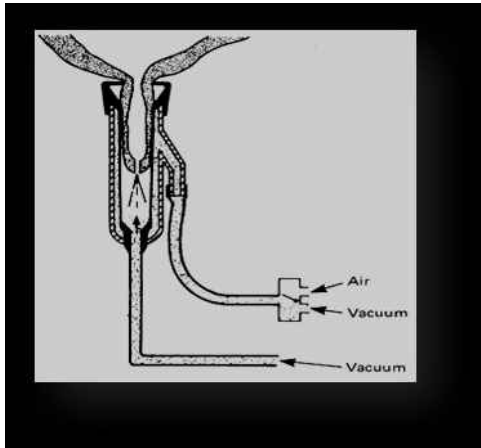


2. Machine milking

It is popularly used now-a-days in most of the western and industrialized countries. They use alternating negative and atmospheric pressure with the help of double chambered teat cup assembly, the continuous partial vacuum inside the inflatable rubber tube that is teat cup liner into which teat is inserted. A partial vacuum and normal atmospheric pressure is alternated in the space between rubber liner and metal shelf of the teat cup by means of pulsator. When negative pressure is applied between



the liner and shelf, milk flows from the teat. When atmospheric pressure enters the chamber, the rubber liner inflation collapses with the result the teat is compressed and massaged. The continuous vacuum would cause congestion and irritation of teats.



Ideal practices used in machine milking

1. Get the cow ready by wiping and massaging the udder and teats for half to one minute. Use the cloth dipped and squeezed out of warm mild antiseptic solution.
2. Strip cup test: use the strip cup, draw the first milk into the cup from each quarter and check it for any abnormality.
3. Put on the teat cups promptly.
4. It is a good practice using a timer to avoid over use of machine.
5. Break the vacuum first. Pull down the teat cups and massage the udder with hand.
6. Apply teat dips.
7. Record the milk weight.
8. After every milking, thoroughly and properly clean the machine.
9. The manufacturer's direction may be followed and service the milking machine.

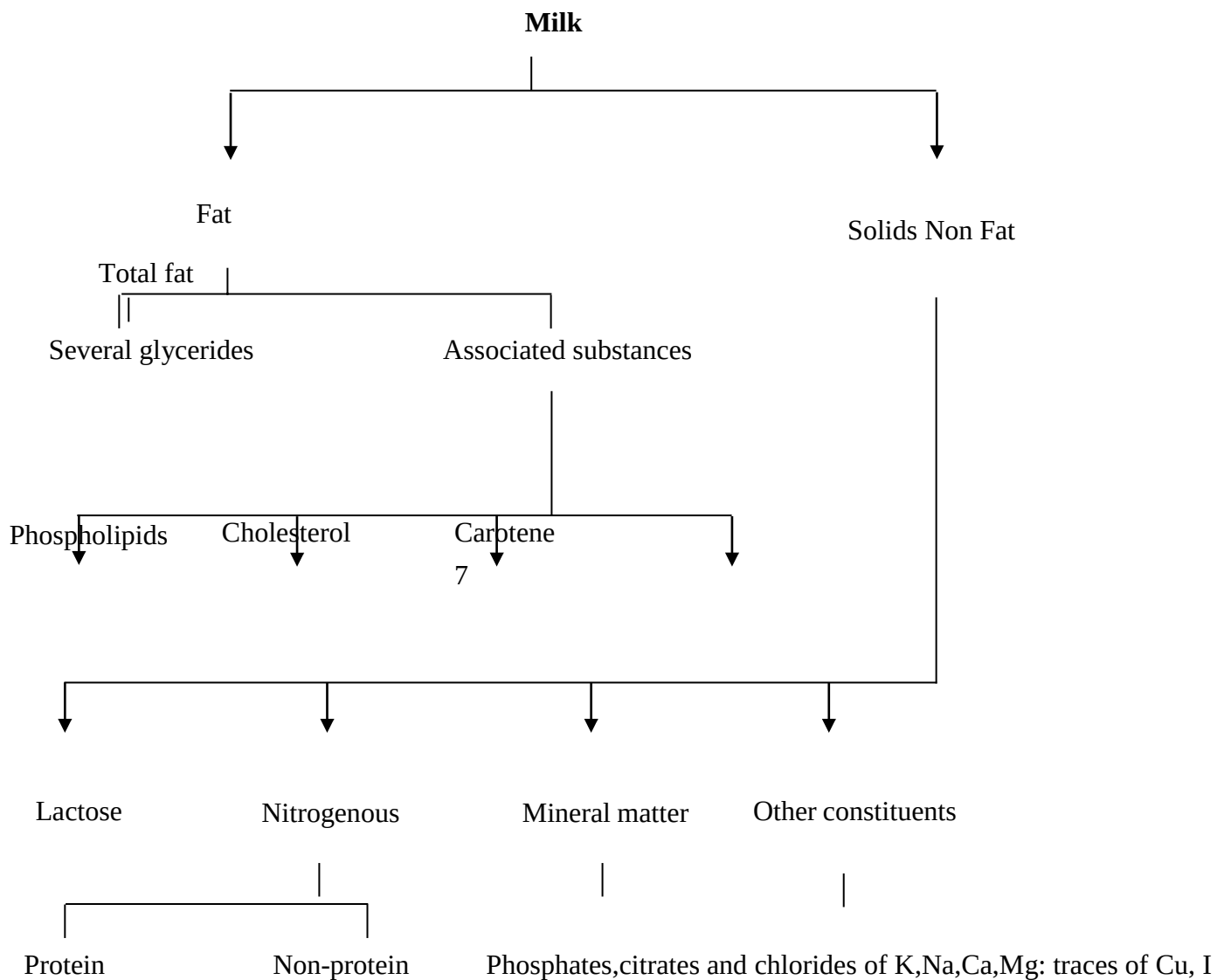
HYGIENIC MILK PRODUCTION / CLEAN MILK PRODUCTION

Milk is the lacteal secretion of the mammary glands of animals. It is obtained generally from the cow or the buffalo during the period following at least 72 hours after calving or until the milk is colostrum free. Milk is a white opaque fluid in which fat is present as an emulsion, protein and some mineral matters in colloidal suspension, and lactose

together with some minerals and soluble proteins in true solution.

Average composition of milk of different mammals (in per cent)

Species	Water	Fat	Protein solids	Total	SNF	Lactose	Ash
Human	87.43	3.75	1.63	12.57	8.82	6.98	0.21
Cow	86.61	4.14	3.58	13.19	9.25	4.96	0.71
Buffalo	82.76	7.38	3.60	17.24	9.86	5.48	0.78
Goat	87.00	4.25	3.52	13.00	7.75	4.27	0.86
Sheep	80.71	7.90	5.23	19.29	11.39	4.81	0.90
Camel	87.61	5.38	2.98	12.39	7.01	3.26	0.70
Mare	89.04	1.59	2.69	10.96	9.37	6.14	0.51
Ass	89.03	2.53	2.01	10.97	8.44	6.07	0.41



Fe, chromium, cobalt, manganese Zinc.

Average composition of milk of some important milch breeds of cows (per cent)

Breed	Total solids	Fat	Protein	SNF	Lactose	Ash
Red Sindhi	13.66	4.90	3.42	8.76	4.81	0.70
Jersey	14.91	5.37	3.73	9.54	4.93	0.70
Friesian	12.26	3.40	3.13	8.86	4.86	0.67
Crossbred	13.13	4.50	3.37	8.63	4.92	0.67

Clean Milk production:

Both pre-and post-secretory management of milk at the farm level should be focused upon for controlling the quality of milk. Milk once secreted becomes the target for transformation by a variety of host organisms at the farm itself. Hence, proper care must be taken regarding preservation of milk, protection of milk constituents, protection against high temperatures and natural calamity. The microbiological quality deserves special attention for stringent export requirements for milk products in global market.

Rural milk collection

Collection of small amounts of milk scattered over long distances, therefore, possess a formidable challenge in maintaining the quality attributes and keeping costs down.

Milk may be collected from individual procedures either by the contractor or by forming village level cooperative societies.

At the village level, milk brought by the individual farmers is first tested for quality. As soon as the milk supply reaches collection centres, it is weighed and a representative sample is drawn for quality grading. The common tests carried out at the point of milk collection are taste and smell, sediment, fat and SNF contents and acidity test. Care should be exercised to bring the milk for chilling/processing within 3 hours of milking otherwise serious deterioration of milk takes place, which affects the quality of products.

Transportation

In the Indian context, most of the milk is transported from rural collection centres to the dairy plant depending upon the volumes of milk handled.

Milk reception at chilling centre

In a distant milk plant, rurally collected milk is first brought to a common chilling centre. At the chilling centre, milk is promptly chilled to 4°C and stored in large tanks of 2,000-10,000 litre capacity. Chilling centre operation is economical only when about 30,000 litre of milk is handled per day. It is then transported, through tankers, to dairy processing unit.

Milk reception at dairy plant

The milk at the dairy plant is received either in cans or road/rail tankers. There are separate reception docks where milk is received through tankers and cans. The milk from each can is automatically or manually emptied into a weighing bowl that indicates the quantity, which is recorded by the operator. The empty cans are further conveyed to can washer where they are automatically washed with detergent and hot water and steam sterilized to be made ready for next collection trip. The milk is pumped through the flow meter into large vertical storage tank (25,000- 1,50,000 litre capacity) called ‘Silo’.

Adulteration: Adulterants in milk

Definition: Addition or removal of legally prohibited substances from the milk with the view to increase quantity and reduce the quality to make extra profit.

Common adulterants:

1. Addition of water
2. Removal of fat.
3. Addition of starch
4. Addition of milk powder
5. Addition of carbonate and bicarbonate.

The practice of adulteration of milk is a reality. Some of the known adulterants are water, salt, sugar, wheat, starch, washing soda, formalin, urea, hydrogen peroxide etc. Some are used for increasing volume and SNF content of milk while others as preservatives to extend shelf life.

Preservatives:

(i) Mercuric chloride or corrosive sublimate. This is very poisonous. It may be added in the form of tablets, which are coloured (usually bright red) to prevent the milk being mistaken for food. (ii) Formalin. This is a 40 per cent solution of formaldehyde. Being in liquid form, it is very convenient to handle. However, it interferes with the fat test. (iii) Potassium dichromate. This is not as effective as the above two, but it is easy to handle in dairy plants because it is available in tablet form.

Thermal processing:

The main purpose of heat treatment of milk is to render it safe for human consumption and to enhance its shelf life. The common pathogenic organisms likely to occur in milk are killed by relatively mild heat treatment. Various categories of heat are,

Pasteurization:

Pasteurization is the processing of exposing the milk to a controlled temperature for a specific time with the object of destroying all the pathogenic bacteria and cooling the milk immediately to a temperature low enough to retard the growth of the surviving bacteria

161°F for 15 seconds and rapidly cooling to 50°F sufficient to kill the most common disease producing bacteria.

Advantages:

1. Pasteurization renders milk safe for consumption
2. It destroys all the common disease producing organism eg. TB, Typhoid, Diptheria, etc. which may be present in milk.
3. Pasteurization destroys approximately 99% of all bacteria and most of the yeast and moulds.
4. Keeping quality is improved facilitating easy transport of milk over long distances.
5. Pasteurized milk or cream – desired type of ripening can be obtained more effectively.
6. Pasteurization eliminates undesirable taints from milk.
7. Products prepared from Pasteurization milk are of more uniform quality.
8. Natural flavour of milk is not affected by Pasteurization.
9. Pasteurization destroys lipase enzyme / which is responsible for rancidity of milk.

Objections to Pasteurization

- a) Organisms developing in Pasteurized milk form harmful products. Streptococcus thermophilus, S.liquifaciens and M.candidus varian and M.Luteus.
- b) Infants do not develop so well on Pasteurized as on raw milk
- c) Products of bacterial growth are not destroyed.
- d) Pasteurization may be used to mask dirty milk.
- e) Pasteurization bring about chemical changes in milk. Cause ppt. of ca and Phosphorus
- f) Pasteurization - partial destruction of vit.E and K, vit.C is destroyed varying degree dependent on the system.

Pasteurization can be achieved either by holding method (batch process) or continuous process. Under batch process the milk is heated to 63°C for 30 minutes in a double-jacketed vat. Heating and cooling is done by spraying or circulating hot water /steam of chilled water between the inner and outer jacket of the vessel. The milk is kept gently agitated mechanically to ensure uniform heating/cooling. The process is called **low temperature long time (LTLT) method**. This method is suitable for small quantities ranging from 200-1000 litre requiring low initial cost of equipment.

High temperature short time (HTST) treatment for pasteurization of milk refers to heating every particle of milk in a continuous flow to a minimum of 72°C for at least 15 seconds followed by immediate cooling to 4°C. The entire process is automated and is ideal for large scale handling of 5,000 lph or higher.

Thermization

This process consists of heating milk below pasteurization temperature to temporarily inhibit bacterial growth. The milk is heated to 63-65°C for 15 seconds and rapidly chilled to 4°C or below to prevent aerobic spore forming bacteria from multiplying after thermization.

Ultra –pasteurization.

Its objective is to enhance or extend the shelf life of the product (milk) by 15 – 30 days. The fundamental principle is to reduce main causes of reinfection of the product during processing and packaging. This is achieved by heating milk to 115-130°C for 2-4 seconds and cooling it to below 4°C.

Ultra-high temperature treatment (UHT)

It is a technique for preserving liquid food products by exposing them for brief intense heating. In short the process is termed as UHT treatment. The heating temperature normally ranges from 135-150°C for 1-6 seconds.

Sterilization:

In this process milk or condensed milk packed in clean containers is usually subjected to high temperature (115-120°C) for 20-30 minutes. The containers may be tin cans (200-400 g capacity) for evaporated/sweetened condensed milk or glass bottle for milk.

Microwave heating:

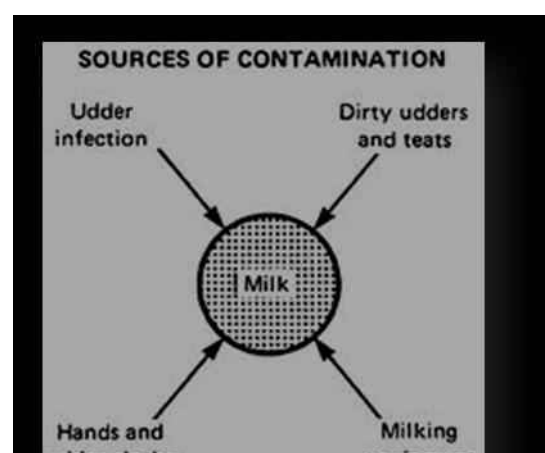
It is a novel method of heating, which greatly reduces the effect of heat penetration lag associate with traditional process of convection or conduction. Microwaves form part of the electromagnetic spectrum (frequency range 915 and 2450 MHz). The heating effect is achieved by transfer of energy to a dipole (in water) within the product.

Different categories of heat treatment.

Treatments	Process	Temperature (° C)	Time (seconds)
Pasteurization	LTLT	63	1800
-	HTST(milk)	72	15-20
-	HTST(cream)	>80	15
Thermization	-	57-68	15
Sterilization	-	115-121	180-780
Ultra-pasteurization	-	115-130	2-4
UHT	-	135-150	1-6

Quality milk production should be the aim for better and sustainable market

- i) Milking schedule should be fixed and definite and there should not be any frequent change in the process. If at all any change it should be gradual.



- ii) Milking parlour should be cleaned at least half an hour before milking. Milking vessels or cans should be thoroughly clean and dry. Ensure that such vessels are round in shape and without any joints. Heating of milking vessel before milking (dry heat application) will improve the keeping quality of milk.
- iii) Udder should be washed with lukewarm water, in which few crystals of potassium permanganate can be added. (Amount sufficient to change the colour)
- iv) Massage (light) the udder while washing.
- v) Ensure that person entrusted with the job of milking is aware of importance of personal hygiene: clean hand and trimmed nails are must.
- vi) Dry the udder with clean cloth from the tip of the teat upward.
- vii) Always ensure that milking is performed with full hand method (Not with folded thumb), quickly, completely and comfortably (Animal should feel comfortable). Milking should be by gentle squeezing of teats NOT by dragging or giving jerks to teats.
- viii) Complete the process from washing to milking within 8 minutes.
- ix) After milking dip the teats in disinfectant solutions e.g. weak iodine solution.
- x) Always milk animals in calm and quite atmosphere preferably play light music (if possible)
- xi) Animals that have recently calved and animals giving more milk should be milked first then the animals giving less milk or in the last phase of lactation. Animals with diseases or with mastitis should be milked separately at the end.
- xii) Animals giving up to 10 liters of milk should be milked twice a day. The Frequency needs to be increased to thrice in case of production ranging from 12-15 liters. Employ two persons for simultaneous milking in case the production is more than 16 liter per day.
- xiii) Milk from animals with medication in teat (s) should not be mixed with other milk.
- xiv) Milk should be strained through muslin cloth (Four layers) or through thin nylon mesh. Cool the milk (by putting ice around the milk can) and transport it to the retail center or collection center of the cooperative society as early as possible.
- xv) Ensure that, water used in cleaning of utensils is clean and free from contamination. Milk is a good media for growth of microorganism. A single contamination may render it unfit for human consumption. Low quality affects shelf life of the milk besides there maybe change in flavor, taste, or nutrient composition.

Sources of contamination and their control

Sources of contamination	Control measures
Interior of the udder	Check for mastitis Discard foremilk
Exterior of the cow, particularly udder and flanks	Wash and wipe the udder Dry milking Use small top milk pail
Barn air and dust	Keep the milk covered
Flies and other vermin	Eliminate the breeding places Fly control with fly sprays, fly traps
The milker	Clean habits Dry milking
Utensils	Clean, sanitize and dry before milking

FACTORS AFFECTING QUALITY OF MILK PRODUCTION

The milk from an individual cow is subject to variation in chemical composition and amount produced is not recognized. Some of the variations from day to day are called “Normal variations” and no causative factor has been associated to them.

Other variations in the quantity have been associated to them due to a large number of factors

Normal Variations

It is not uncommon for fat % to vary as much as % and for milk production to vary as much as %. Usually a low fat % accompanied a low milk production is accounted for failure of “Complete letting down” of milk, the later part of milk is rich in fat. The other constituents of milk also vary from day to day although the magnitude of variation is not so great as that of the milk fat and the amount of milk. In the order of the degree of variation, fat, protein are first, lactose next and salts are the least. The degree of variation for the various constituents is as follows.

S.No	Constituents	Minimum	Maximum
1	Fat	2.60	8.37
2	Protein	2.44	6.48
3	Lactose	2.41	6.11
4	Ash	0.56	0.93
5	Total Solids	10.56	17.9
6	Solid not Fat	7.20	11.90

Factors affecting milk composition

I. Genetic factors

a. Breed and individual cow

Milk composition varies considerably among breeds of dairy cattle: Jersey and Guernsey breeds give milk of higher fat and protein content than Shorthorns and Friesians. Zebu cows can give milk containing up to 7% fat.

b. Variability among cows within a breed

The potential fat content of milk from an individual cow is determined genetically, as are protein and lactose levels. Thus, selective breeding can be used to upgrade milk quality. Heredity also determines the potential milk production of the animal. However, environment and various physiological factors greatly influence the amount and composition of milk that is actually produced. Herd recording of total milk yields and fat and SNF percentages will indicate the most productive cows, and replacement stock should be bred from these.

II. Environmental factors

a. Interval between milking

The fat content of milk varies considerably between the morning and evening milking because there is usually a much shorter interval between the morning and evening milking than between the evening and morning milking. If cows were milked at 12-hour intervals the variation in fat content between milking would be negligible, but this is not practicable on most farms. Normally, SNF content varies little even if the intervals between milking vary.

b. Stage of lactation

The fat, lactose and protein contents of milk vary according to stage of lactation. Solids-not-fat content is usually highest during the first 2 to 3 weeks, after which it decreases slightly. Fat content is high immediately after calving but soon begins to fall, and continues to do so for 10 to 12 weeks, after which it tends to rise again until the end of the lactation.

c. Age

As cows grow older the fat content of their milk decreases by about 0.02 percentage units per lactation. The fall in SNF content is much greater.

d. Feeding regime

Underfeeding reduces both the fat and the SNF content of milk produced, although SNF content is more sensitive to feeding level than fat content. Fat content and fat composition are influenced more by roughage (fibre) intake. The SNF content can fall if the cow is fed a low-energy diet, but is not greatly influenced by protein deficiency, unless the deficiency is acute.

e. Disease

Both fat and SNF contents can be reduced by disease, particularly mastitis.

f. Completeness of milking

The first milk drawn from the udder is low in fat while the last milk (or strippings) is always quite high in fat. Thus it is essential to mix thoroughly all the milk removed, before taking a sample for analysis. The fat left in the udder at the end of a milking is usually picked up during subsequent milking, so there is no net loss of fat.

Pre requisites for good milking

Milking is an art requiring experience and skill. Milking should be done gently, quietly, quickly, cleanly and completely. Cows remaining comfortably yield more milk than a roughly handled and excited cow. Maintenance of clean conditions in the milking barn results in better udder health and producing milk that remains wholesome for longer period. The milking process should be completed within 5 to 7 minutes. Complete milking has to be done. If any residual milk is left it may act as nidus for mastitis causing organism and the overall yield also getting affected.

Preparation for milking

1) Dairy cows and sheds

1. Milking barn should be thoroughly washed and scrubbed after each milking and kept clean and dry before the next milking starts.
2. Dusty feed and silage should be avoided in the milking barn.
3. The hindquarters, thighs and udder should be washed thoroughly before milking.
4. If more hair growth is seen in the udder region, it should be clipped periodically.
5. Buffaloes should be invariably washed before milking.
6. Just before milking udder should be wiped in dry cloth.
7. The teats should be dipped in weak antiseptic solution. In cold weather, warm antiseptic solution can be used.

2) Milkers and pails

1. Milker's hands and milking pails / cans should be thoroughly washed or scrubbed and kept clean.
2. Milkers should wear clean clothes and cover their heads with suitable cap so as to prevent loose hair falling in the milk.
3. The nails should be periodically trimmed and made smooth. Hands should be thoroughly washed and cleaned with antiseptic solutions.

DISEASES OF CATTLE – CLASSIFICATION – SYMPTOMS -

PREVENTING AND CONTROL MEASURES.

Health

The condition in which all the organs and tissues in the system functions normally and harmoniously. Health is fundamental for a sound enterprise.

Disease

Any change from normal state either to single or great extent is called disease stage. Most of the disease can be avoided by proper attention, sanitation, hygiene, nutrition and management practices.

Control of Disease

- ☐ Provide well ventilated and proper housing
- ☐ Provide balanced nutritious diet
- ☐ Strict hygiene and sanitation of animal houses.
- ☐ Adhere regular and routine 'vaccination' schedule
- ☐ Avoid entry of outsiders within the farm-premises.
- ☐ Follow up of latest scientific know how and management practices.
- ☐ Prevention is better than cure and so periodical vaccination
- ☐ Quarantine the newly purchased animal to avoid spread of disease

Classification of disease

Infectious or contagious	Non-infectious or non-contagious
Bacterial a) Anthrax b) Black quarters c) Hemorrhagic septicemia d) T.B e) Brucellosis f) Mastitis	Metabolic Milk fever or Hypocalcaemia Acetonemia or Hypo glycaemia, Ketosis
Viral a) Rinderpest b) Foot and Mouth c) Epimeral fever	Dietary Tympanites or Bloat impaction, Nonspecific enteritis
Parasitic Ecto parasite : Tick, lice and mite Endo parasite : Tape, Round	
Fungal Aflatoxicosis	

BACTERIAL

ANTHRAX

B.anthraxis causes Anthrax in animals. *Bacillus anthracis* spores remain viable for many years in soil, water and animal hides and products. Cattle, sheep and goats are most susceptible to infection.

Symptoms

- ☐ In peracute septicemia death occurs within 2 hours after animal collapsing with convulsions, sudden death in animals that appeared normal is common.
- ☐ In acute septicemia death occurs within 48 to 96 hours clinical signs include fever, anorexia, ruminal stasis, hematuria and blood tinged diarrhea.
- ☐ Pregnant animals may abort and milk production often abruptly decreases.
- ☐ Terminal signs include severe depression, respiratory distress and convulsions.

Diagnosis

- ☐ Examination of blood films from dead animals made by puncturing the superficial vein of the ear or in the region of the foot.

Prevention and Control

- ☐ Prevention of anthrax in animals is aided by active immunization. The organism is susceptible to penicillin-G, tetracyclines, erythromycin and chloramphenicol.

BLACK QUARTER

C. chauvoei causes black quarter or black leg in Cattle. Gram positive, rod shaped with rounded ends. Worldwide distribution in soil and pastures.

Symptoms

The disease usually occurs in young cattle of 6 months to about 2-3 years of age. Crepitating swelling in the hind or fore quarter, lameness, muscles shows trembling with violent twitching. Death usually occurs within 24 hours.

Diagnosis

- ☐ Based on Symptoms

Control and prevention

- ☐ Hyper immune serum (HIS) is used to control explosive outbreaks. Penicillin along with HIS is used to treat the disease.
- ☐ Oxytetracycline & Chlortetracycline can also be employed effectively in early stages.

BOVINE TUBERCULOSIS

- ☐ *Mycobacterium bovis* causes bovine tuberculosis in many animal species and also cause tuberculosis in human

Clinical signs

- ☐ Affected animals become docile, Progressive emaciation, capricious appetite, fluctuating body temperature and rough / sleek hair coat, animal does not put up weight. All these general signs are pronounced following calving.
- ☐ Silent or paroxysmal cough especially during early morning and chilled weather. Chronic cough with dyspnoea, squeaking crackles, enlargement of retropharyngeal lymphnode causes dysphagia and noisy breathing due to pharyngeal obstruction.

Diagnosis

- ☐ Based on history, signs and post mortem lesions

Control and prevention

- ☐ Treatment and vaccination are inappropriate in control programmes for cattle. In many countries, tuberculin testing followed by isolation and slaughter of reactors has been implemented as the basis of national eradication schemes.

JOHNE'S DISEASE

Johne's disease or paratuberculosis is caused by *Mycobacterium avium* subsp. paratuberculosis (also referred as *Mycobacterium johnei*). The disease causes chronic, contagious fatal enteritis, which can affect cattle, sheep, goats, camels and wild ruminants.

Symptoms

In cattle, the disease is characterized by diarrhoea, initially intermittent, dark and semisolid, but becoming persistent and profuse. Progressive weight loss results without loss of appetite, leading to emaciation and eventually death. The mortality rate may approach 100%. Asymptomatic carrier cattle have an increased incidence of mastitis and infertility.

Diagnosis:

- ☐ Microscopical examination by staining the faecal smears with acid-fast stain.
- ☐ Johnin test

Prevention and control

Animals with clinical signs suggestive of paratuberculosis should be isolated. A live vaccine and inactivated adjuvanted vaccines are available.

HAEMORRHAGIC SEPTICEMIA

Pasteurella multocida is small Gram –ve rods or coccobacilli that show *bipolar staining*

Symptoms

- ☐ Fever, a sudden drop in milk yield, signs of abdominal pain, severe diarrhoea and dysentery, respiration becomes rapid and shortly before death the mucous membranes appear cyanotic.

- ☐ In less acute cases there will be odema development in the region of the head, neck and brisket. The nasal discharge may be blood stained or purulent. Death occurs within 2-4 days.

Diagnosis

- ☐ By symptoms and lesions

Control and prevention

- ☐ *Pasteurella* is amenable to Penicillin-G, streptomycin, chloramphenicol, chlortetracycline, sulpha and trimethoprim, enrofloxacin and oxytetracycline.
- ☐ Vaccination

BRUCELLOSIS

Brucella abortus species are obligate intra cellular parasites and cause abortion in last trimester of pregnancy

Symptoms

- ☐ The disease in cattle is almost always caused by *B.abortus*.
- ☐ The incubation period is usually from 30 to 60 days.
- ☐ After bacteraemia the infection localizes in the placenta, if the animal is not pregnant, the infection localizes in udder (interstitial mastitis).
- ☐ In the bull, orchitis and epididymitis.
- ☐ Abortion at 6 months and retained placenta are the cardinal signs.

Diagnosis

Based on symptoms and lesions

Prevention and control

- ☐ The attenuated live vaccine (Strain 19 *B.abortus* biotype 1) is used in female calves 4 to 12 months of age.
- ☐ The adjuvant bacterins is used as booster vaccine.

VIRAL DISEASES

FOOT AND MOUTH DISEASE

Causative agent - Members of the family Picornaviridae, genus Aphthovirus and species Foot and mouth disease virus. Foot and mouth disease (FMD) is the most contagious disease of mammals and cause severe economic loss in susceptible cloven-hoofed animals (cattle, pigs, sheep, goats, and water buffalo).

- ☐ Smallest of the Animal virus : 7 types virus : O,A,C Asia I, SAT 1,2,3
- ☐ Transmission : Direct contact : Thro water : manure : Pasture and cattle attendant

Symptoms

The disease is characterised by the formation of vesicles (fluid-filled blisters) and erosions in the mouth, nose, teats and feet. Initial signs are pyrexia (39.4-

40.6°C), dullness, anorexia, and fall in milk production. These signs are followed by excessive salivation; drooling, serous nasal discharge; shaking, kicking of the feet or lameness; and vesicle (blister) formation in the tongue, dental pad, gums, soft palate, nostrils, muzzle, interdigital space, coronary band, and teats. Pregnant cows may abort, and young calves may die without developing any vesicle. The course of an FMD infection is 2 to 3 weeks. Secondary infection may delay recovery.

Diagnosis

- ☐ Symptom and lesion

Treatment:

- ☐ Antibiotics may be administered to counter bacterial infections.

Prevention & Control:

- ☐ Thorough disinfection of shed, utensils, clothes of attendants.
- ☐ Vaccination – polyvalent – once – 4months or varies with type of vaccine

BOVINE EPIMERAL FEVER

It is caused by the virus bovine epimeral fever, genus epimero virus and family Rhabdoviridae.

- ☐ Biphasic, sometimes triphasic, with peaks of 40-41.5°C (104-107°F) spaced 12-18 hours apart, accelerated heart and respiratory rates, anorexia, ruminal atony, depression, serous or mucoid nasal and ocular discharges, salivation, muscle twitching or waves of shivering, a generalized stiffness or a shifting lameness.
- ☐ There may be submandibular edema or patchy edema on the head. Nasal and ocular discharges, drooling of saliva and periorbital swelling and increased excitability and agitation are other signs.
- ☐ Many animals become recumbent for 12-24 hours. By day three the affected animal is usually standing again and will begin to eat.

METBOLIC DISEASES

MILK FEVER: (Parturient paresis)

Milk fever is a metabolic disease in cows soon after calving. Due to fall in serum calcium level in cows after calving as a result of failure to mobilize calcium reserves and of the development of negative calcium balance in late pregnancy.

Symptoms:

Disease flares up within 72 hours of calving initially the cows show excitement, incoordination of movement muscular tremors in limbs and head, lying in recumbent position with her head directed towards flank. In final stages subnormal temperature, dilatation of the pupil, impalpable pulse, coma and death.

Diagnosis of the disease is based on the occurrence of milk fever in recently calved animals.

Treatment & Control:

Dramatic recovery by intravenous administration of 300-400 ml calcium borogluconate with Vitamin D3 injected intramuscularly. Continued mixing of ½ liter of supernatant lime water for cow may reduce the incidence.

KETOSIS: Acetonaemia

Disturbance of carbohydrate metabolism in high producing dairy cattle that leads to hypoglycaemia and appearance of ketone bodies.

Causes:

Disease is caused by deranged metabolism of carbohydrate and volatile fatty acids resulting in reduced level of sugar in blood (hypoglycemia), increased level of ketones in blood (ketonemia) and in urine (ketonuria)

Symptoms:

Cardinal signs in digestive and nervous type of ketosis usually appear in goodmilkers from 7 days to six weeks after calving. Loss of appetite, rapid loss of weight and marked reduction in the milk yield observed in digestive type of ketosis. In nervous type symptoms include depression, a staring expression, walking in circles, treading with the feet, incoordination of movements, convulsions, sudden falling on the ground with wide expression of bulged eyes. Respiration becomes shallow emanating fruity odour on the breath.

Diagnosis:

Examination of ketone bodies in the urine helps in diagnosing the disease besides the symptoms noticed.

Control and Treatment:

Intravenous administration of 500-1000 ml of 40 per cent glucose, Repeat for 5 days. Cases not responding to glucose therapy, intramuscular injection of 100-200 mg of hydrocortisone or 50 to 200 mg of prednisolone acetate. Concentrate feeding with good fodder during dry period in high yielding cows, ½ to one kg maize or cholam made as gruel mixed with ¼ kg of jaggery or molasses daily to be given to cows nearing parturition.

BLOAT: (TYMPANY)

Bloat is a disease of ruminants in which rumen and reticulum is over distended with the gases of fermentation.

Cause:

Excess intake of fresh legumes and feeding of high grain ration lead to frothy bloat. Obstruction to normal expulsion of gases from rumen by choking the oesophageal passage by corncob, turnip and sugar beet cause free gas bloat.

Symptoms:

Acute form of tympany results in sudden death before rendering any aid to the affected animal. In acute cases, the distension of the rumen occurs quickly, the flank and the whole abdomen is enlarged. On percussion the left flank produces a drum like sound, initially the animal frequently gets up and lies down, kicks at belly and even rolls. Breath becomes difficult and is evidenced by oral breathing, protrusion of tongue and salivation. When the distension of abdomen becomes extreme, the animal exhibits uncoordinated movement, inability to stand, falls all of a sudden. Collapse and death occur quickly. In chronic tympany, the distension of abdomen and intra-abdominal pressure are not serious. The gas is 'free' but retained because of obstruction of the passage thereby preventing normal eructation of gases.

Diagnosis:

Based on characteristic symptoms of distension of abdomen and distress by the affected animal.

Control and Treatment:

In per acute cases puncture the rumen with a sharp knife or with a trocar and canula to expel the gases. Administer orally oil of turpentine 60 ml well mixed with one litre of groundnut oil or gingelly oil or cocounut oil. After six to eight hours administer powdered ginger 30 grams, Asafoetida 30 gram, well mixed to jaggery. Fresh legumes should be wilted and then fed to stall-fed animals. Feed dry roughages before turning the cattle to luxuriant pasture to avoid bloating.

MASTITIS**Introduction**

- ☐ Mastitis is an inflammation of the mammary gland. In which the milk undergo physical, chemical and microbiological changes whereas mammary glandular tissue undergo physical and pathological changes. In which infected milk colour, consistency change and contains more amount of leucocytes.

Etiology

- ☐ Mastitis is caused majorly by Staphylococcus, Streptococcus and coliform bacteria and less importantly by other organism such as other bacteria, viruses, and fungus.

Source of infection

- ☐ S.agalactiae and S.aureus resides primarily in the udder of infected cows.
- ☐ Exposure to uninfected quarters is limited to milking process.
- ☐ Streptococcus uberis, S.dysgalactiae and coliforms are normal inhabitants of the cows environment.

Transmission

Infection occurs via the teat canal – contaminated environment – skin of udder, milking equipment, milker etc.

Clinical signs

- ☐ Per acute form: Pyrexia, anorexia, respiratory distress, swollen, hot and painful udder. Cessation of milk production. Exudate are often blood stained. Mostly, Staph.aureus, Str. Dysgalactiae, Cory.pyogenes, E.coli, and P. aeruginosa.
- ☐ Acute form: Swollen udder, changes in quality of milk. Milk become curd like, yellow,brown fluid with flakes and clots.
- ☐ Subacute form: No changes in the udder tissue.
- ☐ Chronic form: Udder is haemorrhagic, and fibrotic. Swollen and palpable supra mammary lymphnode,. Udder is thick, firm, nodular and atrophic, yellowish or whitefluid with clots and flakes.

Diagnosis

- ☐ Physical examination of the udder.

Treatment

- ☐ Stripping out the milk from the infected quarters. Cleaning of infected quarters with normal saline and distilled water. Infusion of antibiotic therapies immediately after theinfection. Continue use antibiotics as per the antibiogram.
- ☐ Use of inactivated vaccine containing Str.agalactiae, Str.uberis, Staph.pyogenes, Staph.aureus and E.coli in some countries.

Control:

- ☐ Hygienic measures are important.
- ☐ Animals diagnosed positive should be milked at last.
- ☐ Milkers should wash their hands before milking and should use well washed white overalls.
- ☐ A separate clean cloth for each cow is used for washing the udder with a disinfectant.
- ☐ The first stream of milk from each quarter should not be allowed to drop on floor but collected in a separate container. Milkers should not wet their hands with first stream of milk.

Unit III: Sheep and Goat Management

INTRODUCTION TO SHEEP AND GOAT PRODUCTION AND MANAGEMENT

Sheep in India are mostly owned by nomads or semi-nomadic people who constantly move their flocks in search of good pastures and water. Sheep farming is a hereditary occupation with these people. The size of their flock varies from a few to hundreds. The husbandry practices are most primitive. There are certain advantages connected with sheep farming and these are:

1. Sheep requires little concentrate food as compared to cattle. On an average sheep get 80-90 per cent of their protein from forages as against cattle which get about 60- 65 per cent.
2. Sheep are an economical converter of grass into meat and wool.
3. Sheep are the best animal that can be raised on stubbles as well as on grasses and other plants growing between stubbles. In fact there is no substitute for sheep as a class of livestock for utilizing waste land or weed from the field.
4. Unlike goat, sheep do not damage the trees.
5. Sheep require less labour than other kinds of livestock and get along well with low-cost housing and equipment.

Domestication:

Sheep and goats were perhaps the first ruminants to be domesticated around 10,000 B.C.

BREEDS OF SHEEP

Classification of Indigenous breeds based on utility

Apparel wool breeds	Superior carpet Wool breeds	Coarse carpet Wool breeds	Hairy Meat breeds
Hissardale	Chokla	Malpura	Nellore
Niligiri	Nali	Sonadi	Hassan
Kashmir merino	Marwari	Muzzafarnagri	Mecheri
Avivastra	Magra	Jalauni	Kilakarsal
Bharat Merino	Jaisalmeri	Deccani	Madras Red
	Pugal	Bellary	Trichy Black
	Pattanwadi	Coimbatore	Kenguri
	Tibetan	Chottanagpuri	Mandya
	Bonpala	Balangir	Vembur
	Gaddi	Ganjam	
	Rampur Bushari	Bhakarwal	
	Poonchi	Shahabadi	

	Karnah		
	Gurez		
	Changthangi		

Classification of sheep breeds based on agro-ecological regions in India

Northern temperate region	North-western, central arid and semi-arid region	Southern region semi-arid central peninsular area and hot and humid along the coast	Eastern region
Gaddi	Chokla	Deccani (Bellary)	Chottanagpuri
Rampur Bushair	Nali	Nellore	Balangir
Bhakarwal	Marwari	Hassan	Ganjam
Poonchi	Magra	Mandya	Tibetan sheep
Karnah	Jaisalmeri	Mecheri	Bonpala
Gurez	Pugal	Kilakarsal	Garole
Kashmir Merino	Malpura	Vembur	Shahbadi
Changthangi	Sonadi	Coimbatore	
	Pattanwadi	Nilgiri	
	Muzaffarnagri	Ramnad White	
	Jalauni	Madras Red	
	Hissardale	Tiruchi Black	
	Munjal	Kenguri	
	Avivastra		
	Bharat Merino		

Exotic breeds

Fine wool breeds	Mutton breeds	Dual purpose	Pelt breeds
Merino	Suffolk	Corriedale	Karakul
Rambouillet	Southdown		
Polworth	Dorset		

India can be divided on the basis of the agro-ecological conditions and type of sheep into 4 regions viz. I. North-Western, Central arid and semi-arid region II. Southern region, III. Eastern region and IV. Northern temperate region. There are about **44 descript breeds of sheep** available in India.

I. North-western, central arid and semi-arid region

Hissardale:

Hissar and hilly regions of Kulu in Haryana. It was evolved by crossing Bikaneri with Merino rams. Average body weight of rams and ewes are 54 and 34 kg respectively most animals are polled. Colour is predominantly white, although some brown patches may be observed. Wool is of superior quality yield 2-3 kg per annum.

II. Southern region

Important breeds in this region are

Nellore: Nellore, Prakasam and Ongole districts of Andhra Pradesh tall animals with little hair except at brisket, withers and breech. Rams are horned ewes are polled. Long and drooping ears; 86% of animals have wattles male: 36 kg female: 28 kg

Mandya: Mandya district of Karnataka. Relatively small animals colour white - sometimes face is light brown, which may extend up to neck. Compact body with typical "U" shaped conformation from the rear. Ears long, leafy and drooping. Both sexes polled. Coat extremely coarse and hairy adult male: 35 kg, female: 23 kg.

Best mutton type in Tamil Nadu.

1. **Mecheri:** Salem, Namakkal and Coimbatore districts of Tamil Nadu. Medium sized light brown in colour. Both sexes are polled. Body covered by very short hairs. Adult male: 35 kg, female: 22 kg
2. **Kilakarsal or Kilakarisal:** Ramnad, Madurai and Tanjore districts of Tamil Nadu Brown/ dark tan in colour with black spots on head belly and legs. Medium sized ears. Males have thick twisted horns. Most animals have wattle.
3. **Vembur:** Tirunelveli district of Tamil Nadu. Tall animals, coat colour is dark tan with black spots on head, belly and legs. Medium sized drooping ears. Males horned. Ewes polled. Body covered with short hairs. Adult male: 34 kg, female: 27 kg.
4. **Coimbatore:** Coimbatore and Salem districts of Tamil Nadu. Medium sized animals white with black or brown spots. 30% of males polled. Fleece white hairy and open. Adult male: 24 kg, female: 20 kg
5. **Ramnad White:** Ramnad and Sivagangai districts of Tamil Nadu. Medium sized predominantly white. Ears medium sized and directed outward and downward. Males have twisted horns. Ewes polled short and thin tail. Adult male: 31 kg, female: 22 kg
6. **Madras Red:** Chennai and Kancheepuram districts of Tamil Nadu. Body colour predominantly brown, the intensity varying from light tan to dark brown. Some animals may have white markings on forehead, inside the thigh and lower abdomen. Medium sized drooping ears. Tail short and thin. Rams have strong, corrugated and twisted horns. Ewes polled. Body covered with short hairs. Adult male: 35 kg, female: 23 kg

7. **Trichy black:** Trichy, Thiruvannamalai, Dharmapuri districts of Tamil Nadu. Small animals. Body is completely black. Males horned, ewes polled fleece extremely coarse, hairy and open. Ears and tail small. Adult male: 25 kg, female: 18 kg



Kilakarisal



Madras red



Ramnad white



Vembur





Trichy Black



Vembur



Coimbatore



Nellore



Mandya

III. Eastern region

This region consist Bihar, West Bengal, Orissa, Assam, Meghalaya, Arunachal Pradesh, Mizoram, Manipur, Tripura, Nagaland and Sikkim. Most of the wool produced inthis region is suitable only for coarse carpets, blankets and kumblies. The important breeds are Shahabadi, Chhottanagpuri, Ganjam, Balangir, Bonpala and Tibetan.

IV. Northern temperate region

It comprises of Jammu and Kashmir, Himachal Pradesh and hilly regions of Uttar Pradesh. This region has 8% of the total sheep population. The largest population ofcrossbred sheep primarily developed for apparel wool is in this region. The important breeds are Rampur Bushair, Gaddi, Gurez, Karnah, Poonchi and Changthangi.

Exotic breeds of sheep

A. Fine wool breeds:

1. Merino: Native of Spain - origin for most of the wool breeds in the world. Colour-white. Fleece yield male: 4-5 kg and ewes 3-4 kg/annum. Merinos have large number of skin folds.



2. Rambouillet: Developed from Merinos in France. They are large, rugged, fast growing sheep and are good wool producers. Skin is pink. Ewes are good mothers, and prolific. Average wool yield is 4.5 to 5.5kg.



B. Mutton breeds:

1. Native of U.K. large animals with black face, ears and legs. Head and ears are entirely free from wool. Average wool yield 2-3 kg. Mature rams weigh 100-135 kg and ewes from 70-100 kg.



2. Dorset: Native of U.K two types polled and horned Dorsets. Face, ears and legs white in colour and free from wool. Wool yield is 2.75 to 3.25



C. Dual purpose breeds

Corriedale: Native of New Zealand. The parent breeds involved in developing Corriedale are Lincoln, Leicester and Merino. Adult rams: 80 to 100 kg Ewes: 55 to 85 kg. Annual wool production: 4.5 to 5.5 kg. Both sexes are polled. Colour: White may have black spots.



BREEDS OF GOATS

There are as many as 23 well defined breeds of goats in India, though a few other are being identified in different agro-climatic regions. These breeds are adapted to harsh climate, long migration, tropical diseases, poor nutrition and shortage of water. Majority of the goats are mixed and non-descript in different states.

Indigenous Goat Breeds:

1. Classification Indian goat breeds based on utility:

Meat & Skin	Meat, Milk & Skin	Meat, Hair & Skin	Meat, Pashmina & Skin	Milk, Meat & Skin
Black Bengal	Barbari	Bakharwal	Chegu	Beetal
Ganjam	Tellicherry (or) Malabari	Gaddi (Chamba)	Changthangi	Jamnapari (Etawah)
Kannai Adu	Osmanabadi	Gohilwadi		
	Sirohi	Kangan		
	Surti	Kutchi		
	Mehsana	Marwari		
	Zalawadi	Sangamneri		
	Jhakhrana	Khasi (Assam hill goat)		

2. Classification of goat breeds based on agro-ecological regions in India

Northern temperate region	North-western, central arid and semi-arid region	Southern region semi-arid central peninsular area and hot and humid along the coast	Eastern region
Gaddi	Sirohi	Sangamneri	Ganjam
Changthangi	Marwari	Malabari	Bengal
Chegu	Beetal	Osmanabadi	
	Jhakrana	Kannaiadu	
	Barbari		
	Jamnapari		

	Mehsana		
	Gohilwadi		
	Zalawadi		
	Kutchi		
	Surti		

1. Jamnapari: Native of Uttra Pradesh



Coat Colour: White with tan or black markings at neck and ears; beard in both sexes; tuft of long hairs in the buttocks. **Body and legs:** Largest and most elegant of the long-legged goats of India. Pronounced Roman nose having a tuft of hair which results in parrot mouth appearance. **Ears:** Long and pendulous. Live weight: Average, buck 75 and doe 55 kg. Average birth weight is upto 4 kg **Milk:** Large udder and big teats: average yield is 280 kg / 274 days.

2. Beetal: Native of Rajasthan



Coat colour: Predominantly black; red, tan or black, heavily spotted on white also occur; males usually possess beard. **Ears:** Long pendulous drooping, beetal leaf shaped pinna. **Live weight:** Mature buck 70 and doe 46 kg; birth weight 3 kg. **Meat:** Good quality with desirable flavour. **Milk:** Average lactation yield 150 kg .Other qualities: Good quality skin.

3. Sirohi: Native of Rajasthan



Coat colour: Brown, white, and admixture of colours in typical patches; hair coarse and short. **Live weight:** Average, buck 50 and doe 23 kg; birth weight 2.0 kg. **Meat:** Good meat. **Milk:** Average lactation yield 71 kg; lactation length 175 days.

4. Black Bengal: Native of Bengal



Coat colour: Predominantly black, brown/grey and white, soft, glossy and short hairs. **Body and legs:** Dwarf in size, legs short, straight back; beard in both sexes. **Live weight:** Average, buck 15 and doe 12 kg. **Kidding:** Most prolific among Indian breeds. Multiple births is common, two, 3 or 4 kids born at a time. Kidding is twice a year. Average litter size 2.1. **Meat:** Excellent and palatable, quality chevon. **Other qualities:** Fine skin. There is great demand for high-class shoe making.

5. Malabari or Tellicherry: Native of Kerala

Coat colour: Varies from white to complete black; black and white, black and brown, and white with coloured patches occur; both long and shorthaired varieties available. **Body and legs:** Head medium sized, face flat and sometimes with Roman nose; bucks bearded; wattles present.



Ears: Medium in size, directed outward and downward. **Horns;** Mostly horned; small, slightly twisted and pointed. **Liveweight:** Average, buck 39 and doe 31 kg; birth weight 1.6 kg. **Kidding:** Once a year, average age at first kidding is 19-21 months. **Meat:** Average. **Milk:** Average milk yield 65 kg/lactation: average lactation length 172 days. **Other qualities:** Goodskin.

8. Kanni Adu: Native of Thirunelveli area, Tamil Nadu



Coat colour: Predominantly black or black with white or brown spots and hence called as palkanni and chenkanni respectively. **Body and legs:** Tall and stout. **Ears:** Medium. **Horns:** Bucks horned and does polled. **Live weight:** Average, buck 36 and doe 29 kg. **Kidding:** Twice in 18 months, single or twins; age at first kidding 15 months. **Meat:** Good. **Milk:** Enough for kids.

Kodi Aadu:



Distribution: Part of Tuticorin and adjoining areas of Ramanathapuram, Tanjore and Pudukottai districts. They are tall, long animals with slender body. White in colour with splashes of black or red colour. Based on the colour they are classified as Karum Porai (Blakish) and Chem Porai (Reddish brown). Both sexes are horned. Suitable for coastal areas. Used as leader in sheep flock while grazing. Adult body weight: Buck: 39.5kg and Doe 32.2 kg.

Salem Black: Native of Salem area, Tamil Nadu

Home tract: Salem, Dharmapuri and Vellore district of Tamil Nadu. Salem Black goats are tall animal with a lean body and a completely black coat. The head is medium in length with a medium to broad forehead. The ears are leaf like and pendulous. Both male and female are horned. The tail is thin, medium in length and curled upwards. The adult body weight is 38.5 (male) and 29.5(female) kg.



Pashmina goats:

1. Chegu:

Predominantly white but greyish red and mixed colours seen. Long hair below which under coat of delicate fibre (cashmere or pashm). Average, buck 39 and doe 26 kg. Birth weight 2.0 kg.



2. Changthangi

Predominantly white (50) and rest brown, grey and black. Undercoat white/grey; yields warm delicate fibre, pashmina (cashmere, pashm). Average, buck 30 and doe 20 kg; average birth weight 2.1 kg.



Exotic goat breeds

1. Toggenberg: It originated in the Toggenberg Valley in north-eastern Switzerland. The breed is usually, but not always, hornless. The doe weigh more than 80 kg. The average milk production is 5.5 kg per day; the butter-fat content of milk is 3.4 per cent.



2. Saanen:

This breed originated in the Saanen Valley of Switzerland. The goats of this breed are white or light cream. Doe Weigh 65 kg. Bucks weigh 95 kg or more. The average milk yield ranges from 2 to 5 kg per day during a lactation period of 8 to 10 months. The average butter-fat content of milk is 3.5 per cent.



3. **Alpine:** This breed originated in the Alps Normally does not weigh less than 85 kg. French Apline females are excellent milkers and have horns. The butter- fat content of milk is 3 to 4 per cent.



4. **Nubian:** This breed originated in Nubia (north-eastern Africa), but is also found in Egypt and Ethiopia. Anglo-Nubian is a big animal with fine skin and glossy coat, pendulous ears



and Roman nose. Butter-fat percentage is decidedly higher. Anglo-Nubian is known as the **Jersey cow** of the goat world. Bucks weigh 65 kg to 80 kg and does 50 kg to 60 kg. The peak milk production in a day is more than 6.5 kg. The breed average for butter-fat content of milk is 4.5 per cent.

5. **Angora:** This breed originated in Turkey or Asia Minor. This produces valuable textile fibre commercially known as mohair. The average weight of fleece is about 1.2 kg. A good specimen yields upto 6 kg.



6. Boer



Native to South Africa. Boer goat is large, long-legged goat. White with a reddish-brown head and neck. White blaze. Body sometimes has brown patches. Powerful head with compressed nose. Well-developed broad brisket and well sprung ribs. The average weight of male ranged between 80-90 kg and female 50-70 kg.

ECONOMIC TRAITS OF SHEEP

1. **Tupping percentage:** Number of ewes covered to the number of ewes put to a ram
2. **Lambing percentage:** No of ewes lambd to the number of ewes put to a ram (or) Number of ewes lambd to the number of ewes covered.
3. **Weaning percentage:** Number of lambs weaned to no of lambs born alive (or) Number of lambs weaned to the number of ewes put to a ram
4. **Twinning percentage:** Number of twin births to the total number of births
5. **Lambing rate:** Number of lambs born to number of ewes lambing
6. **Lambing interval:** Interval in days between two successive lambing.
7. **Birth weight:** Weight of a lamb in kg measured within 24 hours after birth.
8. **Weaning weight:** Weight of a lamb in kg at the time of weaning (90 days).
9. **Weight at market age:** Weight of the lambs in kg at the age (6, 9 and 12 months).

10. Pre weaning weight gain: Rate of daily gain in weight (gm/day) from the date of birth to weaning period

11. Post weaning weight gain: Rate of daily gain in weight (gm/day) from the date of weaning to one year

12. Pre weaning mortality: Number of lambs died during the period 0-90 days to the Number of lambs born alive.

13. Post weaning mortality: Number of lambs died during the period 90-365 days to the Number of lambs at 90 days

14. Adult: Number of adult sheep died to the number started at one year

ECONOMIC TRAITS OF GOAT

1. Topping percentage: Number of does covered to the number of does put to a buck

2. Kidding percentage: Number of does kidded to the number of does put to a buck (or) Number of does kidded to the number of does covered

3. Weaning percentage: Number of kids weaned to the number of does put to a buck (or) Number of kids weaned to no of kids born alive

4. Twinning percentage: Number of twin births to the total number of births

5. Kidding rate: Number of kids born to number of does kidding

6. Kidding interval: Interval in days between two successive kidding

7. Birth weight: Weight of a kid in kg measured within 24 hours after birth

8. Weaning weight: Weight of a kid in kg at the time of weaning (90 days)

9. Weight at market age: Weight of the kid in kg at the age of 6, 9 and 12 months of age

10. Pre weaning weight gain: Rate of daily gain in weight (gm/day) from the date of birth to weaning period

11. Post weaning weight gain: Rate of daily gain in weight (gm/day) from the date of weaning to one year

12. Pre weaning mortality: Number of kids died during the period 0-90 days to the number of kids born alive

13. Post weaning mortality: Number of kids died during the period 90-365 days to the number of kids at 90 days

14. Adult mortality: Number of adult sheep died to the number started at one year

SYSTEM OF PRODUCTION

I. Extensive: Mentioned in the system of production

II. Semi- intensive: Mentioned in the system of production

III .Intensive production systems:

Intensive production systems involve either grazing on crops or cultivated pastures at a very high stocking density or zero-grazing.

1. Concrete / Solid floor

It is a common floor with solid surface made out of different materials such as cement concrete, vitrified paving brick, building brick, stones and gravel. Such solid floors should be laid properly for good drainage. A slope of 1/40 to 1/60 is desirable towards the dung channel. Proper cleaning and disinfection are essential to control diseases.

2. Slatted floor

Floor set with wood and reinforced concrete bars provide a required gap between them and are used in house for intensive animal production. It has the great advantage of controlling disease by breaking the contact between animals and excreta. The excreta, both liquid and solid passes downward from the wooden floor immediately after it is voided through the gap provided. The excreta collected underneath the floor are suitably disposed by mechanical means. The slatted floor is ideal for raising animals in germ free condition. The distance between two planks (slat) is known as slot and it should be 1 to 1.5 cm. The plank width 7.5 to 10cm and thickness 2.5 to 4cm. Slatted floor should be raised 1 meter above the floor level.

HOUSING

Space allowance

Adult goat	: 1.2.5 – 1.5 m ²
Pregnant doe and bucks	: 2 m ² .
Kids – 3 – 6 months	: 0.7 to 0.9 m ²
6month -1 year	: 1m ²

HOUSING OF SHEEP AND GOATS

- Not expensive
- Adequate space
- Proper ventilation
- Good drainage
- Plenty of light
- Protection from predators and adverse climate
- Dry floor.
- Shed should be in east west direction

SHEEP

Normally sheep do not require elaborate housing facility, but minimum provisions require to increase productivity especially protection against inclement weather conditions

(sun, rain, winds). Shed could be built along the wall of the house. Further protection could be provided with thatched material and bamboos.

Minimum floor space requirement per animal (BIS standard)

S. No	Type of animals	Minimum floor space per animal (m ²)
1	Ram or buck in groups	1.8
2	Ram or buck individual	3.2
3	Lamb or kids in group	0.4
4	Weaner in groups	0.8
5	Yearling or goatlings	0.9
6	Ewe or doe in groups	1.0
7	Ewe with lamb	1.5

Essential appliances required:

- Feeders .
- Water troughs

Type of animal	Space per animal (cm)	Width of manger/water trough(cm)	Depth of manger/water trough (cm)	Height of inner wall of manger/water trough (cm)
Sheep and goat	40-50	50	30	35
Kid/lamb	30-35	50	20	25

CARE AND MANAGEMENT OF YOUNG AND ADULT STOCK:

Pregnant, parturient and lactating ewes/does

1. The ewes/does in advanced stage of pregnancy should be separated from the flock and effective care should be taken in their feeding.
2. Extra feed during the later part of pregnancy (3-4 weeks before parturition) will be beneficial for the condition of the pre-parturient ewes which will help in improving milkproduction of ewes and birth weight and growth of lambs.
3. Ewe/doe in advanced stage of pregnancy should be kept in a separate lambing corals or shed 4-6 days before partition and maximum comfort like soft clean bedding and individual lambing pen should be provided.
4. The ewe/doe should be protected from chilly weather condition.

Care at lambing/kidding

A ewe/doe about to lamb prefers to leave the flock if she is restless. The udder is often distended and external genitals are in flushed and flaccid condition. Generally in a healthy ewe partition is normal.

1. New born lambs after being licked by the mother generally stand on their leg and start seeking for teats and suckle milk. If they are not able to do so after sufficient time, provide help to them in suckling colostrum, the first milk, which is very essential for health and survivability of lambs.
2. The new born should be protected from cold, wind and rain.
3. Artificial milk feeding or arrangement of foster mother should be done for disowned or orphan lambs. Goats can serve as excellent foster mother, but ewes which have lost their lambs early after birth may also be utilized.
4. Ligate, cut and antiseptically dress the naval cord of the lamb.
5. The lambs/kids should not be handled too frequently immediately after birth and let the dams lick and recognize them properly. The first 1 to 2 hours after birth is the vital period for establishment of bond between the newborn and the mother. Hence the dam should be kept in a calm place without disturbance from stray dogs and other animals.
6. Allow newborn lambs to be with their mother all the 24 hours for first week or so.
7. Feed sufficient quantity of good quality hay and concentrate to the lactating ewes for meeting nutritional requirements for early lactation.

Care of lambs/kids

The lambs should be taken care to the maximum extent during early period of life.

1. Kid/lambs should be ensured of taking sufficient colostrum within 1 hour after birth to more resistance against diseases.
2. Take care of indifferent mothers and arrange suckling of lambs/kids by restraining such type of ewes.
3. If possible, make available green leguminous fodders or fresh tree leaves to lambs to nibble during sucking period.
4. Lambs/kids may be ear-tagged or tattooed on the ear for identification.
5. During castration keep the lambs on perfectly dry, clean and hygienic site so as to minimize the risk of losses from tetanus and other infection.
6. Salt lick or mineral block should be kept in the lamb pen to avoid licking of soil/floor.

Weaning and care of weaners:

1. Weaning should preferably be done at 90 days, although in breeds with low milk production or where re-breeding is desired it can be done around 60 days.
2. Supplementary feeding and good clean pasture for growing weaners should be provided.
3. Weaned lambs should be drenched against gastro-intestinal parasites by first month and vaccinated against enterotoxaemia and sheep pox.
4. They should be protected against vagaries of climate and predation.

BREEDING MANAGEMENT OF SHEEP AND GOAT**Selection of ewe and ram for breeding**

Breeding ewe of indigenous breeds should be 18 to 24 months depending upon their body condition. Females should be mated only when they reach 70 per cent of the average adult body weight. The libido of the ram is very important to achieve desirable breeding.

Sex ratio

In India the rams are usually turned out in the flock during mating season at the rate of 2 to 3 per cent of the ewes all through day and night under optimum breeding conditions. The males should be replaced or exchanged once in two years to avoid inbreeding.

Preparation of ewe or ram for breeding**Flushing**

Flushing is feeding of extra concentrate to ewes/goats prior to onset of breeding season, normally 3 or 4 weeks before breeding. This increases the ovulation rate of ewes, so that the number of twins and triplets increase. Flushing can be done by supplementing 250 g of concentrate daily or 500 g of good quality legume hay per head per day. Flushing increases the lambing/kidding rate by 10 to 20 per cent.

Estrous cycle

Ewes and does will exhibit estrus or heat at regular intervals during the breeding season. Estrus is the fertile period and if the female does not conceive, it is repeated every 16-17 days on average in ewes (range 14-19 days) and 19-21 days in does (range 17-24 days). These intervals may be 1-2 days shorter in young animals. The estrus period lasts for about 24-36 hours in ewes and 34-38 hours in does.

Estrus /signs of estrous

The signs of estrous are much more marked in the doe than in the ewe. The outward signs are:

- Tail wagging
- Mounting over other animal
- Seeking male
- Frequent bleating
- Standing for mating
- Generally the young females don't show aggressive estrous behavior The

males are attracted by smell, sight and sound and exhibit following behavior

- Sniffing the vulva
- Extending neck with curling of upper lip
- Biting the side of the ewe/doe and wool pulling
- Paw the ewe – raising and lowering of one front leg in a stiff legged striking motion
- Mount and mate

SYSTEM OF MATING

1. Natural mating

Hand mating

In this system the ewes or does in heat are mated with desired ram or buck. The females are allowed to mate one by one. In this system a ram or buck may not be allowed to mate more than three ewes/does a day. This method of hand mating ensured the expected time of lambing.

Pen mating

In this mating system the ewes / does are divided into batches varying from 20-25 ewes/does. Males are turned into the flock only during the night time and separated during day time, which gives enough rest for the male. This system of mating also prevents the disturbance ewes by the male during grazing hours.

Flock mating

This system of mating is practiced normally by the farmers. In this system males are allowed to run along with the females throughout the day and night. The male may lose its most of its body reserve in chasing the ewes and they may lose their body condition

II. ARTIFICIAL BREEDING

Artificial Insemination

Artificial insemination offer the best means of distributing germplasm from nucleus breeding flock to many small flocks within each eco system. Fresh as well as frozen semen is used. The speculum method of insemination is used for ewes and does.

FEEDING MANAGEMENT

FEEDING MANAGEMENT OF SHEEP

Sheep have small muzzle and split upper lip helping them to nibble small blades of grass on pasture. At present, extensive grazing on marginal grasslands is widely practiced. The animals will graze on crop stubbles, weeds and grasses on fallow as well as rangelands. This will meet the requirements of the animals only partially. Their diet may be supplemented with cultivated fodders, grains and oil cakes, especially during the critical period of production cycle when the nutrient supply is inadequate.

Feeding the Breeding Ewe

The feeding of the ewe will be discussed under the following headings,

- 1. Flushing:** Already discussed in breeding management
- 2. Feeding during breeding season:** The ration provided during flushing will continue during breeding season also.
- 3. Feeding during early and mid-pregnancy:** Good feeding during gestation is the keystone for a healthy strong lamb crop. If feeding is inadequate or defective, weak or dead lambs will result
- 4. Feeding during late pregnancy:** Their ration must be supplemented with available green fodder fed at the rate of 5 kg per head per day. During the last one month of pregnancy concentrate may be fed at the rate of 225 g per head per day.
- 5. Feeding at lambing time:** As lambing time approaches or immediately after lambing, the grain allowance should be materially reduced but good quality dry roughage be fed free choice. In general, bulky and laxative feedstuffs may be included in the ration during the first few days. A mixture of wheat bran and barely or oats or maize at 1: 1 proportion is excellent.
- 6. Feeding lactating ewes:** Ration for ewes must be supplemented to maintain adequate milk production, which is necessary for rapid growth of lambs. If they are provided good pasture, the requirements are more or less met.
- 7. Feeding of ewes from the time lambs are weaned until flushing time:** Ewes may be maintained entirely on pasture. Poor quality pastures and other roughages of low quality can be advantageously utilized during this period.
- 8. Feeding rams for breeding:** The common practice is to allow the rams to graze with the ewes, which will allow the rams to get the same ration as the ewes. If separate feeding is practiced for the ram, it may be given 300-500 g of concentrate mixture consisting of three parts oats or barely, one part maize and one part wheat per day.

Lamb Feeding

- 1. Feeding suckling lambs:** This is the early part of a lamb's life in which it is dependent on its mother's milk to a considerable degree for its nutrition. This period ends when the lambs are weaned.

2. Feeding early-weaned and orphan lambs: Lambs are usually weaned at three months of age. Similarly, some lambs may be orphaned due to the death of ewe or due to disowning by the mother. If legume hay or good quality pasture is not available and if only poor roughage is fed their grain ration should be supplemented with a protein cum vitamin supplement with approximately 12 per cent digestible crude protein.

Complete pelleted ration consisting of roughage and concentrate, both mixed and made into pellets has been found to be advantageous. Recommended rations for the creep feeders and early weaners are (i) Maize 40 per cent, oats 30 per cent, barley 30 per cent plus lucerne hay fed *ad libitum*

Concentrate Mixtures for Supplementing Lamb Grazing

During summer:	
Groundnut cake	20 per cent
Wheat bran	35 per cent
Crushed gram	10 per cent
Oats/Barley	35 per cent
During winter:	
Wheat bran	25 per cent
Oats/Barley/Jowar	50 per cent
Groundnut cake	25 per cent

Rate of feeding concentrate per day:

S. No.	Body weight	When legume fodder is available	When non legume fodder is available
1.	Up to 12 kg	25-50 g	200-300 g
2.	12-15 kg	50 g	300 g
3.	15-25 kg	100 g	400 g
4.	25-35 kg	150 g	600 g

Sheep grazing

Grazing lands in India are over-grazed and generally in very poor condition. There is very little attention paid at present to develop the pastures. Pasture should be divided into compartments for rotational grazing. It should be managed to support 10 to 12 ewes and their lambs, per acre. It will be desirable to change the pasture for grazing periodically as sheep consume less due to the monotony of grazing if they have to graze on the pasture continuously.

FEEDING MANAGEMENT OF GOATS

Feeding Habits

1. Goats are sensitive animals with peculiar feeding habits (browsing). They are fastidious about cleanliness and like frequent change in the feed. Feeds given must be clean and fresh, since goats eat nothing that is dirty or foul smelling.
2. They dislike wet, stale or trampled fodder. For this reason it is advisable to feed them in hay-racks or hang the feed in bundles from a peg in a wall or from a branch of a tree.
3. Double-sided portable hay-racks are the most suitable
4. Convenient for stall feeding. It is preferable to serve them small quantities at a time; when served in large quantities at a time, they waste a lot of it by trampling.
5. Goats are ruminants. They are very fond of leguminous fodders. They do not relish fodders like sorghum (*sorghum vulgare pers*) and maize (*zea mays* L.), silage or straw. Goats do not relish hay prepared from forest grasses, even if cut in early stages, but very much relish hay prepared from leguminous crop.
6. Some of the common green roughages liked by the goats are: Lucerne, berseem, Napier grass, cowpea, cabbage, cauliflower leaves, shrubs and weeds of different kinds;
7. Leaves of trees such as babul, neem, subabul, glyricidia, agathi, malbari and dry leaves of trees are most liked by the goat.

1. Nutrients required

The nutrients needed may be divided into maintenance, production (for milk, meat and hair production) and pregnancy requirements.

2. Maintenance ration:

Goats have higher basal metabolic rate than cattle; therefore, their maintenance requirements are higher than those of cattle. The requirement by weight is calculated and an additional feed of about 25 to 30 per cent for maintenance is allowed. For its size the goat can consume substantially more feed than cattle or sheep, viz. 6.5 to 11 per cent of its body weight in dry matter when compared with 2.5 to 3 per cent for cattle or sheep. This means that the goat can satisfy its maintenance requirement and produce milk from forage alone.

3. Production ration:

The nutritional requirements of a goat weighing 50 kg and yielding 2 litres of milk with 4 per cent fat may be met by feeding 400 g of concentrate mixture and 5 kg of leguminous fodder. The ration should have 12 to 15 per cent protein content.

4. Pregnancy ration:

A week before she kids, the doe should be provided with more succulent type of food. For three or four days after kidding, the level of diet should be lowered and made more fibrous. After this period the feeding should be done at a normal rate.

5. Feeding of young stock:

Feeding schedule for kids should be such that a weekly growth rate of 0.6 kg is obtained. The kid should be fed 50 to 100 ml of colostrums, four to five times a day, depending on its birth weight for three days. The composition of concentrate mixture (in parts) should be: gram, 20; maize, 22; groundnut-cake, 35; wheat bran, 20; mineral mixture, 2.5; and common Salt, 0.5.

Mineral mixture:

Goats require slightly larger quantities of calcium than sheep. The mineral mixture may be included in the concentrate ration at the rate of 0.2 per cent.

Common salt:

The provision of salt licks is very important for goats as they secrete a good amount of sodium and chloride ions in milk. Salt to the extent of 2 per cent may also be mixed with the daily grain ration of goats.

Vitamins:

Goats need particularly vitamins A, D and E. The microbes in the rumen synthesize most of the other needed vitamins. Synthetic vitamins A and D may be included in the ration of growing kids.

DISEASES OF SHEEP AND GOAT

TETANUS

Causative Agent: *C.tetani*.

Symptoms

- ☐ Initial symptoms include mild stiffness with head, neck and tail and unwillingness to move. Twitching of muscles and animal fall over to one side and unable to rise.
- ☐ In the terminal stages the rigidity of muscles extend from the limbs to the trunk, nostrils get dilated, ears erect, nictitating membrane protruded and mastication becomes impossible because the mouth cannot be opened – hence called **Locked Jaw**. Respiration becomes shallow and rapid before final respiratory failure.

Diagnosis

- ☐ Based on symptom and lesions.

Prevention and Control:

- Antibiotics
- Immunization – passive, active or combined.

- For prevention, the farm animals should be vaccinated routinely with tetanus toxoid.

ENEROTOXAEMIA

Causative Agent: *Clostridium perfringens* type D

Symptoms

- Sudden death mostly lambs with good body condition followed by monsoon.
- It damages vascular endothelium (including blood vessels in the brain) leading to fluid loss, edema, off feed, grinding of teeth, bloody diarrhea.

Diagnosis

- Based on symptom and lesions.
- Large numbers of Gram-positive rods are suggestive of *C. welchii*.

Control and Prevention

- Before the lambing season the ewes are vaccinated.
- Lambs can also be vaccinated at 4 weeks of age.

PESTE DES PETITS RUMINANTS

Peste des petits ruminants (PPR), also known as goat plague, is caused by a paramyxovirus of the Morbillivirus genus. PPR is an acute or subacute viral disease of goats and sheep characterized by fever, erosive stomatitis, conjunctivitis, gastroenteritis, and pneumonia. Goats are usually more severely affected than sheep (hence called goat plague).

SYMPTOMS

Fever (40° to 41°C), watery discharge is from the eyes, nose and mouth, the mucous membranes of the mouth and eyes become much reddened. Pin-point greyish areas on the gums, dental pad, palate, lips, inner aspects of the cheeks and upper surface of the tongue, diarrhoea and difficult to breath

DIAGNOSIS

- Based on clinical, pathological, and epizootiological findings.

CONTROL

- Control of PPR outbreaks relies on movement control (quarantine) combined with the use of focused ("ring") vaccination and prophylactic immunization in high-risk populations.
- There is no treatment for PPR. However, mortality rates may be decreased by the use of drugs that control the bacterial and parasitic complications.

SHEEP AND GOAT POX

Sheep pox and goat pox are viral diseases of sheep and goats characterised by fever, generalized papules or nodules, vesicles (rarely), internal lesions (particularly in the lungs), and death. Both diseases are caused by strains of the genus Capripoxvirus.

SYMPTOMS

Fever, depression, polypnoea, conjunctivitis, lacrimation, rhinitis, oedema of eyelids and photophobia. The cutaneous eruption begin with erythematous areas especially noticeable in hair or wool-free parts of the body, papules and nodules formation. Animals may recover within 20-30 days. Death is frequent when complications like abortion and secondary infections.

DIAGNOSIS

- ☐ Based on the characteristic symptoms and pox lesions in the hairless parts of the animal.

CONTAGIOUS ETHYMA

Contagious pustular dermatitis otherwise known as Scabby mouth, sore mouth or orf is a highly contagious, viral disease of sheep and goats. It is an acute infectious disease of sheep characterized by the formation of vesicles, pustules, and finally thick scabs on the lips, nostrils, face, eyelids, teats, udders, feet, and occasionally inside the mouth. Lambs are generally more susceptible than adults. The disease in sheep and goats is normally self-limiting, and clears without treatment within three to four weeks. Antibiotics are not necessary, unless secondary bacterial infection occurs

BLUE TONGUE

- ☐ Bluetongue (BT) is a non-contagious, athropod-borne viral disease of sheep and other domestic and wild ruminants, such as goats, cattle and deer.
- ☐ The infection in animals is transmitted through the bite of Culicoides insects.
- ☐ . The incubation period of BT in sheep is usually 7-10 days.
- ☐ **Sheep**
 - Rise in temperature to 40-42°C., excessive salivation and frothing at the mouth, hyperemia and swelling of the buccal and nasal mucosa followed by erosions and ulcerations. Extensive ulcerations on the dental pad and dorsal surface of the tongue. Hyperemia is also observed around the coronarybands of the hooves and lameness. Swelling and bluish discolouration of tongue. Signs include abortions, stillbirths, and weak “dummy lamb” live births.
- ☐ **Goats**
 - Bluetongue infection of goats is an inapparent infection similar to that described for cattle.

DIAGNOSIS

- ☐ Based on the characteristic symptoms and lesions.

PREVENTION AND CONTROL

- ☐ Control is mainly aimed at vector control-water management (reduction of *Culicoides* breeding sites), use of insecticides and larvacides (spraying breeding areas), and insect repellents in which animals are dipped.
- ☐ Vaccination should be practiced only in endemic areas.

Unit IV: Management of Swine

SWINE PRODUCTION

Introduction

Swine rearing in India is carried out under variety of adverse social, climatic, and environmental conditions. Swine industry has remained underdeveloped mainly due to religious taboos and prejudices. Swine rearing in our country is mainly in the hands of socially and economically weaker illiterate people. Great emphasis has been laid on the improvement of the productivity of pigs by implementing crossbreeding programme with exotic breeds of swine to develop animals of larger sized litter, efficient feed conversion, higher dressing percentage and better quality pork

Census

As per 2007 census 11.13 million pigs were present in India among this 2.38 millions pigs were crossbred and 8.74 million pigs are indigenous desi pigs.

ADVANTAGE OF PIG FARMING

1. Swine are prolific producer. The litter size, on an average is 8 and a sow can be managed to raise 2 or more litter per year. Gilts can be bred at the age of 8 months, farrow at 1 year of age and if well fed and kept healthy may weigh more than 80kg by the time of 5 ½ -6 months of age.
2. Gestation period is less (114 days or 3months, 3weeks, 3days)
3. The piglets are ready to market by 6th month of age
4. They have better feed efficiency and produce 1kg of pork with 3 to 3.5 kg of feed.
5. The dressing percentage of swine is 65-80% because skin is also used for eating. The percentage of edible meat in pig is higher because of smaller bones.
6. Pigs convert inedible feed, forage, and certain grain by-products obtained from oil milling, meat by-products, garbage, garden waste, vegetable market waste and dairy by products like skim milk into valuable nutritious meat
7. Compared to other species pork contain less quantity of cholesterol and more energy value
8. The initial investment is small in a swine enterprise and the return is quick due to its short life span.
9. Pork fat is more valuable than fats produced by other animals. It is a good source of polyunsaturated fatty acids.

DISADVANTAGE

1. Religious taboo [Muslims will not eat pork]
2. The pig is in direct competition with man for food.

3. Pigs are sensitive to unfavourable rations. This mixed with careless management creates problems.
4. Swine are highly susceptible to diseases and parasites, the most common parasite being the roundworm *Ascaris lumbricoides*. Management practices to maintain a disease free herd is important for a successful swine enterprise.
5. Skilled attention is required at the time of farrowing to avoid loss of piglets
6. Eating pork is considered as of 'low to the dignity'
7. Well established farms to supply the stocks are less in number

Scientific name of modern pig: ***Sus domesticus***

Indigenous domesticated pigs

They have no distinct breed features. Their characteristics vary with topography and climatic conditions from region to region. They are black, brown, grey, rusty grey or blending/admixture of two or more colours. They are traditionally raised by weaker section of the community.

Exotic/improved breeds of pigs

Classification of swine breeds based on utility	
Lard type 1. Duroc 2. Poland china 3. Chester white 4. Berkshire 5. Hampshire 6. Spotted Poland china	Bacon type 1. Tamworth 2. Yorkshire
Exotic breeds experienced in India	
1. Large White Yorkshire 2. Middle white Yorkshire 3. Saddle back 4. Tamworth 5. Berkshire	
Classification based on colour	
White breed	Yorkshire, Landrace, Chester White
Black with some white	Hampshire, Berkshire, Poland china
Black and white	Spot
Red breed	Duroc and Tamworth
Red with some white	Hereford

IMPORTANT BREEDS OF PIGS

1. Large White Yorkshire

The large White Yorkshire is a popular English Bacon breed, which had its origin Yorkshire and neighboring countries in northern England. Yorkshire sows are noted as ***good mothers***.

- Bacon type breed
- The body colour is solid white with occasional black pigmented spots called ‘freckles’
- Other distinguishing characteristic are erect ear, snout of medium length, large in size and slightly dished face
- Skin is pink coloured and is free from wrinkles with long and moderately fine coat. Neck is long and full to the shoulder. Back is slightly arched
- Matured body weight boar: 300-400kg and sow 230-320kg
- Most extensively used exotic breed in India.. It is an excellent breed for the purpose of crossbreeding



2. Middle White Yorkshire

This breed was evolved by crossing the Large White Yorkshire with smaller breed of Yorkshire extraction. The breed is accepted as excellent pork pig, reaching slaughter weight early and with a high percentage of lean meat to bone. The breed is hardy, grows rapidly, but is not as prolific as the Large White Yorkshire. Extensively used to upgrade desi pigs as it is smaller in size. Adult body weight boar: 250-340kg and sow 180-270kg.

3. Landrace

The origin of this breed is Denmark, where it has been bred and fed to produce the highest quality bacon in the world.

- Large, long body. The breed is white in colour, although black skin spots ‘freckles’ rather common.
- White bristles in white body
- The breed is characterized by its long, deep side; square ham. The legs are relatively short. The carcass is more lean than that of the meat.

- Matured body weight boar 270-360kg and sow 200-320kg.



ECONOMIC TRAITS

- Litter Size
- Weight at Birth.
- Weaning Weight.
- Litter size at Weaning.
- Growth rate:
 1. Birth to Weaning : Weaning wt.- birth wt./ 56
 2. Weaning to 154 days: Weight at 154 days-Weaning weight/ 98 days
 3. Weight at 154 days to 210 days: Wt.at 210 days- Wt. at 154 days / 154 days.
- Feed efficiency: Feed consumed per kg of live weight.
- Mortality percentage.

Comparison between desi Vs. Exotic (India)

	Desi	Exotic
Litter size at birth	7.5	10
Birth weight (Kg.)	0.91	1.4
Weaning weight (kg.)	4.1	13.5
Weaning percentage	54	78.5
Dressing percentage	66	68
Maturity (Months)	14	8 - 10
Growth rate (gm)	70-100	over 300 gm.
Back fat thickness – (cm)	3-7	4-5

HOUSING OF PIG

The dwelling place of pig is called "sty". A sort of semi-indoor system of swinefarming in loose houses is generally suitable for India.

Location

The piggery unit should be located well away from the human dwelling places.

Types

- Indoor system
- Semi-indoor system
- Extensive system

Functional units of sty**Boar sty**

Each pen shall accommodate not more than one animal.

Dry sow and gilt sty

It is meant for growing gilt or dry sow. 5-10 pigs can be accommodated in each pen depending upon the stage of pregnancy and age.

Farrowing sty

Pregnant sows are transferred to these sties 7-10 days before the expected date of farrowing and left here up to weaning of their litters (56 days). Each pen shall accommodate not more than one animal with the litter. Number of farrowing pen required is 10 % of the breedable female. Each pen should have the facilities of guard rail and creep area.

Guard rail

The guard rails are made up of tubular iron rod of 2 to 2.5 inches thickness and fixed 25 to 30cm away from the wall and above the floor. So that the piglets can run under the rail when the sow lies down so that the crushing of piglets can be avoided in between the sow and walls and also to avoid eating of creep ration by sows.

Creep area

In corner of the pen, creep area should be provided with the help of stone slab should be fitted in the corner of the pen, 25-30cm above the floor level. In that area piglet only can enter and get the creep feed. Lighting can also provided in creep area to give warmth to young ones.

Weaner sty

Piglets that are weaned are housed in these pens up to six months of age. Each pen may accommodate 30 piglets of age group between two to six months.

Fattening sty

These are for housing pigs that are being transferred for the market. Each pen may have 16 to 32 animals kept for fattening purpose.

Sty for sick animal

Normally a minimum accommodation for 5 percent of the stock shall be provided for housing the sick animals.

Floor space requirements

S. No.	Type pf animals or category	Covered area (m2)	Open area(m2)	No. of animals housed
i.	Boar	6-7 (2.5x2.5to2.5x3)	8.8-12	1
ii.	Farrowing sow	7-9 (2.5x3to3x3)	8.8-12	1
iii.	Weaner/ fattening pigs	0.9-1.8 (0.8x1.2to1.2x1.5)	0.9-1.8	30
iv.	Dry sow/gilt	1.8-2.7 (1.2x1.5to1.8x1.5)	1.8-2.7	5-10

Feeding and water space requirement

	Feeding	Watering
Adult Pig	60-75 cm	6-7.5 cm
Young pig	25-35 cm	2.5-3.5 cm

SWINE NUTRITION

The nutritional requirements of swine vary with sex, age and physiological status of the animal. According to the nutritional requirements, swine can be classified into eight groups as follows;

- a) **Young pigs:** Pre-starter pigs (2-5 kg) and Starter pigs (5-15 kg)
- b) **Weaners:** Grower pigs (15-35 kg) and Growing and finishing hogs (35-90 kg)
- c) **Adult males:** Boars
- d) **Adult females:** Gilts, Pregnant sows and gilts, and Lactating sows

Feeding pre-starter pigs:

Feeding pre-starter rations to piglets increases growth rate of pigs and reduced mortality among them. The pre-starter ration should be rich in protein (24%) and vitamins and low in fibre content. This feeding starts usually when the pigs are one week old, weighing 2-2.1 kg. This extends usually to a period of two weeks during which each pig consumes one and half to two kilograms of pre starter ration and gains about three kilograms, to weigh 5.0 kg at the end of the period.

Feeding starter pigs:

The starter ration should contain 20-22 percent protein. It is fed to pigs between 5.0-15.0 kg live weight. This period generally ends when the pigs are 8 weeks old. They consume nine to eleven kilogram of starter ration to gain about ten kilograms.

Feeding grower pigs:

When the average weight of pigs is 15 kg, the starter ration is replaced by grower ration which has only 18 percent protein. Pigs consume 50-60 kg of grower ration to weigh about 35 kg at the end of this period.

Nutritional anaemia in piglets: Pigs, which are reared on concrete floors with little opportunity to eat some soil, which will supply iron and copper, may develop anaemia. Anaemia can be controlled by the following methods.

- (1) Providing clean sand or clean soil in the shed itself.
- (2) By allowing them access to clean area of soil.
- (3) By giving iron and copper pills.
- (4) The udder of the sow may be brushed with a solution of ferrous sulfate and copper sulfate. By injecting 2 ml of iron dextrons or iron dextrins at three days' and three weeks' age.

Growing and finishing pigs:

This is the stage during which the weaned pigs grow from 35 kg to 60 kg body weight. Pigs above 60 kg live weight and until marketing are known as finishing pigs. The protein requirement is only 13 per cent. Under hot climatic conditions it will be advantageous to market/slaughter pigs between 70-75 kg.

Feeding adult male pigs:

The boar may be given almost the same kind of ration fed to the gilts during the breeding period.

Feeding gilts:

Gilts, like boars, should be fed limited amount of nutrients to keep them thrifty but not fat. If they are already fat, there will be no scope for flushing during the breeding season. Gilts meant for breeding should be removed from the growing-fattening group when they are between 45-57 kg body weights.

Flushing:

Litter size is affected by the condition of the gilts or sows at the breeding time. In order to ensure maximum results, sows and gilts are fed a more liberal ration from about two weeks before breeding until after they are bred. This should produce a gain, from 0.7 kg to 0.9 kg per day, and ensure good health. Such a procedure is known as flushing. The ration for flushing should be well balanced and fed in required quantities. Gilts require flushing ration of 25 to 30 gm for every kg body weight.

Feeding pregnant sows:

Pregnant sows should get a ration which contains, protein 14 per cent, calcium 0.7 per cent and common salt 0.5 per cent. Sows should gain 27 to 36 kg during gestation period, compared to 32 to 41 kg by gilts.

Feeding sows at the time of parturition and immediately after parturition:

About four to five days before the farrowing, the sow's ration should be reduced to nearly half. They should be prevented from becoming constipated by including laxative feeds like wheat bran, linseed meal, etc. in the ration. On the day of farrowing, it is better not to give any feed at all. But enough of clean fresh water should be available throughout the period. It is best to feed 50 per cent of the ration on the first day and to increase the ration by 200 to 300 grams daily until she is full fed.

Feeding lactating sow:

Sows receiving adequate ration produce 2.5 to 3.6 kg milk per day. A bulky ration may be continued up to the second week of lactation. The dry-lot lactation ration should contain 15 percent crude protein for both first litter gilts and sows. If on the other hand they are put on a legume pasture, 13 percent crude protein will be sufficient in the ration. Lactation rations should contain 0.7 percent calcium, 0.6 percent phosphorous and 0.5 percent common salt. Lactating sows required 14 to 23 litres of water per head per day.

CREEP RATION

Creep feed: Creep feed is used to provide special feed for the young; also the feed should be provided for the young within a special enclosure

CARE AND MANAGEMENT OF ADULT AND YOUNG

STOCKSELECTION OF GILTS / BOAR

- The gilt should be selected at market weight viz. when animal weigh about 90 kg
- The mother of the pig to be selected should have larger litter of 8 piglets or more
- The gilt or boar should have reached a body weight of about 90 kg in about six month
- The pig should have adequate length and depth of body, thick well-muscled hams and should be firm and trim; it should have a prominent neck
- The pig should be free from leg defects
- The gilt should have minimum of **12 evenly spaced** functional teats. An animal with blind teat should be avoided
- The pig should be free from diseases and physical defects

BREEDING MANAGEMENT

Guide line for normal reproduction of pigs

AGE AT PUBERTY	6-7 MONTHS
Breeding age of gilts	8 -9 months
Breeding weight of gilts	80-90 kg
Breeding age of boar	1 - 1 ½ years
Sex ratio	1:10

Estrus cycle	19-23 days average 21 days
Estrus period	2-3 days
Mating time	Gilt first day and sow second day of onset of heat
Number of service per conception	Two at an interval of 12-24 hours
Gestation period	114 days (3 months, 3 weeks, 3 days)
Suckling period	56 days
Average litter size at birth	10-12
Average litter weight at birth	12-14 kg
Birth weight	1.2-1.3 kg
Average litter size at weaning	8-10
Average litter weight at weaning	120 kg in gilts and 150 kg in sows
Weaning weight	12-13 kg
Occurrence of heat after weaning	4-7 days
Volume of ejaculate	200 ml
Average number of sperms per ml	100,000
Average age to castrate pigs	2-3 weeks
Market age of fattening pigs	5 and 7th months
Market weight at 6 months	70-75 kg in Indian condition
Farrowing interval	7-7 ½ months
Farrowing rate	1.8-1.9
Average life of sow	6 litters

Detection of heat

- Vulval swelling, reddening of vulva is noticed 2-6 days prior to oestrus.
- The animal will be restless and the increased activity level and a typical grunt, frequently mount on other sows and allow them to be mounted, reduced appetite and frequent urination,
- ‘Standing heat reflex’ will be exhibited during oestrus, which consist of immobilization, arching back are noticed.
- Another important symptom which known as **standing heat test or riding test or back pressure test** in is the sow remains immobile with erected ear when normal pressure is applied to lumbo sacral region. This test can be performed in front of boar or adjacent to a boar pen. This method is the ideal method for heat detection in pigs.

MATING MANAGEMENT

Hand mating:

In this method bring the sows to the boar for individual service when they are in standing heat. It most common method of mating in pure bred herds.

Pen mating:

In this method one boar is allowed in the sow pen (8-10) for detecting heat and mating. There is a chance for over usage of boar.

Flock or herd mating:

In this method the boar is permitted for run with the sows. Mature boar can usually satisfactorily serve 15-20 sows in pasture mating system.

- After detecting heat, the sow and boar should be introduced in a familiar area so as to facilitate better mating.
- If new boar is introduced in to the herd, he should not be used until after a minimum period of 30 days of isolation.
- Mating should be done early in the morning or late in the evening
- Sow ovulate 16-48 hours after the onset of estrous and it will loss its fertility rate rapidly, hence it is better to bred at least twice preferably during the latter half of the first day or early on the second day of estrous which will increase the conception rate and increase the litter size .
- Sow normally come to heat one to four days after farrowing but it is better to mate during the first heat after weaning the litter
- Ovulation rate increases during successive estrous [up to fifth] thus it is advantageous to delay breeding of gilts until the second or third estrous

SIGNS OF PREGNANCY AND FARROWING

- Female become more docile during pregnancy and the physical activity will be reduced, the development of mammary system is another indication of pregnancy. Normally gilts will show some signs of udder development after 2 months of pregnancy. A sow normally shows signs of new udder development some 2 to 3 weeks before farrowing.
- 1 to 3 days before farrowing the pitch of voice may be lowered and the sow may begin to prepare nest for her litter
- Normally farrowing appears to follow a diurnal pattern, more frequently in late afternoon and most frequently at night
- Duration of farrowing varies from 2 to 24 hours in normal parturition. The sow often takes rest in between expulsion of piglets.

MANAGEMENT OF SOW AT FARROWING

- The sow should be transferred to farrowing pen 1 week prior to expected date of farrowing. Before moving to farrowing pen, the sow should be cleaned and spraying against ecto parasite also is important.

- The farrowing pen should be fitted with 'guard rails' or 'farrowing crate' to protect the young piglets from crushing by the mother. The rails should be 9 inches from the floor and 12 inches from the wall
- The temperature regulatory mechanism is ill-developed in newborn piglets. The baby piglet's body temperature falls very rapidly when environmental temperature falls. Normally they have subnormal body temperature during first 30 minutes and it returns to normal during next 48 hours. Hence the pen should be maintained at **24 to 28°C** until the piglets are three to four days old. This can be achieved by providing artificial lighting using electric bulbs or infrared bulbs.
- The placenta drops one after another and within two hours this will complete
- The piglets should be allowed to suckle after birth. Normally they suckle 8 to 10 times in 24 hours in the initial period.

Management of lactating sows

- During the lactation period heavy nutrient loss may occur through the milk. Hence up to 1 month of lactation high nutrient diet is essential.
- Ad libitum clean drinking water should be provided at all times.
- Keeping the farrowing sty clean, cool and dry.
- If the litter size is large, the litter may be split into two or three groups, based on their size, allow the smaller group first followed by the larger. This type of milk feeding is called "split up suckling" or split milk feeding

Care and management of piglets from birth to weaning

Management of suckling piglets

- To prevent 'naval ill' the naval cord should be tied off and it should be cut 3-5 cm distal to the legation and this portion should be dipped in a solution of 2% iodine or 70% ethyl alcohol.
- The 'needle teeth' should be clipped. The baby piglets have four pair of sharp teeth two on each jaw called needle teeth or wolf teeth. It is advisable to clip these needle teeth shortly after birth [within 10 days].
- The piglets should be ear-tattooed/ear tagged immediately after birth
- Castration of male piglets should be done a week or 10 days before weaning.
- Cutting of tails is advisable whenever pigs are to be raised in total confinement. This can be done at the time of needle teeth clipping.

Orphan Pigs:

The 'orphan pig' results when the sow dies after farrowing, mastitis, lactation failure or litter larger than the sow is able to raise. Two possibilities are there to raise 'orphan pigs'

[1] Adaptation by a foster sow [2] Raising by cow milk or sow milk replacer

First the piglets should receive colostrum. Cow colostrum apparently provides some immunity.

Adaptation by a foster sow:

- ☐ If another sow has farrowed within a short period before or after birth of the orphaned litter, the piglets may be transferred to her. This transfer must be done within a few days after farrowing because those mammary glands that are not used soon stop their milk production.
- ☐ To insure acceptance of new pigs, the sow should be separated from her own litter while the new pigs are combined, and a disinfectant or other odorous materials should be sprinkled on all the pigs to disguise odour.

COMMON DISEASES AND THEIR PREVENTION AND CONTROL

MEASURE CLASSICAL SWINE FEVER

- ☐ Classical swine fever (CSF) is a contagious viral disease of pigs. CSF is caused by a virus belonging to the family Flaviviridae and the genus **pestivirus**.

SYMPTOMS

- ☐ Acute infection
 - In acute form the pigs appear sick, inactive and drowsy with arched back. Some pigs stand with droopy head and straight tail. Huddling, vomiting, high fever, anorexia and constipation. Conjunctivitis, staggering gait, posterior weakness and purple discoloration of abdominal skin
 - In last stage of the infection, pigs will become recumbent, and convulsions may occur shortly before death. Severe diarrhoea will also occur during last stages.
- ☐ Chronic form
 - Dullness, capricious appetite, pyrexia and diarrhoea for up to 1 month. Weight loss, hair loss, dermatitis and discoloration of abdomen or ears are the other symptoms. A chronically infected pig may have a disproportionately large head relative to the small trunk.

DIAGNOSIS

- ☐ Based on high morbidity and mortality, high fever, diarrhoea. Kidney and lymph node lesions will help in field diagnosis.

VACCINATION

- ☐ Modified live vaccines (MLV) are used to control CSF.

SWINE INFLUENZA

- ☐ Swine influenza is a highly contagious respiratory viral infection of pigs caused by swine influenza virus, characterized by coughing, sneezing, nasal discharge,

Elevated rectal temperatures, lethargy, difficult breathing, depressed appetite and rarely associated with reproductive disorders such as abortion.

- ☐ The first clinical signs are fever (40.5-41.5°C), puffy eyes, anorexia leading to loss of weight, depression, prostration and huddling leading to weakness. These signs are followed by sudden onset of acute respiratory signs, which include paroxysmal coughing, sneezing, irregular abdominal breathing and ocular and nasal discharges.
- ☐ In breeding stock, abortions, infertility, production of small weak litters and increased stillbirths.

PREVENTION AND COTROL

- ☐ Good husbandry practices including All-in/All-out to limit the spread of the disease, provision of fresh clean drinking water, avoiding ducks and turkey contamination's/contact including staff and proper use of disinfectants to clean infected buildings.

FOOT AND MOUTH DISEASE

- ☐ Viral disease of pigs caused by FMD virus of the genus Aphthovirus. Characterized by fever (40-40.6°C), anorexia, reluctance to move, and scream when forced to move. These signs are followed by vesicles on the coronary band, heels, inter digital space and on the snout. Mouth lesions are not too common and when they occur are smaller and of shorter duration than in cattle and tend to be a "dry"-type lesion. There is no drooling. Sows may abort. Piglets may die without showing any clinical sign.

DIAGNOSIS

- ☐ Based on symptom and lesions

Control and eradication program

- ☐ Prevention of movement of animals and animal products in the area affected.
- ☐ Destroy carcasses
- ☐ Disinfect vehicles leaving the infected area.
- ☐ Perform vaccination.

SWINE ERYSEPLAS

- ☐ A bacterial disease caused by *Erysipelothrix rhusiopathiae*.

Symptoms

- ☐ Erysipelas occurs in pigs of all ages, but pigs from 2 months to one year age are highly susceptible.

Acute

- ☐ The acute disease is characterized by high fever, inappetence, depression, a rapid course of illness, and death within 2-3 days in untreated animals.
- ☐ Some animals may show a stiff gait and reluctance to stand or move, and urticarial cutaneous lesions may develop.

- The diamond shaped raised skin lesions are pathognomonic. Pregnant sows may abort.

Chronic

- In the chronic form arthritis is more common.
- The hock, stifle, elbow and carpal joints are most likely to be affected resulting in severe lameness. The mitral valves are involved in valvular endocarditis. Diamond shaped skin lesions are pathognomonic.

Diagnosis can be achieved by

- Based on symptom and lesions
- Gram-positive rods in acute cases and Gram-positive filaments in chronic cases.

Treatment

- In addition to hyper immune serum, treatment with antibiotics such as penicillin and tetracyclines are effective.

PIGLET MORTALITY

The major task in pig husbandry is avoiding piglet mortality and raising piglets successfully up to weaning. After weaning the mortality is comparatively less

Pre-partum death may be due to deficiency of iron, which can be prevented by injection of sow with iron. Intrapartum death is due to anoxia induced by lack of placental blood flow associated with uterine contraction or premature rupture of umbilical cord. This intra-partum death occurs mostly in aged sows.

Pre-weaning mortality

The pre-weaning mortality ranges between 12-30%.

Causes of mortality

Sl. No	Reasons	%
1	Still birth	17.4
2	Eaten by the sow	0.50
3	Genetic defect	1.60
4	Over laid (crushing by mother)	66.30
5	Enteritis	2.20
6	Pneumonia	0.50
7	Unknown	11.50

Starvation and overlying by the pig

70% of death is due to starvation and overlying by the sow, which can be avoided by

1. Improving birth weight and vigor of newborn piglets
2. Minimizing risk of chilling or hypothermia
3. Minimizing agalactia

It is a part of complex condition of **MMA** (M= mastitis, M= metritis & A=agalactia). The MMA syndrome can involve metabolic, bacterial and hormonal factor with stress plays a part. Since its main effect is loss of milk in the first three days after farrowing, the condition contributes to piglet loss from starvation.

Reason and control of MMA

1. Elevated temperature of sow is associated with this condition, hence regular monitoring of sows rectal temperature and treatment with antibiotic and oxytocin is essential to avoid this condition. Such conditions which are not detected earlier, it can be noticed by loss of body condition of piglets and it is very difficult to recover the condition quickly. In refractory case prompt provision of an alternative source of food for piglet by foster sow or artificial feeding will minimize the loss

2. The udder and teat of sow should be dry and kept hygiene to avoid such problem

Piglet anemia

The newborn piglet has only limited reserve of iron in the liver for hemoglobin synthesis. This is due to poor placental transfer of iron to foetus. The sow milk is very low in iron and the suckling pig should be supplemented with iron during first few days to prevent piglet anaemia.

Symptom

- a. Pale in the region of ears and belly
- b. Listlessness
- c. Rapid breathing
- d. Often exhibit diarrhea

Controlled

1. Placing fresh, clean earth in the piglets pen each day
2. Using soil drenched with a solution made from 500 gm ferrous sulphate, 75 gm copper sulphate and 3 litter of water
3. Daily administration of 4 ml of 1.8 percent ferrous sulphate solution
4. The daily painting of the mother's udder with ferrous sulphate solution and sugar [0.5 kg of ferrous sulphate in 10 liter of water]
5. All these methods are labour intensive and the safest and easiest method of combating piglet **anaemia is to inject the piglet with 100-150 mg of iron in the form of iron dextron** 3 days after birth. If necessary a second and slightly smaller injection can be made some 3 weeks later

Unit V: Poultry Management

POULTRY PRODUCTION

The term “Poultry” indicates all domesticated avian species such as chicken, ducks, turkeys, Japanese quail, guinea fowls, geese, pigeons, ostrich, emu etc. Although very often used as synonymous to chicken.

Economic importance of Poultry

1.Chicken:

Egg production (Layers)	Meat production (Broilers)
320 eggs (17 to 72) weeks of age	1.8 to 2.0 kg at 5 weeks of age
India: ranks fourth in the world I China II. USA III. Japan IV. India (In India – I Andhra Pradesh; II TamilNadu; III Maharashtra) Per capita consumption: 52eggs/individual/Year	India : Ranks fifth in poultry meat production I. USA II. China III.Brazil IV.Mexico V. India (In India – I Andhra Pradesh; II Tamil Nadu; III Maharashtra) Per capita consumption: 2.0 kg/individual/Year

2. Waterfowl (7% of poultry population in India)

Ducks		Goose
Egg production Morethan 300 eggs (22 to 72 weeks of age)	Meat production (Green duck production) 2.0 to 2.5 kg at 8 weeks of age	1. Egg production: 30 to 50 eggs per year 2. Meat production; 4 to 6kg at 7 months 3.Cant liver production
West Bengal, Orrisa, Andhra Pradesh, Tamil Nadu, Kerala and Assam		1.8 to 2.0 kg at 5 weeks of age

3. Turkeys

Egg production: 60 to 90 eggs per year

Meat production: Male - 7 to10kg at 7 months of age

Female- 4 to 6kg at 7 months of age

4. Japanese quail

1. Egg production: 200 to 240 eggs from 6 to 45 weeks of age
2. Meat production: Male – 180 to 200g at 5 weeks of age
Female – 200 to 250g at 5 weeks of age

5. Ostrich and Emu

1. Skin / leather
2. Feather / plumage
3. Fat / oil – High medicinal value
4. Cornea
5. Tendon for transplantation

Common terms used in Poultry

Species	Young	Adult	
		Male	Female
Chicken	Chick	Cock	Hen
Duck	Duckling	Drake	Duck
Turkey	Poult	Tom	Hen
Quail	Chick	Cock	Hen
Guinea fowl	Keet	Cock	Hen
Goose	Gosling	Gander	Goose
Pigeon	Squab	Pigeon	Pigeon
Swan	Cygnets	Swan	Swan

Incubation period, Chromosome numbers and age at sexual maturity of different species of poultry:

S. No.	Species	Incubation period (days)	Chromosome number (pairs)	Age at sexual maturity (weeks)
1.	Chicken	21	39	18-20
2.	Duck	28	40	28-30
3.	Muscovy duck	33-35	40	28-30
4.	Goose	28-32	40	28-30
5.	Guinea fowl	27-28	39	28-32
6.	Turkey	28	40	28-30
7.	Quail	17-18	39	6-7
8.	Pigeon	18	39	10-12
9.	Ostrich	42	40	52

10.	Emu	52-55	40	52
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Definitions

Bill: The beak of duck.

Broiler / Fryer: It is a bird of about 5 weeks of age of either sex (straight-run chicks) with an average body weight of 1.8 to 2.0 kg with a flexible breast bone cartilage, pliable and tender meat.

Brooder: Metallic or wooden equipment used for artificial brooding of baby chicks.

Candling: The process of visual examination of an intact egg, by holding between the eye and a lighted candle or other source of light to determine the interior quality, shell soundness or stage of embryonic development.

Capon: Surgically castrated male less than 10 months of age.

Clutch: Eggs laid by a hen on consecutive days without break are referred as 'Clutch'.

Cull: Unproductive hen.

Culling: Elimination of unproductive or undesirable bird.

Debeaking / Beak trimming: Removal of a portion of

beak **Desi fowl:** Indigenous or native fowl.

Desnooding: Removal of snood in turkeys

Down time: Period between culling of one batch and introduction of new batch of birds in the shed.

Dubbing: The process of removal of comb and wattles at day-old age with the help of scissors.

Egg tooth: Small horny protrusion on the tip of chick's beak used to pip the egg shell when hatching. It drops off soon after chicks' escapes from the egg.

Green Duck: Ducks used for meat production which are marketed from 9 to 16 weeks of age.

Hatchery: A building where incubators are kept for hatching purpose.

Layer: A mature female fowl kept for egg laying purpose.

Livability: Ability of individual bird to live and remain vigorous and productive.

Moulting: Molting is the act or process of shedding and renewing feathers.

Pause: Interval between clutches.

Photoperiod: Period of illumination by both natural and artificial means.

Pipping: The act of pecking the shell and shell membrane by the chick in order to break the shell and hatch out of the shell.

Roaster: A broiler grown upto the age of 9 – 11 weeks to a body weight of 2.5 to 3.0 kg

Snood: A small muscular structure hanging from the base of beak in turkeys is called "Snood"

CLASS, BREEDS, VARIETIES AND STRAINS OF POULTRY

India, Bangladesh, Myanmar (Burma), Thailand, Sri Lanka and Malaysia were the home of the ancient jungle fowls from which the modern day breeds and commercial strains of chicken were evolved.

The four varieties of jungle fowl in ancient times were,

- 1) *G. gallus* - Red jungle fowl
- 2) *G. lafayetti* - Ceylon jungle fowl
- 3) *G. sonnerati* - Grey jungle fowl
- 4) *G. varius* - Java or black or green jungle fowl

The binomial nomenclature for domestic fowl (chicken) is *Gallus gallus domesticus*.

The binomial nomenclatures of other species of poultry are,

- 1) Ducks (Domestic) - *Anas platyrhynchos*
- 2) Geese - *Anser anser*
- 3) Guinea fowl - *Numida meleagris*
- 4) Turkey - *Meleagris gallopavo*
- 5) Japanese quail - *Coturnix coturnix japonica*
- 6) Pigeon - *Columba livia*
- 7) Ostrich - *Strutio camelus*
- 8) Emu - *Dromaius novaehollandiae*

Class: In poultry it is used to designate group of birds that have been developed in certain geographical areas or regions

For example: American, English, Mediterranean and Asiatic

Breed: It refers to group of birds with common ancestors and having similarity in shape, confirmation, growth, temperament, shell colour of eggs etc.

For example: *Leghorn, Rhode Island Red, Australorp, Aseel etc.*

Variety: A variety is a subdivision of a breed, distinguished either by colour, comb type feather pattern etc from other group of the same breed.

For example: *White Leghorn, Brown Leghorn, White Plymouth Rock,*

Barred Plymouth Rock, Black Minorca etc.

Strains: Strains are closely related inbred flocks with definite economic characters. A strain is the name given by a breeder who has done breeding on the birds and introduced certain economic characters in the birds. A breed or a variety may have several strains and may be phenotypically alike but often differ on their production performance depending upon breeding history.

For example: *Babcock, Bovans, Hyline, Hisex, Lohmann (Layer strain)*

Cobb, Hubbard, Ross, Hybro (Broiler strain)

BREEDS OF POULTRY

Breeds of chicken:

Class	Breed	Variety
<u>American class</u> 1. Clean shank 2. Yellow skin 3. Red ear lobe 4. Dual purpose bird 5. Medium size 6. Brown shelled eggs	Plymouth Rock	Barred White Buff Patridged Silver penciled
	Rhode Island Red	Single comb Rose Comb
	New Hamp Shire	Single comb
<u>Asiatic class</u> 1. Feathered shank 2. Yellow skin 3. Red ear lobe 4. Meat purpose 5. Massive size 6. Brown shelled eggs	Brahma	Light brahma Dark brahma
	Long shan	Black White
	Cochin	Black White Buff Pat ridged
<u>Mediterranean class</u> 1. Clean shank 2. Yellow/white skin 3. White ear lobe 4. Egg purpose bird 5. Small size 6. White shelled eggs	Leghorn	Single & Rose comb (white) Single & Rose comb (Light brown)Single & Rose comb (Dark brown) Single comb (Buff)
	Minorca	Single & Rose comb (white) Single & Rose comb (Black)Singlecomb (Buff)
<u>English class</u> 1. Clean shank 2. White skin 3. Red ear lobe 4. Dual purpose bird 5. Medium size 6. Brown shelled eggs	Australorp (Black Shank)	Black
	Cornish (Yellow skin and Pea comb)	White Dark

Indian chicken breeds:

1. Kadaknath (Block coloured meat)
2. Aseel (Used for fighting purpose)
3. Chittagong and 4. Busra

Note:

1. *Araucana* is an American breed, which lays blue or green coloured eggs.
2. *Aseel* is an Indigenous game fowl used for fighting purpose.
3. *Frizzle* is an asiatic breed, which is notable for its curved feather formation. Each feather is curled towards the head of the bird.
4. *Silkie* is an Asian breed, which has persistent broody character.
5. *White Leghorn* is the most popular egg type breed.

Breeds of Ducks:

Duck breeds are classified either as meat-type or egg-type:

- 1) Meat-type breeds (Green duck production) - Aylesbury, Muscovy, Pekin, Rouen, Cayuga, Swedish and Call ducks.
- 2) Egg-type breeds - Indian Runner and Campbell.

Note:

- 1) *Mule ducks* - These are hybrid ducks produced by mating female muscovies to mallard-type drakes. Mule ducks produce satisfactory meat yields, but they are sterile.
- 2) *Indian Runner ducks* are characteristically stand erect and their carriage is almost perpendicular.
- 3) *Khaki Campbell ducks* are prolific egg layers and can lay around 365 eggs in a laying year.

Breeds of Turkeys:

American Poultry Association recognized breeds of turkey are

1.	Broad breasted bronze	5.	Bourbon Red
2.	Broad breasted white	6.	Narragansett
3.	Beltsville small white	7.	Black and
4.	White Holland	8.	Slate

Among these *Broad-breasted bronze*, *Broad-breasted white* and *Beltsville small white* turkeys are popular in commercial point of view.

Breeds of Geese:

Important breeds of geese are

1.	Toulouse	6.	Buff
2.	Embden	7.	Pilgrim
3.	African	8.	Sebastopol and
4.	Chinese	9.	Egyptian
5.	Canada		

Breeds of Guinea fowl:

Popular breeds of Guinea fowl are

- 1) Pearl
- 2) White and
- 3) Lavender

Breeds / Varieties of Japanese quail:

Important breeds/varieties of Japanese quail are,

- 1) Manchurian Golden
- 2) British Range
- 3) English White and
- 4) Tuxedo

Commercial layer strains developed by private sectors in India:

S.No.	Breeder Name	Commercial strains	Remarks
1	Babcock	B300 / ISA white	70% of Indian share
2	Hendrix	Bovans white	10% of Indian share
3	Hy-Line	Hy-Line Brown	Popular strain in world;The egg size is smaller
		Hy-Line White	
4	Dekalb	Dekalb XL (white)	USA bird;Largest selling white egg bird in the world.
5	Lohmann	1) Lohmann White	10% of Indian share
		2) Lohmann Brown	

Commercial broiler strains developed by private sectors in India:

S.No.	Breeder Name	Commercial strains	Remarks
1	Cobb	1) Cobb 100	40-50% of Indian market
		2) Cobb 400	
2	Hubbard	Hubbard Hi-Y	

3	Ross	Ross 308	Next to cobb in India; 35-40% of Indian market.
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POULTRY HOUSE CONSTRUCTION

Need for poultry house

- 1) To protect birds from adverse climatic conditions
- 2) To ensure easy and economic operation
- 3) To ensure scientific feeding in a controlled manner
- 4) To facilitate proper micro-climatic conditions in a near vicinity of bird
- 5) For effective disease control measures
- 6) To ensure proper supervision

Different types of poultry houses

1) Brooder / chick house

It is used to brood and rear egg-type chicks from 0 to 8 weeks of age.

2) Grower house

It is used to grow egg-type birds from 9 to 18 weeks of age.

3) Brooder cum grower house

Here, the birds are reared from 0 to 18 weeks of age (entire brooding and growing period of egg-type chicken).

4) Layer house

In which birds over 18 weeks of age are reared, usually up to 72 weeks of age.

5) Broiler house

In which broilers are reared up to 6 weeks of age.

6) Breeder house

In which both male and female breeders are maintained at appropriate sex ratio.

7) Environmentally controlled (EC) house

In which, entire environment is manipulated in such a way that is optimum for the birds growth.

Optimal environmental conditions for rearing broilers

Temperature	- 22-30°C (70-85°F)
Relative Humidity	- 30-60 %
Ammonia level	- Less than 25 ppm
Litter moisture	- 15-25%
Air flow	- 10-30 metres/minute

Location

- 1) Poultry house should be located away from residential and industrial area.

- 2) It should have proper road facilities
- 3) It should have the basic amenities like water and electricity.
- 4) Availability of farm labourers at relatively cheaper wages.
- 5) Poultry house should be located in an elevated area and there should not be any water-logging.
- 6) It should have proper ventilation.

Layout of poultry farm

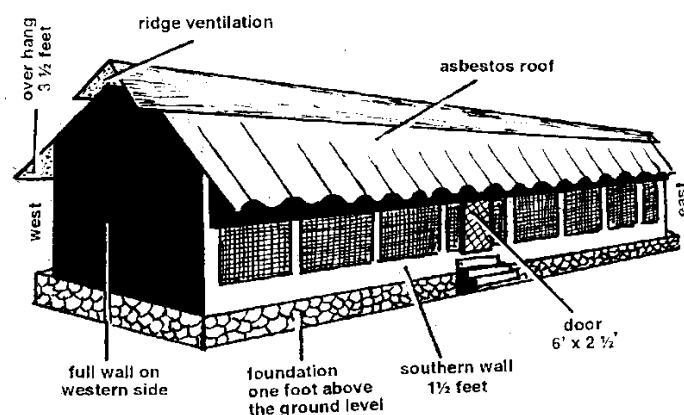
The basic principles to be observed for layout are,

- 1) The sheds should be so located that the fresh air first passes through the brooder shed, followed by grower and layer sheds. This prevents the spread of diseases from layer houses to brooder house.
- 2) There should be a minimum distance of 50-100 feet between chick and grower shed and the distance between grower and layer sheds should be of minimum 100 metre.
- 3) The egg store room, office room and the feed store room should be located near entrance to minimize the movement of people around the poultry sheds.
- 4) The disposal pit and sick room should be constructed only at the extreme end of the site.

DETAILS OF OPEN-SIDED POULTRY HOUSE CONSTRUCTION

1) House Orientation (Direction)

The poultry house should be located in such a way that long axis is in east-west direction. This will prevent the direct sunshine over the birds.



A model poultry house

2) Size

Each broiler require one square foot of floor space while a layer requires two square feet of floor space under deep-litter system of rearing. So the size of the house depends on the number of birds to be reared.

3) Length

The length of the house can be of any extent. The number of birds reared and availability of the land determines the length of poultry house.

4) Width

The open sided poultry houses in tropical countries should have a width not more than 22 to 25 feet in order to allow ample ventilation and aeration at the mid-portion. In environmentally controlled poultry houses, the width of the house may be even 40 feet or more since the ventilation is controlled with the help of exhaust fans.

5) Height

The height of the sides from foundation to the roof line should be 6 to 7 feet (eaves height) and at the centre 10 to 12 feet. In case of cage houses, the height is decided by the type of cage arrangements (3 tier or 4 tier).

6) Floor

The floor should be made of concrete with rat proof device and free from dampness. The floor of the house should be extended 1.5 feet outside the wall on all sides to prevent rat and snake problems.

7) Doors

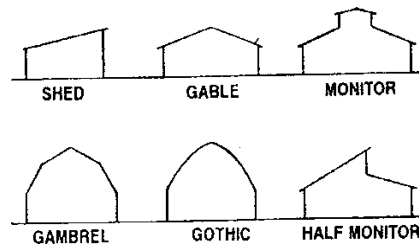
The door must be open outside in case of deep-litter poultry houses. The size of door is preferably 6 x 2.5 feet. At the entry, a foot bath should be constructed to fill with a disinfectant.

8) Side walls

The side wall should be of 1-1.5 feet height, and generally at the level of bird's back height. This side wall protects the bird during rainy days or chill climate and also provides sufficient ventilation. In case of cage houses, no side wall is needed.

9) Roof

The roof of the poultry house may be thatched, tiled, asbestos or concrete one depending upon the cost involvement. Different types of roofs are Shed, Gable, half-monitor, full-monitor (Monitor), Flat concrete, Gambrel, Gothic etc. Gable type is mostly preferred in tropical countries like India.



10) Overhang

The overhang of the roof should not be less than 3.5 feet in order to prevent the entry of rain water into the shed.

SYSTEMS OF POULTRY HOUSING

Poultry can be housed under different systems based on following factors,

- 1) Availability of land
- 2) Cost of land
- 3) Type of farming activity
- 4) Climatic condition
- 5) Labour availability

Broadly, poultry housing systems are classified into three systems:

- 1) Free range or extensive system
- 2) Semi-intensive system
- 3) Intensive system
 - a) Deep-litter system
 - b) Cage system

1) Free range system

This system is adopted only when adequate land is available to ensure desired stocking density by avoiding overcrowding. We can rear about 250 adult birds per hectare. Foraging is the major source of feeding for birds. Shelter is usually provided by temporary roofing supported by ordinary poles. This system is most preferred for organic egg production.

Advantages

- 1) Less capital investment
- 2) Cost of housing is least.
- 3) Feed requirements are less since birds can consume fairly good amount of feed from grass land.
- 4) Fertility of soil can be maintained.

Disadvantages

- 1) The scientific management practices cannot be adopted.
- 2) Eggs are lost when laid inside the dense grasses unless special nests are provided.
- 3) Losses due to predatory animals are more.
- 4) Wild birds may bring diseases unless proper care is taken.

2) Semi-intensive system

As the name indicates birds are half-way reared in houses and half-way on ground or range, i.e. birds are confined to houses in night or as per need and they are also given access to runs. The houses are with solid floors while runs are fields only. The stocking density rate on an average for adult birds is 750 per hectare. The feeding and watering facilities are provided in the pen.

Advantages

- 1) More economical use of land compared to free range system
- 2) Protection of birds from extreme climatic conditions
- 3) Control over scientific operation is some extent possible

Disadvantages

- 1) High cost for fencing.
- 2) Need for routine cleaning and removal of litter material from the pen.

3) Intensive system

Birds are totally confined to houses either on ground / floor or on wire-netting floor in cages or on slats. It is the most efficient, convenient and economical system for modern poultry production with huge numbers.

Advantages

- 1) Minimum land is required for farming.
- 2) Farms can be located near market area.
- 3) Day-to-day management is easier.
- 4) The production performance is higher as more energy is saved due to restricted movements.
- 5) Scientific management practices like breeding, feeding, medication, culling etc. can be applied easily and accurately.
- 6) The sick birds can be detected, isolated and treated easily.

Disadvantages

- 1) Birds' welfare is affected. They cannot perform the natural behaviour like roosting, spreading wings, scratching the floor with legs etc.
- 2) Since they are not exposed to outside sunlight and feed sources, all the nutrients should be provided in balanced manner to avoid nutritionally deficient diseases.
- 3) Chances for spreading of diseases are more.

DEEP LITTER SYSTEM

In this system the birds are kept inside the house all the time. Arrangement for feed, water and nest are made inside the house. The birds are kept on suitable litter material of about 3” to 5” depth. Usually paddy husk, saw dust, ground nut hulls, chopped paddy straw or wood shavings are used as litter materials. The litter is spread on the floor in layers of 2” height every fortnightly till the required is achieved.

Advantages

- 1) Vit B2 and Vit B12 are made available to birds from the litter material by the bacterial action.
- 2) The welfare of birds is maintained to some extent
- 3) The deep litter manure is a useful fertilizer.
- 4) Lesser nuisance from flies when compared to cage system.

Disadvantages

- 1) Because of the direct contact between bird and litter, bacterial and parasitic disease may be a problem.
- 2) Respiratory problems may emerge due to dust from the litter.
- 3) The cost of litter is an additional expenditure on production cost.
- 4) Faults in ventilation can have more serious consequences than in the cage system

Litter management

Qualities of good litter material

- 1) It must absorb moisture from droppings quickly
- 2) It must release moisture and dries up rapidly
- 3) It must have least tendency to form cakes
- 4) It must be free from mould growth
- 5) It must be free from sharp objects and other objectionable materials.
- 6) It must be non-toxic, inert and compressible.
- 7) It should possess good insulating properties and protect chicks from extremes of climate.
- 8) It must be cheap and locally available
- 9) It must be biodegradable, supply some nutrients to the birds and form a good quality manure later.
- 10) It must have medium particle size, soft and light in weight.

Quality of a litter material is tested by picking a hand full of litter material and presses it tightly. A good litter material is the one which breaks up when hand is open and the litter material with more moisture will form a ball in the hand. Too much dried litter material causes dustiness in the farm.

A concrete floor is advised for a perfect disinfection at the end of each batch. About 3-5” of litter should be put into the house initially and more should be added till a depth of about 8-12” is reached. Before adding fresh litter material, remove the caked up and wet litter material. At the beginning the litter contains only 12% moisture. If the moisture level exceeds 30% due to poor ventilation, too many birds, irregular stirring or damp floor, the litter will cease to function efficiently. Sometime many birds will bear “balls” made of litter materials at their claw tips. Ball formation is a sign of bad litter management. Built-up litter has a very important role in providing warmth to the birds in winter. In winter the litter depth can be increased to 10-12” while in summer it may be reduced to 2.5-4”.

CAGE SYSTEM

This system involves rearing of poultry on raised wire netting floor in smaller compartments, called cages, either fitted with stands on floor. It has been proved very efficient for laying operations, right from day-old to till disposal. At present, 75% of commercial layers in the world are kept in cages. Feeders and waterers are attached to cages from outside except nipple waterers, for which pipeline is installed through or above cages. Auto-operated feeding trolleys and egg collection belts can also be used in this rearing system. The droppings are either collected in trays underneath cages or on belts or on the floor or deep pit under cages, depending on type of cages.

Advantages

- 1) Minimum floor space is needed
- 2) More number of eggs per hen can be received
- 3) Less feed wastage
- 4) Better feed efficiency
- 5) Protection from internal parasites and soil borne illnesses
- 6) Sick and unproductive birds can be easily identified and eliminated.
- 7) Clean eggs production
- 8) Vices like egg eating, pecking is minimal.
- 9) Broodiness is minimal
- 10) No need of litter material
- 11) Artificial Insemination (AI) can be adopted.

Disadvantages

- 1) High initial investment cost.
- 2) Handling of manure may be problem. Generally, flies become a greater nuisance.
- 3) The incidence of blood spots in egg is more

- 4) Problem of cage layer fatigue. (It is a condition, in which laying birds in cages develop lameness. It may be due to Ca and P deficiency but the exact reason is not known)
- 5) In case of broilers, incidence of breast blisters is more, especially when the broilers weight is more than 1.5 kg.

Types of cages

Based on the number of birds in a cage, it is classified as

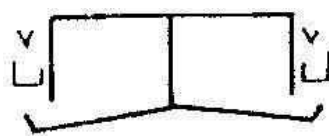
- 1) Single or individual bird cage (Only one bird in a cage)
- 2) Multiple bird cage (From 2 to 10 birds, usually 3 or 4 birds per cage)
- 3) Colony cages (Holding birds more than 11 per cage)

Based on the number of rows

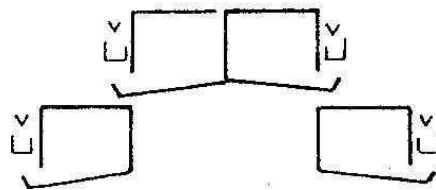
- 1) Single-deck
- 2) Double-deck
- 3) Triple-deck
- 4) Four-deck
- 5) Flat-deck

Based on arrangement of cages

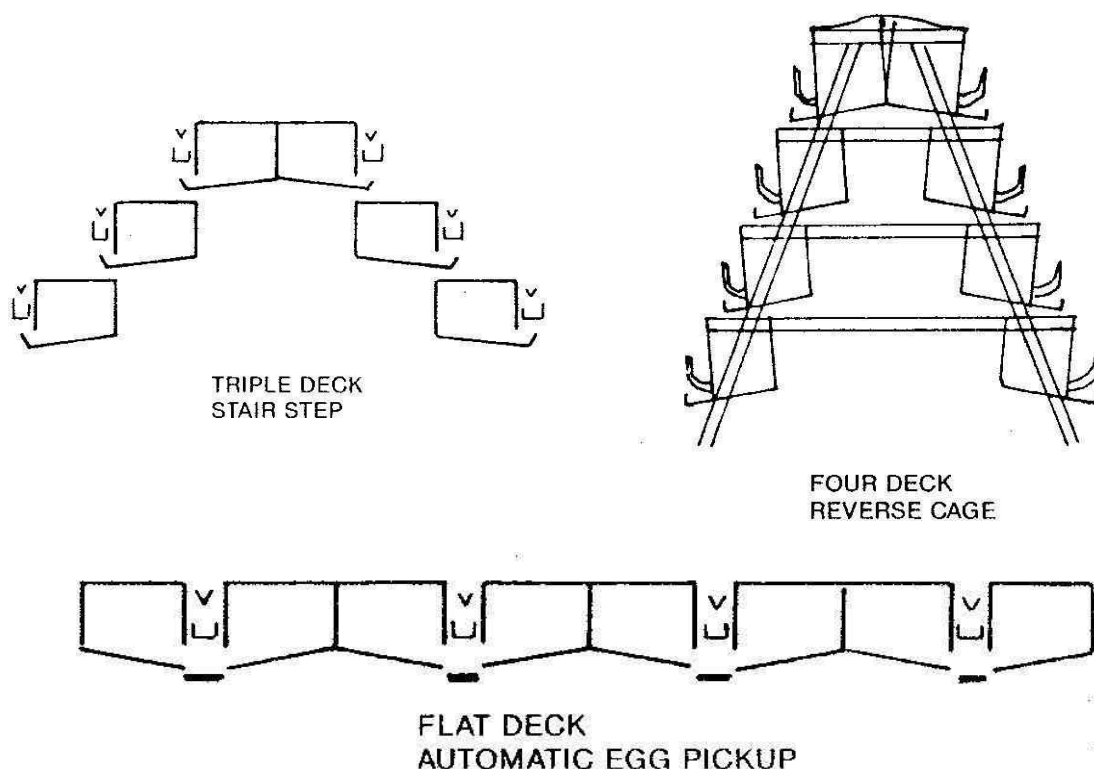
- 1) Stair-step cages
 - a) M-type cages
 - b) L-type cages
- 2) Battery cages (Vertical cages)



SINGLE DECK



**DOUBLE DECK
STAIR STEP**



Based on the type of bird reared

- 1) Brooder / chick cages
- 2) Grower cages
- 3) Layer cages
- 4) Breeder cages
- 5) Broiler cages

1) Brooder cage / chick cage

Specifications:

Front feeding length	: 60 inch
Front & Back height	: 12 inch
Depth	: 36 inch

No. of chicks (0-8 weeks) accommodated per box : 60

Chick cages are arranged either as single deck or double deck system. The feeders and waterers are arranged on outside. Now-a-days nipple drinker system is followed from day-old itself. Newspaper may be spread over the cage floor for first 7 or 10 days. Feed is usually provided inside the cage during the first week of age.

2) Grower cage

Specifications:

Front feeding length	: 30 inch
Front & Back height	: 15 inch
Depth	: 18 inch

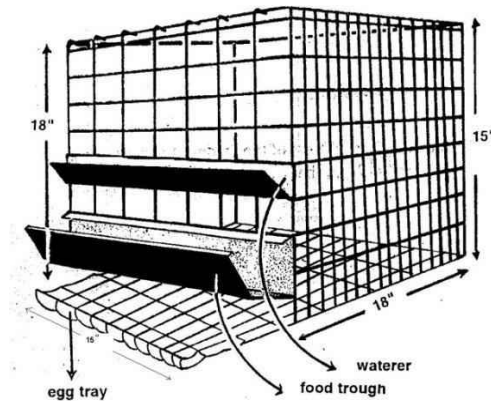
No. of growers (9 to 18 weeks) accommodated per box : 10

3) Layer cage

Two types of stair-step layer cages are commonly used in open-sided poultry houses

- a) Conventional cages
- b) Reverse cages (Californian cages)

a) Conventional cages



Specifications for each box: Front feeding length : 15 inch
Front height : 18 inch
Back height : 15 inch
Depth : 18 inch

b) Reverse cages

Specifications for each box: Front feeding length : 18 inch
Front height : 18 inch
Back height : 15 inch
Depth : 15 inch

These cages can hold 3 to 4 birds. They are arranged either in 2-tier or 3-tier. A slope of 1/6 is provided in conventional cages, where as in reverse cages the slope is 1/5.

Elevated cage layer house

The height of the shed is raised by 6-7 feet using concrete pillars. The distance between two pillars is 10 feet. Two feet wide concrete platforms are made over the pillars. When 3 'M' type cages are arranged 4 platforms are needed. In case of 2 'M' and 2 'L' type cages are arranged 3 platforms are needed. When constructing platforms projecting angles or iron rods to be provided to fix the cages. The inter-platform distance is 6-7 feet depending upon the type of the cages used. The total height of the house is 20-25 feet and the width is 30-33 feet. This type of houses provides sufficient ventilation in tropical countries.

Floor space requirements

Type	Age (in weeks)	Deep-litter (ft ²)	Cages (ft ²)
Egg-type chicken	0-8	0.60	0.20
	9-18	1.25	0.30
	>18	1.50	0.50
Meat-type chicken	0-4	0.30	-
	4-8	0.75	-

BROODING OF CHICKS

Brooding is the art and science of rearing baby chicks. A newly hatched chick does not develop the thermoregulatory mechanism fully and takes about two weeks to develop this mechanism and homeostasis. Therefore, they cannot maintain the body temperature properly for the first few weeks of life. Brooding can be classified into natural and artificial brooding.

Natural brooding

It is done with the help of broody hens after hatching, up to 3 to 4 weeks of age.

Artificial brooding

In artificial brooding large number of baby chicks are reared in the absence of broody hen. Equipments used for brooding are called brooders. Brooder comprises of three elements:

- 1) Heating source
- 2) Reflectors
- 3) Brooder guard

Heating source may be electrical; gases like natural gas, LPG and methane, liquid fuel like kerosene, solid fuel like coal, wood can be used as a heating material.

1) Charcoal pot:

Where electricity is not available, ordinary charcoal pot are used to provide supplementary heat to chicks. These pot are covered with plate / pans to dissipate the heat.

2) Gas brooder:

Natural gas, LPG or methane is connected to heating element which is hanged 3 to 5 feet above the chick to provide heat.

3) Electrical bulb:

One electrical brooder can be used for 300 to 400 chicks at the rate of 1 watt/bird.

4) Infra-red bulbs:

One 250 watts IR bulb can provide brooding for about 150 to 250 chicks.

5) Reflectors:

These reflectors are called Hovers. Flat type hover – These hovers are provided with heating element, heating mechanism and pilot lamp. Canopy type hover – These reflectors are in concave shape consisting of ordinary electrical bulb and thermostat mechanism.

6) Brooder guard / chick guard

They are used to prevent chicks from straying too far away from heat supply until they learn the source of heat. We have to provide brooder guard with a diameter of 5 feet, height of the brooder should not exceed 1.5 feet. For this purpose, we can use materials like cardboard sheet, GI sheet, wire mesh, and mat etc. depending upon the season of brooding. During winter season, brooding is done for 5-6 days. In summer season it is 2-3 weeks.

Receiving of chicks

- 1) After culling the previous adult birds, clean and disinfect the poultry house.
- 2) 3 to 4 weeks interval may be provided between 2 batches as down time.
- 3) Form a circle of about 5 feet diameter with brooder guard. The 5 feet diameter brooder can hold about 200 to 250 chicks.
- 4) At the centre of brooder guard, provide any one of heat source like IR bulb, ordinary incandescent bulb or gas brooders.
- 5) Spread litter material about 2" height in a circle and then spread old newspaper over the litter material.
- 6) Arrange feeders and waterers alternatively like cart-wheel fashion.
- 7) Check the brooder for proper temperature 24 hours prior to arrival of chicks.
- 8) Switch on the brooder heating source several hours before the arrival of the chicks in order to maintain required brooding temperature.
- 9) Spread ground maize or rava or fine mash / crumble feed on the old newspaper for 1 or 2 days. Afterwards, they will learn to consume feed from the feeder.
- 10) Provide electrolyte, glucose and vitamins in the drinking water for first 2 to 3 days to overcome stress. After arrival of chicks, moist the beak and leave the chicks under heating source.
- 11) Maintain a brooder temperature of 90 to 95°F for the first week and then reduce 5°F every week until it reaches the room temperature.
- 12) Watch the behaviour of chicks in order to find out whether temperature provided is correct or less or more. In case of too much temperature, we can reduce the heat by reducing the power of the bulb or we can raise the heating element. In case of too low temperature, we have to supplement more heating source or we can further down the heating element. In case of chill weather or chill breeze, we can provide curtains towards the wind direction.

13) Remove the old newspaper after 3 days and destroy it by burning. If necessary, spread another set of newspaper.

14) Remove brooder guard after 7 to 10 days depending upon the season. While removing the brooder guard, see that the corners of the sheds are rounded in order to avoid mortality due to huddling.

15) Change the feeders and waterers according to age and requirement.

16) 24 hours lighting programme may be adopted during 0-8 weeks of age. One hour darkness may be provided to train the chicks in case of any power failure.

17) Medication programme: First and Second day – Electrolytes and vitamins. 3rd to 7th day – Antibiotics. (Other medications as and when required)

GROWER MANAGEMENT

1) Proper cleaning and disinfection of grower house is needed before introduction of grower birds.

2) Provide sufficient floor space, feeding space and water space.

3) Spread litter material to a height of 4" in case of deep-litter system.

4) Arrange feeder and waterers in the grower house.

5) Change the feeder and waterer according to the need.

6) Adopt restricted feeding programme during growing period to prevent fattening of pullers and early sexual maturity and thus to improve egg production.

7) Follow good litter management to avoid diseases like coccidiosis.

8) Only 12 hours lighting programme is sufficient in case of open-sided houses. No artificial light is needed.

9) Aim for the uniformity of the flock. Sample weights are taken once in a week to find out the average body weight as per the breeder suggestions.

10) Follow strictly the recommended vaccination, medication and other management programs like deworming, debeaking etc for the growers.

Restricted feeding

It is adopted during growing period of layers or breeders. There are two types of restricted feeding.

1) Quantitative feed restriction

In which the amount of feed is reduced below the normal requirement of birds. This can be done on day-to-day basis or skip-a-day programme or skip-two days in a week programme. But this restriction depends on the matching of the flock average body weight with standard body weight provided by the breeder.

2) Qualitative feed restriction

In which the quality of the feed is reduced below the standard requirement of the bird. This can be done by including unconventional feeds or lesser nutrient feed ingredients in place of high protein or high energy diet. Here the quantity of allotment to the bird is not restricted.

During restricted feeding programme, provide more number of feeders and see that all the birds are taking feed simultaneously or otherwise dominant birds will take more amount of feed and the weaker will be subjected feed deprivation and hence the uniformity will be affected.

Advantages of feed restriction during growing period

1) A considerable saving on feed cost because, only 80% of the calculated feed requirement will be offered.

2) They are likely to consume less feed per dozen eggs even during laying period when they are offered *ad libitum* feed.

3) The pullets accumulate less fat and therefore produce more eggs.

4) It is easier to identify weaker birds at an early age during feed restriction. Culling of such birds helps not only saving feed but also promoting layer house survivability because, healthier birds will be moving to laying house.

5) Layers feed-restricted during growing period have been found to produce heavier eggs in longer clutches than those fed *ad libitum*.

Uniformity

At a given age, growing pullets should have average body weight very closer breeder recommendations and at least 70% of the birds' weight within 10% of flock average. Points to be considered for getting uniformity among growing pullets are,

- 1) Receive chicks of uniform weight.
- 2) Provide proper feeding, watering and floor space.
- 3) Change the feeder and waterer according to the age.
- 4) The height of the feeder and waterer should be at the back height of the bird.
- 5) Provide proper energy in the diet.
- 6) Sample weights of the pullets are taken at regular intervals and change the feed accordingly.
- 7) Provide proper feeding space, so as to all birds consume feed simultaneously.

LAYER MANAGEMENT

Points to be considered during layer management are,

- 1) Proper cleaning and disinfection of layer house.

- 2) Provide proper floor space, feeding space and watering space both in deep-litter and cage system
- 3) In deep-litter system, floor space of 2 sq.ft. per bird and feeding space of 5" per bird are provided
- 4) In cage system 4 birds/box of 18" x 15" cage floor space is provided (0.46 sq. ft per bird)
- 5) 6 feet linear feeder can be used for every 30 layers or 18" diameter circular feeder of 4-5 no for every 100 birds.
- 6) Provide 18" diameter plastic waterer of 2 numbers for every 100 birds.
- 7) Spread litter material, in case of deep-litter system up to 6" thickness.
- 8) Arrange feeder and waterer in the poultry house to the height of birds' back.
- 9) Grill size may be changed according to the size of the birds head.
- 10) Provide nest box for every 5 layers about a week before the first egg is laid.
- 11) There are three types of nest: 1) Individual nest – One nest box is sufficient for 4-5 birds. 2) Community nest – This will accommodate 50-60 birds. 3) Trap nest – This will accommodate 1 bird at a time and is used for academic and breeding studies.
- 12) The nest should be provided with litter material. The litter material has to be replaced at least once in a week to prevent contamination of the eggs. During night hours the nest should be closed to prevent sitting of birds in the nest.
- 13) In deep-litter system, the litter material should be raked in the evening daily after egg collection is over. The litter should be treated chemically at least once in a month or whenever necessary in case of wet litter problem to prevent ammonia emission in the house.
- 14) Provide 16 hours light during laying period.
- 15) Provide well-balanced layer mash. Phase feeding may be followed for layers according to age, level of production and climatic factors. The average feed consumption during laying period ranges from 100-110 gram.
- 16) Feed consumption during winter increases and during summer, feed consumption decreases. Summer and winter management should be followed for better flock percentages.
- 17) Deworming should be done regularly at an interval of 6-8 weeks depending on the worm load, especially when reared under deep-litter system.
- 18) Collect the eggs at least 5 times a day in deep-litter system and twice a day in cage system.
- 19) Cull the unproductive layers regularly.
- 20) Broody hen may be separated and this broodiness may be eliminated by,

- a) Providing diet containing more protein, vitamin and minerals.
- b) Light may be provided through out night.
- c) Birds may be placed in cages
- d) Birds may be subjected to stress
- e) Liver stimulant may be provided.

Vaccination schedule for layers

S.No.	Age	Name of the vaccine	Route of administration
1.	1st day	Marek's Disease vaccine	Subcutaneous injection at Hatchery
2.	7th day	Ranikhet Disease F Strain/Lasota. RD killed.	Eye drop or Nasal drop.0.2 ml S/C.on the same day
3.	14 to 16 days (II week)	Infectious Bursal disease(live) IBD(killed)	Eye drop 0.2 ml. S / C on the same day
4.	21 to 24 th day (III week)	Infectious Bronchitis	Eye drop
5.	30 to 35 days	Ranikhet disease-Lasota strain	Eye drop
6.	42 to 45 days	Infectious Bursal disease (live)	Eye drop
7.	56 to 70 days (8-10 th week)	Ranikhet disease "K" (Mesogenic)	Subcutaneous
8.	84 to 91 days (12 - 13th week)	Fowl Pox vaccine	Wing web puncture or Intramuscular
9.	91 to 98 days (13 to 14th week)	Infectious Bronchitis Vaccine	Through Drinking Water
10.	126 to 133 days	Ranikhet disease K" (Mesogenic)	Subcutaneous Injection
11.	After peak production , every 8 Weeks	Ranikhet Disease Vaccine "Lasota"	Through Drinking Water

Takes:

After week of fowl pox vaccination, an inflamed area is seen at the site of vaccination. It is called as takes.

BROILER MANAGEMENT

Definition

It is a bird of about 8 weeks of age of either sex (straight-run chicks) with an average body weight of 1.5 to 2.0 kg with a flexible breast bone cartilage, pliable and tender meat.

Housing systems

Broilers can be housed on deep-litter, slatted or wire floor or cages. However, cage, slat and wire floor rearing of broilers are not as popular as litter floor rearing, due to problems like breast blisters, leg weakness and higher initial investment.

Rearing systems

The systems of rearing refer to either single batch at a time (all-in all-out system) or multiple batches of brooding and rearing of broilers.

All-in all-out system

Under all-in all-out system, the farm will have only one batch of broilers, belonging to the same hatch at any time. Sufficient chicks will be purchased to accommodate the entire farm capacity, reared and marketed in a single lot.

This system is more hygienic, lesser sub-clinical infections and horizontal spreading of diseases and thereby lesser mortality rate, better growth rate and improved feed efficiency. However, this system is not suitable for large scale farming and needs higher fixed and working capital per bird.

Multiple batch system

The multiple batch system consists of rearing of more than one batch of chicks at any time, with a batch interval of 1 to 4 weeks. Here, the farmer is buying day-old chicks and selling grown up broilers at weekly, fortnightly, once in three weeks or at monthly intervals. The chicks are reared for five to six weeks of age, or until they attain the desired body weight and sold for table.

The ideal system for India at present is having 5 to 6 batches of broilers at any time, with weekly interval between batches and “direct retail marketing”. Here, the birds will be marketed daily, from 40 to 54 days of age, based on their body weight, i.e. heavier birds will be sold earlier; giving a chance for weaker birds to have a compensatory growth.

Floor space, feeder space and waterer space

The floor space requirement of broilers varies depending on their body weight at the time of marketing, housing systems, marketing age and ambient temperature. The feeder and waterer space also varies depending on the environmental temperature and health condition of the birds. The following is the approximate floor, feeder and waterer space requirement for broilers.

Age	Floor space/ bird	Feeder space/ bird	Waterer space / bird
Up to 18 days	450 cm ² (0.5 sq.ft.)	3 cm	1.5 cm
From 19 days to 42 days	1000 cm ² (1.1 sq.ft.)	6-7 cm	3 cm

Brooding and rearing of broilers

It is similar to that for egg-type chicks.

Cage rearing of broilers

Broilers can also be reared on cages. Broiler cages are similar to that of grower cages. To prevent the breast blisters, the bottom of the cage may be coated with some plastic materials. The floor space requirement in cages is 50% of the floor space needed in deep-litter. The relative advantages and disadvantages of cage rearing of broilers are,

Advantages

- 1) Higher density of rearing possible
- 2) Easy to catch the birds at market time and hence reduces bruising
- 3) No expenditure on litter
- 4) No incidences of coccidiosis
- 5) Reduced cannibalism
- 6) Cleaning and disinfection easier
- 7) Better growth and feed efficiency

Disadvantages

- 1) Higher incidences of breast-blisters which increases carcass condemnations
- 2) Higher incidences of crooked keel
- 3) Wing bones will be more brittle which will be a disadvantage for the processor also.
- 4) Birds are not having access to the unidentified growth factors in deep-litter system.
- 5) Cleaning faecal-trays is not labour friendly.
- 6) High initial investment on cages.
- 7) Birds will be uncomfortable especially during summer

Feeding

Generally, three types of feed are offered to broilers from day-old to marketing.

0-2 Weeks – Broiler Pre-starter mash / crumble

3-4 Weeks – Broiler Starter mash

5-6 Weeks – Broiler Finisher mash

Lighting

In open sided broiler houses, the most commonly recommended lighting programme is 24 hours light during brooding period, followed by 23 hours light and one hour darkness per day, until marketing. This one hour darkness is to train the birds to acclimatize for darkness, in case of power failure, which may cause panic and trampling.

Vaccination schedule

S. No.	Age	Vaccine	Route of administration
1	First day	Marek's (at hatchery)	S/C at neck
2	5-7 th day	RDV F1	I/O or I/N
3	14 th day	IBD Vaccine	I/O or I/N
4	21 st day	RDV Lasota	Drinking water
5	28 th day	IBD Vaccine (Booster)	Drinking water

NUTRIENT REQUIREMENT OF POULTRY

Poultry being simple stomached species, cannot synthesize most of the nutrients required for them and so the nutrients become dietary essentials chicken has to be fed adequate quantities of balanced diet for its growth, livability and to exhibit its genetic potential to the full extent. Poultry differ from other species of livestock in body temperature and digestion

Poultry feed is composed of 60-65% Energy giving materials 30-35% of Protein source 2-8% Minerals source and above all water. Which is considered as the Principal nutrient should be pure, whole some, free from physical impurities, toxic substances and Bacterial contamination. Water: feed ratio 2.2: 1. It is variable with age, climate, feed and physiological activity. Excess energy is stored as body fat.

Energy source

Yellow maize

cumbu

chulam

Rice polish.

This energy materials constitute 50% of the ration.

Protein :

Protein is required by the bird for 1. Growth 2. Maintenance of body tissues 3. Production.

Both vegetable and animal proteins are used in the feed.

Vegetable Protein:

Ground nut oil cake

Soyabean oil cake

Gingelly oil cake

Sunflower oil cake

Mustard oil cake

This is added at 15-25% in the ration. It is always advisable to add two or more for better balancing.

Animal Protein:

Fish meal, meat meal, Blood meal & worm pupa meal out of this fish meal is ideal this forms 5-10% of the ration.

Fibre

Grain by products like bran are included from 10-30% for fibre, bowel movement and minerals. If molasses available it can also be added for energy at 5-7% levels, which is a cheap source of energy, reduces dustiness, improves palatability.

Mineral mixture for poultry

Included at 2.3% it is advisable to use salt free mineral mixture because fish meal available in our country is salted. Calcium supplements such as shell grit, calcite, limestone, etc. are used at 4-5% levels.

Standard requirement of nutrients:

	Chick	Grower	layer mash	Broiler starter	Broiler Finisher
Crude Protein % min.	22	16	18	23	20
Crude fibre % max.	7	8	10	6	6
Calcium % min.	1	0.8	2.75	1	1
Total phosphorus % min	0.7	0.6	0.75	0.7	0.7
Metabolizable Energy Kcal/kg.	2800	2600	2700-2750	3000	3100
Lysine (%min.)	1.0	0.7	0.5	1.0	1.0
Methionine (% min)	0.35	0.25	0.25	0.35	0.35

Approximate feed intake by commercial chicken in tropics

	Egg type (gm.)	Broiler (G.)
1 st week	10	15
2 nd week	15	25
3 rd week	20	35
4 th week	25	50
5 th week	30	65
6 th week	35	85
7 th week	45	105
8 th week	50	120

9-12 weeks – 40

13-16 weeks – 50 (Restricted feeding)

17-20 weeks – 60

During laying

0% Egg Production	80 gm.	70 %	115
25% Egg Production	95 gm.	80 %	120
50% Egg Production	105gm.	Over 80%	12
60% Egg Production	110 gm.		

Guide lines for feed management

- Purchase quality ingredients / feed with least moisture and devoid of adulteration.
- If own feed is mixed formula may be modified depending upon the cost and availability of ingredients
- If agricultural farm is attached farm grown grain can be used
- Purchase one month or two months requirements.
- Screen the feed store room against rodents, sparrows other vermins.
- Observe the feed intake by the birds during summer, winter.
- If moisture level exceeds 15% (except molasses) during hot season it may cause growth of fungus and precipitate problems like aflatoxicosis.
- The feeder should not be filled to minimize the feed wastage.

FACTORS INFLUENCING THE NUTRIENT REQUIREMENTS

There are many factors which influence the nutrient requirement of poultry, they are

- i. Species variation
- ii. Breed and strain (genetic factor)
- iii. Age of the bird

- iv. Purpose of the bird
- v. Environmental temperature (season)
- vi. Water consumption
- vii. Quality of nutrients
- viii. Disease conditions
- ix. Freshness of feed
- x. Intensity of production
- xi. Value added products
- xii. Nutrient inter relationships

i. Species variation

The vit-B complex requirement is high for Japanese quails and turkey when compared to chicken. If its requirement is high, the protein requirement is also high.

Geese and emu can utilize high fibre and low energy diets as they can
Synthesize many B complex vitamins.

Ducks can utilize fish scales.

ii. Breed and strain (Genetic factor)

Heavy breeds require less percentage of protein than light breeds.

White leghorn requires more methionine than Australorp.

Lysine and arginine requirement also differ from breed to breed and strain to strain.

In general, there is a specific difference among different breeds and strains for most of the nutrients.

iii. Age of the bird

Poultry during first 21 days of life have higher requirements for some nutrients than during the growing period. The requirement of vitamin-A increases with age. The requirement of riboflavin is similar during growing and breeding period and less during growth and production of table egg. Calcium requirement during egg production is 3-4 times more than during the starting and growing period. However, trace mineral requirement is similar throughout the life.

iv. Purpose of the bird

Breeders must receive a high dietary level of vitamins than table egg producers. The egg must be fortified with vitamins to ensure better hatchability and livability of newly hatched chicks. The three most critical vitamins needed in large amounts in breeder ration are riboflavin (B2), vit-A and vit-B12. Breeders 2-3 times more of riboflavin, 20% more of vit-A and 2 times that of vit-B12 than for commercial layers.

v. Environmental temperature (season)

When the environmental temperature increases, the requirement for protein increases and for energy decreases. Since there is reduction in feed intake in warm weather, all nutrients should be increased in proportion to the expected reduced feed intake.

vi. Water consumption

The water consumption of broilers and layers has been found to be a function of feed consumption. Broilers consume 1.5 times more water than feed whereas layers consume 2 times more water than feed consumption.

vii. Quality of nutrients

Nutrient requirements depend on qualities like biological value of ingredients and presence of toxins.

viii. Disease condition

Disease condition like FLKS influences the nutrient requirement. To control this, energy content of the diet has to be reduced and protein content has to be increased. Lipotropic factors like choline chloride have to be included in the diet. In coccidiosis, high protein adversely affects the birds, so the protein level can be reduced. In general, most of the diseases cause lower feed consumption and hence care has to be taken to include the required nutrients depending upon the feed consumption.

ix. Freshness of the feed

Fresh feed is consumed well than stale/old feed. In fresh feed nutrient loss is less.

x. Intensity of production

Birds producing more than 90% require 20% protein

Birds producing 80% requires 18% protein

Birds producing 70% requires 16% protein

xi. Value added products

For production of designer eggs birds need vit-E, selenium, carotenoid pigments and omega 3 fatty acids.

xii. Nutrient inter relationship

Energy – protein interrelationship

Vit-D3 – Ca – P inter relationship

Choline – methionine – vit-B12 interrelationship.

INCUBATION AND HATCHING

Incubation is the act of bringing an egg to hatching. It may be either natural or artificial.

Natural incubation

In this method, broody hens are made to sit over hatching eggs to hatch out chicks. A hen can sit over hatching eggs to a longer period without feed and water due to some hormonal changes in its body. *Prolactin* is responsible for broodiness in chicken.

Artificial incubation

In this method, mechanical equipments (incubators) are used for hatching purpose. Incubators are placed in a building called hatchery.

DIFFERENT TYPES OF INCUBATORS

I. Based on air circulation:

- 1) Still air incubators
- 2) Forced draft incubators

II. Based on heating source:

- 1) Hot air incubator
- 2) Hot water incubator

III. Based on fuel used

- 1) Gas operated incubator
- 2) Oil operated incubator
- 3) Electrically operated incubators.

STEPS INVOLVED IN COMMERCIAL HATCHERY OPERATIONS

Receiving cleaned hatching eggs



Fumigation of eggs for sanitation

(3x concentration for 20 minutes)

(3x means 60g KMnO₄ and 120 ml formalin for 100 cu.ft.)



Storage in egg holding room

(65°F temperature and 75% R.H.)



Pre-incubation warming at Room temperature

(4 to 6 hours)



Loading eggs in the setter

(For first 18 days)

☐

Candling of eggs for removing infertile eggs

☐

Transfer to the hatcher

(Last 3 days)

☐

Pulling out the chicks

(at 90% dry
condition)

☐

Sexing

(Only for layer chicks) (Vent sexing / feather sexing)

☐

Grading

☐

Vaccination

(Marek's vaccine – s/c on neck – 0.25 ml)

☐

Delivery

Location

(Plastic / cardboard chick box with proper ventilation and
comfort)

HATCHERY DESIGN

The chick hatcheries are modern buildings that provide separate rooms for each hatchery operations, but each room has its individual requirements. The hatchery area should be a separate unit with its own entrance and exit, unassociated with those of the poultry farm. The hatchery should be situated at least 1000 ft from poultry houses.

Size of the hatchery

The size of the hatchery is based on the egg capacity of the setters and hatchers, number of eggs that can be set each week and number of chicks hatched each week. Also, necessary space to be allotted for future expansion.

Hatchery design

Hatchery should be constructed in such a manner that the hatching eggs may be taken in one end and the chicks removed at the other. There should not be no backtracking

Hatchery rooms / Structures

Hatchery rooms must be adequate in size. Usually, hatcheries of medium size will hatch chicks twice a week, but large hatcheries will hatch more than two hatches per week. Consequently, hatching schedules will affect the size of some rooms in the hatchery. The Floor

space of the various hatchery rooms when there are two hatches per week are,

1) Shower room

To maintain bio-security it is essential that all persons entering the premises shower and change into clean clothing in an adjoining room. It is the only entrance and exit, and the hatchery becomes an isolated unit as far as human beings are concerned.

2) Hatching eggs receiving counter

Employees delivering hatching eggs to the hatchery must not enter the hatchery in the course of their duties. Eggs should be delivered to the hatchery through a specialized door.

3) Fumigation room

The fumigation room should be as small as possible in order to reduce the amount of fumigant used. A fan should be used to circulate the air and exhaust the fumigant.

4) Egg holding (Egg-cooler) room

Egg holding room should be about 8 ft high, insulated, slowly ventilated, with complete air movement, cooled, and humidified. The room must be refrigerated to maintain a temperature of 65 °F. A forced-air type of refrigeration unit is required in order to keep a uniform temperature throughout the room.

5) Pre-incubation warming room

Here eggs are kept for the purpose of drying the ‘sweat’ over eggs. It can be achieved by providing sufficient number of ceiling fans in this room.

5) Setter room

Setters (incubators) are kept in this room. The size of the setter room will depend on the make of the equipment used. The exact room size involves the aisle and working area necessary to move the eggs and chicks in and out of the machines. A minimum space of 3 ft should be allotted between the sides of adjacent setters and from wall to sides or back of the setters. Similarly, minimum of 10 ft should be allotted in front of two setters when kept face-to-face arrangements.

6) Egg candling (Dark) room

This room is usually constructed in between setter and hatcher room for candling eggs. Candling is usually practiced when eggs are transferred from setter to hatcher. Provisions should be made to dark the room to facilitate easy candling.

7) Hatcher room

Hatchers are kept in this room. Here sufficient spaces are to be allowed around hatcher similar to that of setter room. Since it is prone for contamination with fluffs and debris at the time of hatching, the door towards setter room is to be tightly closed unless the necessity arises.

8) Chick-holding room

Next to hatcher room, chick-holding room is present. A relative humidity of 65% is

maintained to prevent excessive chick dehydration. Here, the chicks are sex-separated, graded, vaccinated and placed in chick boxes.

9) Wash room

After chicks are boxed, the trays are washed in a tray washer in the washroom. Necessary pipelines with high-pressure pumps are kept in this room.

10) Clean room

After the trays are washed, they are placed in their trolleys and moved to the adjacent clean room.

PRINCIPLES OF INCUBATION

Five major functions are involved in the incubation and hatching of poultry eggs. They are:

- ❖ Temperature
- ❖ Humidity
- ❖ Ventilation (Oxygen and Carbon dioxide level and air velocity)
- ❖ Position of eggs
- ❖ Turning of eggs

1. Temperature

Temperature is the most critical environmental concern during incubation. Embryo starts developing when the temperature exceeds the Physiological Zero. *Physiological zero* is the temperature below which embryonic growth is arrested and above which it is reinitiated. The physiological zero for chicken eggs is about 75°F (24°C). The optimum temperature for chicken egg in the setter (for first 18 days) ranges from 99.50 to 99.75 ° F and in the hatcher (last 3 days) is 98.5 ° F.

2. Humidity

Incubation humidity determines the rate of moisture loss from eggs during incubation. In general, the humidity is recorded as relative humidity by comparing the temperatures recorded by wet-bulb and dry-bulb thermometers. Recommended incubation relative humidity for the first 18 days ranging between 55 and 60% (in setter) and for the last 3 days ranging between 65 and 75%.

3. Ventilation

Ventilation is important in incubators and hatchers because fresh oxygenated air is needed for the respiration (oxygen intake and carbon dioxide given off) of developing embryos from egg setting until chick removal from the incubator. Generally the oxygen content of the air in the setter remains at about 21%. For every 1% drop in oxygen there is 5% reduction in hatchability.

Carbon dioxide is a natural by-product of metabolic processes during embryonic development and is released through the shell. The tolerance level of CO₂ for the first 4

days in the setter is 0.3%. CO₂ levels above 0.5% in the setter reduce hatchability and completely lethal at 5.0%.

4. Position of eggs

Artificially incubating eggs should be held with their large ends up. It is natural for the head of the chick to develop in the large end of the egg near the air cell, and for the developing embryo to orient itself so that the head is uppermost. Eggs positioned horizontally will incubate and hatch normally as long as they are turned frequently. Under normal circumstances eggs are set with large end up for the first 18 days (in setter) and in horizontal position for the last 3 days (in hatcher).

5. Turning of eggs

Birds, including chickens and quail, turn their eggs during nest incubation. Nature provides nesting birds with the instinct of turning eggs during incubation. Similarly eggs to be turned at least 8 times a day. Turning of eggs during incubation prevents the developing embryo adhering to the extra-embryonic membranes and reduces the possibility of embryo mortality. In large commercial incubators the eggs are turned automatically each hour i.e. 24 times a day. Most eggs are turned to a position of 45° from vertical, and then reversed in the opposite direction to 45° from vertical.

HANDLING OF HATCHING EGGS AND STORAGE

The quality of hatching egg cannot be improved after lay but one can reduce the loss in hatching egg quality by adopting some standard procedures.

Maintaining egg quality in the breeder house

Use of enough clean, dry and mold-free nesting material can avoid cracked and dirty eggs. Similarly hens to be trained to use nests to lay eggs instead of laying on floors by providing sufficient number of nest boxes well in advance before the laying starts. The frequency of hatching egg collection is very important to maintain quality. Hatching eggs should be collected at least 4 times a day. Hatching eggs are susceptible to contamination and every effort must be made to reduce the microbial load.

Selection of hatching eggs

Eggs that are cracked, dirty or misshapen are usually not used for hatching. Very small or very large eggs do not hatch as well as eggs in the middle size range. Eggs with thin or very porous shells are not likely to hatch well because of excessive losses of water during incubation.

Reducing contamination of hatching eggs

Fumigation with formaldehyde gas is an effective method of sanitizing hatching eggs. Solutions containing quaternary ammonium compounds, formalin, hydrogen peroxide or phenols may be moderately effective in reducing the microbial load over hatching eggs. DO

NOT wash eggs unless necessary. If it is necessary to wash eggs always use a damp cloth with water warmer than the egg.

Storage of hatching eggs

In normal hatchery operations, eggs cannot be set immediately after they are laid. Many hatcheries set eggs once or twice in a week. If hatching eggs are stored up to 1 week, hatching eggs should be kept in an egg holding room with the temperature of 65°F and the relative humidity of 75%.

HATCHERY OPERATIONS

The operation of a chick hatchery involves the production of the largest number of quality chicks possible from the hatching eggs received at the hatchery.

The sequences of hatchery operations followed in commercial hatcheries are,

- | | |
|-------------------------------|--------------------------|
| 1) Securing hatching eggs | 9) Pulling the hatch |
| 2) Traying hatching eggs | 10) Hardening |
| 3) Fumigation | 11) Grading |
| 4) Cold Storage | 12) Sexing |
| 5) Warm eggs prior to setting | 13) Vaccination |
| 6) Loading of eggs | 14) Chick delivery |
| 7) Candling | 15) Washing and cleaning |
| 8) Transfer of eggs | 16) Disposal of waste |

1) Securing hatching eggs

Hatcheries can get the hatching eggs from any one of the following ways:

- 1) From own breeder flock
- 2) From other breeder flocks
- 3) From other hatcheries

2) Traying hatch eggs

The eggs from the breeder flocks should be transferred to the egg setter trays in the hatchery immediately after receiving.

3) Fumigation of hatching eggs

After traying, eggs are to be kept in the fumigation chamber for fumigation. Fumigating with 3x concentration of formaldehyde for 20 minutes will kill about 97.5 to 99.5% of the organisms on the shells. One 'x' concentration means 20 g of KMnO₄ with 40 ml of formalin for 100 cubic feet (3x means 60 g of KmnO₄ + 120 ml of formalin for 100 cubicfeet).

4) Cold Storage

When the eggs are not set immediately after receiving, they should be kept in cold rook at the temperature of 65 °F and 75% relative humidity.

5) Warm eggs prior to setting

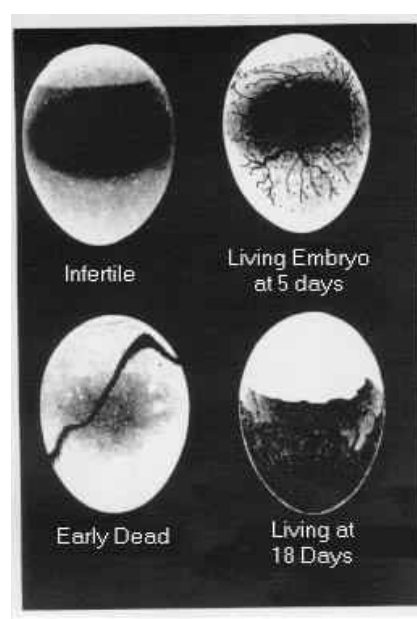
Approximately 6 hours prior to placing eggs in the setter they should be moved from the egg-cooler room to normal room temperature. Here, atmospheric air condenses over eggshell and form water droplets over eggshell, which is called as ‘*Sweating*’.

6) Loading of eggs

Placing of eggs in the setter is called ‘*Loading of eggs*’. Eggs can be set in the setter either all-in all-out basis or batch basis. Most of the commercial hatcheries are practicing batch system of loading eggs in the setter that will minimize the initial time taken to reach normal incubation temperature in the setter.

7) Candling

Candling is a process in which eggs are kept in front of a light source to find out the defects in eggshell, embryonic development etc. Under commercial operations, candling is done when the eggs are transferred from setter to hatcher (at 19th day for chicken eggs). There are two methods of candling that can be used. The fastest method involves the use of *a table or mass candler*. An entire tray of hatching eggs may be placed on the mass candler and examined with one observation. Candling with a *spot candler or individual candler* is a little slower, but it is more accurate.



8) Transfer of eggs

In modern incubators, eggs are transferred from setter to hatcher at 19th day of incubation (for chicken egg) or when approximately 1% of the eggs are slightly pipped. In general, one-seventh of total incubation period is needed to keep eggs in the hatcher.

9) Pulling the hatch

The process of removing the chicks from the hatcher is often called *pulling the hatch*. Chicks should be removed from the hatcher as soon as all are hatched and about 95% are dry. Excessive drying in the hatcher should be avoided.

10) Hardening the chicks

When the chicks are first placed in the chick boxes they are soft in the abdomen, are not completely fluffed out, and do not stand well. They must be “hardened” by leaving them in the boxes for 4 or 5 hours. Such hardening makes it easier to grade the chicks for quality, and the chicks are more easily vent-sexed.

11) Grading the chicks

No chick below the minimum standard must be allowed to go to a customer. Some standards for quality are, 1) No chick deformities 2) No unhealed navels 3) Above a minimum weight 4) Not dehydrated and 5) Stand up well.

12) Sexing the chicks

Layer type day-old chicks are needed to be sex separated either by vent sexing or auto-sexing (feather sexing). In case of meat-type day-old chicks sexing is not practiced.

13) Vaccination

Most chicks are vaccinated against Marek's disease in hatchery before delivery. Most common method of vaccination of day-old chicks is by subcutaneous method in the nape of the neck.

14) Chick delivery

Baby chicks should reach the customer's farm early in the morning. Not only the weather is cooler during this part of the day but the early arrival allows a full day for close observation of the chicks by the caretaker.

15) Washing and cleaning

Cleaning the hatchery between hatches is of primary importance. The process must be complete. Except for the setters and setter room, every piece of equipment must be thoroughly vacuumed, scrubbed, disinfected and fumigated.

16) Disposal of waste

Hatchery wastes include infertile and non-hatched eggs and dead and cull chicks that should be disposed in such a manner not to create problem to the neighbors and also not to contaminate the hatchery premises.

IMPORTANT INFECTIOUS DISEASES OF POULTRY

Disease is a condition caused by living factors like viruses, bacteria or parasites or non-living factors such as deficiencies, toxins and other physical or chemical agents. Diseases can be classified depending on the causative factors. Diseases cause severe loss in poultry production, not only from death of birds but also due to loss in production. Some of the common poultry diseases and the symptoms of each are described below

I. BACTERIAL

DISEASES

Coli bacillosis

This is a common bacterial disease caused by *E.coli* organism. It can manifest in different forms such as infection of the yolk sac, colisepticaemia, etc. lethargy and diarrhea will be noticed. Control of infection of the yolk sac depends on sanitary conditions at the hatchery and provision of appropriate warmth during brooding. The practice of efficient

Water sanitation methods and the use of clean drinkers are advised. Provide adequate ventilation and manage the litter properly.

Salmonellosis

This disease is transmitted vertically from parent flocks and affects baby chicks with a high mortality. Huddling, pasting of vent feathers and whitish diarrhea are noticed. Ensure salmonella free chicks from the hatchery by using sterilized fish meal and meat meal. Pelleting of feed will also be helpful. Add antibiotics like furazolidone to the feed to control salmonellosis.

Coryza

Symptoms if this disease is swollen face and eyes filled with cheesy substance. Respiratory distress and poor feed intake are noticeable. Avoid overcrowding and wet litter and reduce the ammonia level by proper ventilation.

Fowl Cholera

Comb and wattle of the affected birds become swollen and bluish. High mortality and morbidity are noticed with a watery nasal discharge and saliva drooping. Pin- point blood spots on the heart and duodenum and white spots on the liver are present. White diarrhea is also noticed. Control the rodents; ensure proper management, good ventilation and stress-free conditions.

Chronic Respiratory Disease

It is caused by micoplasma. Respiratory distress, the sound of rales, sneezing, poor weight gain and nasal discharge are symptoms of the disease. Infected chicks transmit the through feed and water. Avoid overcrowding, improve ventilation, and ensure hatchery sanitation and screening of parent stock. Mixing tiamutin or tylosine in the feed will be helpful.

II. VIRAL DISEASES

These are prevented by carrying out protective vaccinations at recommended age; treatment will not be effective.

Newcastle Disease

It is the most dreadful disease in poultry with a high mortality rate in all age groups. Respiratory distress, paralysis of legs and wings, greenish diarrhea and twisting of neck are noticed. Control IBD and give toxin free feed. Multiple vaccinations should be given as suggested and routine screening of sera samples in large farms are advisable.

Infectious Bronchitis (IB)

Respiratory distress, rales and gurgling sounds are evident. Prevent the disease by vaccination.

Infectious Anaemia

Young chicks are affected with a mortality of up to 20 percent and the disease predisposes adults to other diseases. Vaccination of the parent stock is required.

Infectious Bursal Disease (IBD or Gumboro disease)

Young birds are affected which show symptoms of listlessness, huddling, whitish pasty diarrhea and pecking at vents. The consequences of this disease are 10-60 percent mortality and breakdown of immunity leading to outbreak of other diseases. Vaccinate young birds at the recommended age and include immune stimulants like Vit-E in the feed and the feed should be free from toxin.

Marek's disease

This disease is characterized by poor growth and paralysis of one leg which is stretched backwards. Birds under 16 weeks of age are affected, so unable to take feed followed by death. Vaccination should be done at day old in the hatchery.

Infectious Laryngo Tracheitis (ILT)

Symptoms like respiratory distress, rales, extended neck for inhalation, nasal discharge, low egg production and mortality up to 30 percent are noticeable. Secondary infections can be controlled by antibiotics.

Inclusion Body Hepatitis (Hydro pericardium syndrome or Leechi disease)

This condition affects broilers at 4-5 weeks of age and a 30-60 percent mortality is experienced. Straw-coloured fluid around the heart is noticed. Vaccines are available against this disease.

Fowl pox

Raised skin lesions appear at the corners of the beak, as well as at the comb and wattle. The mortality level is low, but the recovered birds perform poorly. Vaccination at appropriate age is suggested.

III. FUNGAL DISEASES

Brooder pneumonia

Young chicks of 1-2 weeks of age are affected. A high mortality rate of 10-50 percent is noted with symptoms of gasping, respiratory distress and strong thirst. Yellowish white pin-point filaments on lungs are present. Avoid using mouldy, fungus infested litter material. Use thoroughly cleaned drinkers and feeders.

Mycotoxiosis

Caused by mycotoxins produced by fungi in feed ingredients. The symptoms are reduced feed intake, poor growth rate or egg production, breakdown of immunity and vaccination failures are the consequences. Avoid using mouldy feed ingredients, use toxin

binders like zeolites, aluminium silicate or charcoal in the feed. Add liver tonics and withdraw fungal infested feed.

GENERAL DISEASE CONTROL MEASURES:

1. Buy chicks from reputed disease free companies.
2. Adhere strictly to vaccination programme.
3. Keep the houses dry cool and well ventilated.
4. Rodent and fly proof.
5. Sanitation of litter, feeder and waterer.
6. Follow medication schedules.
7. P.M. disposal through burial or incineration of the waste and dead.
8. Earmark areas for specific age group.
9. Screening visitors.
10. Foot baths with sanitizers.
11. All in all out system.
12. Personnel sanitation.

VACCINATION

It is correctly pointed out that "Prevention is better than cure". Many viral diseases cannot be treated but can be controlled only by preventive vaccination.

1) Routes of administration

Administration through 1) Drinking Water: It is time and labour saving method. Vaccine is reconstituted in cold drinking water along with skim milk powder at the rate of 4 gram per litre of water and used immediately. For example RDV Lasota Vaccine.

2) Intra ocular -Intra nasal instillation.

The vaccine is reconstituted in normal Saline solution. One drop of diluted vaccine is applied to the nostrils or eye. Ex: RDVF. The virus particle gets absorbed in the mucous membrane and immunization is obtained.

3) Spray Vaccine

Spray or mist spraying is done in chick boxes in the hatcheries. Small drops of equal size is sprayed and the boxes are allowed for 10 to 15 minutes for drying. Drying should not be done near light or by hot air.

4) Wing web puncture method

Fowl pox vaccine is reconstituted in 50% glycerol saline and taken in forked needle and vaccination is done by puncturing through wing web. Care should be taken that muscle, nerve and blood vessels are damaged by the vaccination.

5) Feather Follicle Method

Pigeon pox vaccine is reconstituted with 50% glycerol saline. After plucking of the feather follicles in the internal thigh region, with the help of a glass rod, the vaccine is

smear and rubbed. After 5 days the birds have to be examined for "Takes". Takes are cellular reaction taking place in the nervous system.

6) Subcutaneous injection

Ranikhet K vaccine is reconstituted with normal saline and 0.5ml is given between two layers of skin in the wing web region without damaging nerves, blood vessels and muscle. The vaccine should be protected in ice box during vaccination and should be used within one hour.

Debeaking

It is recommended to debeak the layer birds to control feather pecking and cannibalism, bullying. It is carried out by means of electrocautery. It is important to remove only one third of the upper beak taking care to avoid tongue. It is usually practiced at the age of 10-14 days and repeated at the age of 14-16 weeks. Debeaking should never be done with penknife.

Overcrowding, inadequate space for standing/ feeding/ watering and resting, starvation, external wounds, less fiber diet and deficiency of vitamins and minerals may predispose the birds to cannibalism.

Procedure

The bird has to be restrained by holding wings and legs by left hand and the tongue is pushed backwards by opening mouth and introducing index finger so that the tongue is not cut. The upper beak is cut to 1/3rd of its length and the lower beak is slightly trimmed. After debeaking vitamins and antibiotics are to be administered for 3-5 days to avoid stress and secondary infections.

Deworming

It is the process of removing worms from digestive tract of the birds. The tapeworm passes segments and is consumed by intermediate host (earthworm, cockroach) where intermediate stage gets developed and passed out, which in turn is consumed by host. The eggs or ova of round worms are passed in the droppings which is picked by other birds directly or indirectly with the help of chance carriers (personnel, insects, flies, ants, etc). Sometimes wild birds such as crows may serve as source of infestation.

Birds show the following symptoms when they are infested with worms

- Dullness- weakness, emaciation
- Paralysis-due to toxins produced from worms
- Enteritis-diarrhea with blood
- Anemia-due to sucking of blood by worms.
- Drop in egg production.

If infestation is on a larger scale there may be mechanical block of intestinal lumen and sometimes rupture occurs. This may also result due to intestinal stasis of food particles.

Deworming is practiced at intervals of 45 days in layer birds and also before RDVK vaccination. Deworming is done against tape worms only on absolute necessity.

Delicing

Is the process of removing of external parasites like ticks, mites and fleas which suck the blood from the bird. The following symptoms are observed during external parasitic infestation: itching, restlessness, external wounds, loss of body weight, weakness, anemia and drop in production.

Procedure

The dipping of the birds in sunny days has to be done with the following chemicals to remove the external parasites.

Sumathion or malathion -5ml in 100ml of water. The bird has to be immersed in the chemical solution avoiding eye and mouth. The dipped one has to be dried in a separate enclosure. The feeders, waterers and building should be sprayed with this chemical solution to remove the external parasites. After dipping, to relieve stress to the bird vitamin A, B complex has to be given to improve the health of the birds.